

Theory of Mathematics

Volume I: Linear Algebra

Contents

I	Vector Spaces	1
1	Scalars, Vectors and Linear Combinations	2
1.1	Fields and scalars	3
1.2	Vector spaces	3
1.3	Linear combinations	3
1.4	Concrete models	3
1.5	Affine versus linear geometry	3
1.6	Complex scalars and realification	4
1.7	Core structural results	4
1.8	Worked examples	4
1.9	Diagnostic checklist	4
1.10	Solved dossiers	5
1.11	Exercises with complete solutions	6
1.12	Graded supplementary exercises	7
2	Subspaces, Span and Linear Independence	11
2.1	Subspaces	11
2.2	Intersections and sums	12
2.3	Span	12
2.4	Linear dependence	12
2.5	Redundancy and exchange	12
2.6	Homogeneous systems	12
2.7	Core structural results	13
2.8	Worked examples	13
2.9	Diagnostic checklist	13
2.10	Solved dossiers	13
2.11	Exercises with complete solutions	15
2.12	Graded supplementary exercises	16
3	Bases and Dimension	19
3.1	Bases	19
3.2	Coordinates relative to a basis	20
3.3	Extension and reduction	20
3.4	Dimension	20
3.5	Dimension of subspaces	20
3.6	Dimension formula	20
3.7	Core structural results	21
3.8	Worked examples	21
3.9	Diagnostic checklist	21
3.10	Solved dossiers	21

3.11	Exercises with complete solutions	23
3.12	Graded supplementary exercises	24
4	Coordinates and Change of Basis	27
4.1	Coordinate isomorphisms	27
4.2	Transition matrices	28
4.3	Inverse transitions	28
4.4	Coordinate covariants	28
4.5	Dual coordinates	28
4.6	Conditioning of bases	28
4.7	Core structural results	29
4.8	Worked examples	29
4.9	Diagnostic checklist	29
4.10	Solved dossiers	29
4.11	Exercises with complete solutions	31
4.12	Graded supplementary exercises	32
5	Linear Transformations	35
5.1	Definition and first consequences	35
5.2	Determination by a basis	36
5.3	Algebra of operators	36
5.4	Composition	36
5.5	Inverse maps	36
5.6	Important operators	37
5.7	Core structural results	37
5.8	Worked examples	37
5.9	Diagnostic checklist	37
5.10	Solved dossiers	37
5.11	Exercises with complete solutions	39
5.12	Graded supplementary exercises	40
6	Kernels, Images and Isomorphisms	43
6.1	Kernel	43
6.2	Image and rank	44
6.3	Rank–nullity	44
6.4	Injective versus surjective	44
6.5	Quotients	44
6.6	First isomorphism theorem	44
6.7	Core structural results	45
6.8	Worked examples	45
6.9	Diagnostic checklist	45
6.10	Solved dossiers	45
6.11	Exercises with complete solutions	47
6.12	Graded supplementary exercises	48
II	Matrices and Operators	51
7	Matrix Representation of Linear Maps	52
7.1	Matrix of a linear map	52
7.2	Column rule	53

7.3	Composition and multiplication	53
7.4	Identity and inverses	53
7.5	Change of basis for operators	53
7.6	Rank and row reduction	54
7.7	Core structural results	54
7.8	Worked examples	54
7.9	Diagnostic checklist	54
7.10	Solved dossiers	54
7.11	Exercises with complete solutions	56
7.12	Graded supplementary exercises	57
8	Determinants and Trace	60
8.1	Alternating multilinear characterization	60
8.2	Permutation formula	60
8.3	Row operations	61
8.4	Invertibility	61
8.5	Geometric meaning	61
8.6	Trace	61
8.7	Core structural results	61
8.8	Worked examples	62
8.9	Diagnostic checklist	62
8.10	Solved dossiers	62
8.11	Exercises with complete solutions	63
8.12	Graded supplementary exercises	65
9	Eigenvalues and Eigenvectors	68
9.1	Eigenvectors and eigenvalues	68
9.2	Eigenspaces	69
9.3	Characteristic polynomial	69
9.4	Algebraic and geometric multiplicity	69
9.5	Independence of eigenvectors	69
9.6	Field dependence	69
9.7	Core structural results	70
9.8	Worked examples	70
9.9	Diagnostic checklist	70
9.10	Solved dossiers	70
9.11	Exercises with complete solutions	72
9.12	Graded supplementary exercises	73
10	Invariant Subspaces and Triangularization	76
10.1	Invariant subspaces	76
10.2	Quotient operators	77
10.3	Invariant flags	77
10.4	Triangular matrices and flags	77
10.5	Triangularization theorem	77
10.6	Block triangular form	77
10.7	Core structural results	78
10.8	Worked examples	78
10.9	Diagnostic checklist	78
10.10	Solved dossiers	78
10.11	Exercises with complete solutions	80

10.12	Graded supplementary exercises	81
11	Diagonalization and Minimal Polynomials	84
11.1	Diagonalizable operators	84
11.2	Direct sums of eigenspaces	85
11.3	Minimal polynomial	85
11.4	Divisibility property	85
11.5	Square-free criterion	85
11.6	Spectral projectors by interpolation	86
11.7	Core structural results	86
11.8	Worked examples	86
11.9	Diagnostic checklist	86
11.10	Solved dossiers	86
11.11	Exercises with complete solutions	88
11.12	Graded supplementary exercises	89
12	Canonical Forms	92
12.1	Generalized eigenvectors	92
12.2	Nilpotent operators	92
12.3	Jordan blocks	93
12.4	Jordan canonical form	93
12.5	Primary decomposition	93
12.6	Rational canonical form	93
12.7	Core structural results	94
12.8	Worked examples	94
12.9	Diagnostic checklist	94
12.10	Solved dossiers	94
12.11	Exercises with complete solutions	96
12.12	Graded supplementary exercises	97
III	Euclidean and Hilbert-Space Geometry	100
13	Inner Products and Orthogonality	101
13.1	Inner products	101
13.2	Norm and distance	101
13.3	Cauchy–Schwarz	102
13.4	Orthogonality	102
13.5	Orthogonal complements	102
13.6	Parallelogram and polarization	102
13.7	Core structural results	103
13.8	Worked examples	103
13.9	Diagnostic checklist	103
13.10	Solved dossiers	103
13.11	Exercises with complete solutions	105
13.12	Graded supplementary exercises	106
14	Gram–Schmidt and Orthogonal Projection	109
14.1	Orthonormal lists	109
14.2	Gram–Schmidt process	110
14.3	Projection onto a subspace	110

14.4	Best approximation	110
14.5	QR factorization	110
14.6	Least squares	110
14.7	Core structural results	111
14.8	Worked examples	111
14.9	Diagnostic checklist	111
14.10	Solved dossiers	112
14.11	Exercises with complete solutions	113
14.12	Graded supplementary exercises	114
15	Orthogonal and Unitary Operators	117
15.1	Isometries	117
15.2	Orthogonal and unitary matrices	117
15.3	Equivalent characterizations	118
15.4	Determinant and orientation	118
15.5	Reflections and Householder maps	118
15.6	Rotations and planar blocks	118
15.7	Core structural results	119
15.8	Worked examples	119
15.9	Diagnostic checklist	119
15.10	Solved dossiers	119
15.11	Exercises with complete solutions	120
15.12	Graded supplementary exercises	122
16	The Spectral Theorem	125
16.1	Adjoint	125
16.2	Self-adjoint and normal operators	125
16.3	Reality of the spectrum	126
16.4	Spectral theorem	126
16.5	Functional calculus	126
16.6	Positive operators	127
16.7	Core structural results	127
16.8	Worked examples	127
16.9	Diagnostic checklist	127
16.10	Solved dossiers	127
16.11	Exercises with complete solutions	129
16.12	Graded supplementary exercises	130
17	Quadratic Forms	133
17.1	Quadratic and bilinear forms	133
17.2	Matrix representation	134
17.3	Polarization	134
17.4	Congruence	134
17.5	Sylvester law of inertia	134
17.6	Definiteness	134
17.7	Core structural results	135
17.8	Worked examples	135
17.9	Diagnostic checklist	135
17.10	Solved dossiers	135
17.11	Exercises with complete solutions	137
17.12	Graded supplementary exercises	138

18 Singular-Value Decomposition	141
18.1 Singular values	141
18.2 SVD theorem	142
18.3 Construction from the spectral theorem	142
18.4 Rank and norms	142
18.5 Pseudoinverse	142
18.6 Low-rank approximation	143
18.7 Core structural results	143
18.8 Worked examples	143
18.9 Diagnostic checklist	143
18.10 Solved dossiers	144
18.11 Exercises with complete solutions	145
18.12 Graded supplementary exercises	146

Part I

Vector Spaces

Chapter 1

Scalars, Vectors and Linear Combinations

Linear algebra begins with two operations: adding vectors and scaling them by elements of a field. The point of the abstract definition is not to hide geometry but to identify exactly which arguments depend only on those operations. This chapter moves back and forth between concrete coordinate vectors, functions and polynomials so that linear combinations become a working language rather than a formal slogan.

Learning goals

After this chapter, the reader should be able to:

- compute linear combinations in $\mathbb{R}^n$, polynomial spaces, and other concrete vector spaces;
- decide whether a target vector lies in the span of a finite list by solving the coefficient equations;
- rewrite a vector equation as a linear system and interpret its solution set in terms of linear combinations;
- describe the set of all linear combinations of one, two, or three given vectors parametrically;
- construct examples showing how the same abstract vector can have different coordinate descriptions;
- identify when a proposed operation fails one of the vector-space axioms by producing a concrete counterexample;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

1.1 Fields and scalars

We work over a field $\mathbb{F}$, usually $\mathbb{R}$ or $\mathbb{C}$. The field laws allow addition, subtraction, multiplication and division by nonzero scalars. Every vector-space identity ultimately rests on these laws together with the vector-space axioms.

1.2 Vector spaces

A vector space V over $\mathbb{F}$ is an abelian group under addition equipped with scalar multiplication $\mathbb{F} \times V \rightarrow V$ satisfying distributivity, compatibility, and $1v = v$. The zero vector is unique, and $0v = 0$, $a0 = 0$, and $(-1)v = -v$ follow from the axioms.

1.3 Linear combinations

Given $v_1, \dots, v_r \in V$, a linear combination is $a_1v_1 + \dots + a_rv_r$ with $a_i \in \mathbb{F}$. Finite linear combinations are the algebraic mechanism for building new vectors from old ones.

Example 1.1 (Membership by coefficient equations). In $\mathbb{R}^3$, decide whether

$$b = (4, 1, 5)$$

belongs to the span of $v_1 = (1, 0, 1)$ and $v_2 = (0, 1, 1)$. Write

$$av_1 + cv_2 = (a, c, a + c).$$

The first two coordinates force $a = 4$ and $c = 1$, and the third then gives $a + c = 5$, exactly the required value. Hence $b = 4v_1 + v_2$.

The useful method is not guessing coefficients: write a general linear combination, compare coordinates, and then check every remaining equation. The last coordinate acts as a consistency check.

1.4 Concrete models

The spaces $\mathbb{F}^n$, polynomial spaces $\mathbb{F}[x]$, matrix spaces $M_{m,n}(\mathbb{F})$, and function spaces $\mathbb{F}^X$ all satisfy the same vector-space axioms. Coordinatewise formulas are examples of a common structure.

1.5 Affine versus linear geometry

Linear combinations with arbitrary coefficients preserve the origin. Affine combinations impose $a_1 + \dots + a_r = 1$ and describe points in affine hulls. Convex combinations add $a_i \geq 0$ and describe convex hulls.

Example 1.2 (Linear, affine, and convex combinations). Let $p = (0, 0)$ and $q = (2, 0)$. The point $(3, 0)$ can be written as

$$-\frac{1}{2}p + \frac{3}{2}q.$$

The coefficients sum to 1, so this is an affine combination. It is not a convex combination because one coefficient is negative. It is, of course, also a linear combination because linear combinations impose no sum condition.

Thus the same formula separates three notions: linear span records directions through the origin, affine span records the whole line through the two points, and convex span records only the segment between them.

1.6 Complex scalars and realification

A complex vector space is automatically a real vector space after restricting scalars, but its real dimension doubles when the complex dimension is finite. Forgetting part of the scalar field changes the linear relations that are allowed.

Example 1.3 (What changes when the scalar field changes). Consider $V = \mathbb{C}$. Over $\mathbb{C}$, the single vector 1 spans the space because $z = z \cdot 1$, so $\dim_{\mathbb{C}} V = 1$. Over $\mathbb{R}$, the scalar i is no longer available, and one needs 1, i :

$$a + bi = a \cdot 1 + b \cdot i.$$

To check independence over $\mathbb{R}$, if $a + bi = 0$ with real a, b , then both real and imaginary parts vanish. Hence $\dim_{\mathbb{R}} \mathbb{C} = 2$.

The abstract set and addition have not changed; only the allowed scalars have. That alone changes dimension and linear dependence.

1.7 Core structural results

Theorem 1.4 (Uniqueness consequences of the axioms). *For every vector space V , the zero vector and additive inverse of each vector are unique; moreover $0v = 0$, $a0 = 0$, and $(-a)v = -(av)$ for all $a \in \mathbb{F}$.*

Proof. Each identity follows by adding a suitable additive inverse and using distributivity. For example, $0v = (0 + 0)v = 0v + 0v$; cancelling $0v$ gives $0v = 0$. $\square$

Theorem 1.5 (Linear-combination regrouping). *A linear combination of linear combinations is again a linear combination of the original vectors.*

Proof. Substitute each inner expression and use associativity and distributivity to collect the coefficient of each original vector. $\square$

1.8 Worked examples

Example 1.6 (Polynomials). In $\mathbb{R}[x]$, the polynomial $3 - 2x + 5x^3$ is a linear combination of $1, x, x^3$. The same expression may be viewed inside any larger polynomial space without changing its coefficients.

Example 1.7 (Functions). If $f, g : X \rightarrow \mathbb{R}$, then $2f - 3g$ is the function $x \mapsto 2f(x) - 3g(x)$. Function spaces make clear that vectors need not be arrows.

Example 1.8 (Matrices). The matrix $\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ is a linear combination of the four matrix units E_{ij} . Matrix units play the role of coordinate directions.

1.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

1.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 1.9 (A hidden dependence in three vectors). In $\mathbb{R}^3$ let $u = (1, 1, 0)$, $v = (0, 1, 1)$ and $w = (1, 2, 1)$. Determine all triples (a, b, c) such that $au + bv + cw = 0$ and interpret the answer.

Solution. Since $w = u + v$, the relation is $(a + c)u + (b + c)v = 0$. The vectors u, v are independent, so $a + c = b + c = 0$. Hence $(a, b, c) = t(-1, -1, 1)$. The one-dimensional relation space records exactly one redundancy: w contributes no new direction. $\square$

Problem 1.10 (Changing the scalar field). Regard $\mathbb{C}$ first as a vector space over $\mathbb{C}$ and then over $\mathbb{R}$. Compare the linear combination $i \cdot 1$ in the two structures and explain why the dimensions differ.

Solution. Over $\mathbb{C}$, multiplication by i is scalar multiplication, so $i = i \cdot 1$ and $\{1\}$ is a basis. Over $\mathbb{R}$, i is not an available scalar; one needs both 1 and i , and every $z = a + bi$ is the real linear combination $a \cdot 1 + b \cdot i$. Thus the dimensions are 1 over $\mathbb{C}$ and 2 over $\mathbb{R}$. $\square$

Problem 1.11 (Polynomial coordinates without a basis theorem). Express $p(x) = 2 - 3x + 5x^2$ as a linear combination of $q_1 = 1 + x$, $q_2 = x + x^2$, and $q_3 = 1 - x^2$.

Solution. Seek $p = aq_1 + bq_2 + cq_3$. Comparing coefficients gives $a + c = 2$, $a + b = -3$, and $b - c = 5$. The third equation follows from the first two, so there are infinitely many representations: taking $a = t$ gives $b = -3 - t$ and $c = 2 - t$. This nonuniqueness reveals that q_1, q_2, q_3 are dependent. $\square$

Problem 1.12 (Affine but not linear). Show that the line $L = \{(x, y) : x + y = 1\}$ is closed under affine combinations but is not a vector subspace of $\mathbb{R}^2$.

Solution. If $p, q \in L$ and $\alpha + \beta = 1$, then the coordinate sum of $\alpha p + \beta q$ is $\alpha \cdot 1 + \beta \cdot 1 = 1$, so L is affine-combination closed. But $(0, 0) \notin L$, so L cannot be a vector subspace. $\square$

Problem 1.13 (Matrix units as building blocks). Write $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ as a linear combination of the matrix units $E_{11}, E_{12}, E_{21}, E_{22}$ and explain why the representation is unique.

Solution. $A = 2E_{11} - E_{12} + 3E_{21} + 4E_{22}$. If a linear combination of the four matrix units is zero, inspection of the four entries forces every coefficient to vanish. Thus the units are independent, and entrywise comparison also shows they span $M_2(\mathbb{R})$. $\square$

Problem 1.14 (Functions as vectors). Let $f(t) = 1$, $g(t) = t$, and $h(t) = 1 + t$ in the vector space of real functions on $[0, 1]$. Find every relation $af + bg + ch = 0$.

Solution. The identity means $(a + c) + (b + c)t = 0$ for every t . Hence $a + c = 0$ and $b + c = 0$, so $(a, b, c) = s(-1, -1, 1)$. Again $h = f + g$, illustrating that dependence is structural and not tied to coordinate vectors. $\square$

Problem 1.15 (Convex versus affine combinations). Can $(3, 0)$ be written as a convex combination of $(0, 0)$ and $(2, 0)$? Can it be written as an affine combination?

Solution. A convex combination has the form $(2\beta, 0)$ with $\alpha + \beta = 1$ and $\alpha, \beta \geq 0$, so $0 \leq 2\beta \leq 2$; $(3, 0)$ is impossible. For an affine combination, nonnegativity is dropped: choose $\beta = 3/2$ and $\alpha = -1/2$. $\square$

Problem 1.16 (Complex linearity test). Define $T : \mathbb{C} \rightarrow \mathbb{C}$ by $T(z) = \bar{z}$. Is T linear over $\mathbb{R}$? Over $\mathbb{C}$?

Solution. For real scalars a, b , conjugation satisfies $T(az + bw) = aT(z) + bT(w)$, so it is real-linear. But $T(iz) = -i\bar{z}$ while $iT(z) = i\bar{z}$, so it is not complex-linear unless $z = 0$. $\square$

Problem 1.17 (A parameterized linear combination). For which real t does $(1, t, t^2)$ belong to the span of $(1, 0, 1)$ and $(0, 1, 1)$?

Solution. A general span vector is $(a, b, a + b)$. Matching gives $a = 1$, $b = t$, and therefore $t^2 = 1 + t$. Thus t must solve $t^2 - t - 1 = 0$, so $t = (1 \pm \sqrt{5})/2$. $\square$

Problem 1.18 (Why finite sums are enough). Explain why the definition of span uses finite rather than arbitrary infinite linear combinations, even in an infinite-dimensional vector space.

Solution. Vector-space axioms define only binary addition and its finite iteration. No notion of convergence or infinite summation is part of a general vector space. Infinite sums require extra topological or analytic structure, so algebraic span is necessarily built from finite combinations. $\square$

Problem 1.19 (Direct verification from axioms). Prove from the vector-space axioms that if $av = 0$ and $a \neq 0$, then $v = 0$.

Solution. Multiply the equation by a^{-1} . Compatibility of scalar multiplication gives $v = 1v = (a^{-1}a)v = a^{-1}(av) = a^{-1}0 = 0$. This cancellation principle is used constantly in linear algebra. $\square$

Problem 1.20 (A geometric synthesis). Describe the set of all linear combinations of two nonparallel vectors in $\mathbb{R}^3$ and compare it with their affine hull.

Solution. The linear combinations form the plane through the origin spanned by the two vectors. Their affine hull as two points consists only of the line through those two points. The distinction is that span treats the vectors as directions from the origin, whereas affine hull treats them as points with coefficients summing to one. $\square$

1.11 Exercises with complete solutions

Exercise 1.21. Show directly from the axioms that $a(-v) = -(av)$.

Hint. Write the target as $a_1v_1 + \cdots + a_rv_r$ and compare coordinates before solving for the coefficients. $\square$

Solution. Use $v + (-v) = 0$, multiply by a , and use distributivity. The resulting equation $av + a(-v) = 0$ says that $a(-v)$ is the additive inverse of av . $\square$

Exercise 1.22. Write $(2, -1, 3)$ as a linear combination of the standard vectors of $\mathbb{R}^3$.

Hint. Separate the scalar coefficients from the vectors first; the vector equation is a system of scalar equations. $\square$

Solution. $(2, -1, 3) = 2e_1 - e_2 + 3e_3$. $\square$

Exercise 1.23. Give a vector-space structure on the set of real-valued continuous functions on $[0, 1]$.

Hint. Parameterize the coefficients rather than guessing individual vectors in the span. $\square$

Solution. Use pointwise addition and scalar multiplication. Continuity is preserved by both operations, so the operations remain inside the set. $\square$

Exercise 1.24. Why is the unit circle in $\mathbb{R}^2$ not a vector space?

Hint. Test the proposed identity componentwise; one failed coordinate is enough to refute it. $\square$

Solution. It does not contain the zero vector and is not closed under scalar multiplication; for example $2(1, 0) = (2, 0)$ is not on the unit circle. $\square$

Exercise 1.25. Over which scalars is $\mathbb{C}$ naturally a one-dimensional vector space? What is its dimension over $\mathbb{R}$?

Hint. Translate the vector relation into an augmented matrix and row-reduce only after writing the correct columns. $\square$

Solution. It is one-dimensional over $\mathbb{C}$, with basis $\{1\}$, and two-dimensional over $\mathbb{R}$, with basis $\{1, i\}$. $\square$

Exercise 1.26. Find an affine combination of $(0, 0)$ and $(2, 0)$ giving $(3, 0)$.

Hint. Look for a coefficient choice that isolates one of the listed vectors or exposes a contradiction. $\square$

Solution. Take coefficients $-\frac{1}{2}$ and $\frac{3}{2}$; they sum to 1 and give $(3, 0)$. $\square$

Exercise 1.27. Show that the set of all 2×2 real matrices is a vector space.

Hint. Check whether the alleged vector-space operation respects the zero vector before testing the harder axioms. $\square$

Solution. Matrix addition is an abelian group operation and scalar multiplication satisfies the vector-space laws entrywise. $\square$

Exercise 1.28. Explain why a finite sum $\sum_i a_i v_i$ is independent of parentheses.

Hint. Use two different coefficient choices to decide whether the representation is unique. $\square$

Solution. Associativity of vector addition allows any parenthesization; a formal induction on the number of summands proves the claim. $\square$

1.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations**Exercise 1.29** (Linear combination in $\mathbb{R}^3$). Express $(7, -2, 5)$ in the standard basis of $\mathbb{R}^3$.*Hint.* Read the coordinates as the coefficients of e_1, e_2, e_3 . □*Solution.* $(7, -2, 5) = 7e_1 - 2e_2 + 5e_3$. □**Exercise 1.30** (Span membership). Does $(2, 3, 5)$ lie in $\text{span}\{(1, 0, 1), (0, 1, 1)\}$?*Hint.* Write a general span vector $(a, b, a + b)$. □*Solution.* Yes. The first coordinates give $a = 2, b = 3$, and $a + b = 5$. □**Exercise 1.31** (Polynomial combination). Write $2 - 3x + x^2$ as a combination of $1, x, x^2$.*Hint.* Match coefficients by degree. □*Solution.* $2 - 3x + x^2 = 2(1) - 3x + 1(x^2)$. □**Exercise 1.32** (Matrix units). Expand $\begin{pmatrix} 0 & 4 \\ -1 & 2 \end{pmatrix}$ in the matrix units.*Hint.* Each entry is the coefficient of the corresponding E_{ij} . □*Solution.* $4E_{12} - E_{21} + 2E_{22}$. □**Exercise 1.33** (Parameter membership). For which t is $(1, t, t^2)$ in $\text{span}\{(1, 0, 1), (0, 1, 1)\}$?*Hint.* The first two coordinates force the coefficients; use the third as consistency. □*Solution.* One needs $t^2 = 1 + t$, so $t = (1 \pm \sqrt{5})/2$. □**Proofs****Exercise 1.34** (Scalar cancellation). Prove that $av = 0$ and $a \neq 0$ imply $v = 0$.*Hint.* Multiply by the inverse scalar a^{-1} . □*Solution.* $v = (a^{-1}a)v = a^{-1}(av) = 0$. □**Exercise 1.35** (Negative scalar). Prove $(-a)v = -(av)$.*Hint.* Show the two vectors add to zero. □*Solution.* $(-a)v + av = (-a + a)v = 0v = 0$, so $(-a)v$ is the additive inverse of av . □**Exercise 1.36** (Regrouping). Show a linear combination of linear combinations is a linear combination of the original vectors.*Hint.* Substitute the inner sums and collect coefficients. □*Solution.* If $w_j = \sum_i a_{ji}v_i$, then $\sum_j b_j w_j = \sum_i (\sum_j b_j a_{ji})v_i$. □**Exercise 1.37** (Finite-span principle). Explain why algebraic span uses finite sums.*Hint.* Ask which infinite operations are provided by the vector-space axioms. □*Solution.* The axioms define finite iteration of addition only; infinite sums require additional convergence structure. □

Counterexamples

Exercise 1.38 (Translated line). Show $\{(x, y) : x + y = 1\}$ is not a vector subspace.

Hint. Check the zero vector first. □

Solution. $(0, 0)$ does not satisfy $x + y = 1$, so the set is not a subspace. □

Exercise 1.39 (Unit sphere). Give two independent reasons the unit sphere is not a vector space under inherited operations.

Hint. Test zero and scalar closure. □

Solution. It omits 0, and $2v$ is not on the unit sphere when $\|v\| = 1$. □

Exercise 1.40 (Complex conjugation). Show $T(z) = \bar{z}$ is real-linear but not complex-linear.

Hint. Compare $T(iz)$ with $iT(z)$. □

Solution. Real scalars commute with conjugation, but $T(iz) = -i\bar{z} \neq i\bar{z}$ for $z \neq 0$. □

Applications

Exercise 1.41 (Barycentric interpolation). Find the affine midpoint of $p = (1, 2)$ and $q = (5, -2)$ and interpret the coefficients.

Hint. Use coefficients $1/2, 1/2$, which sum to one. □

Solution. The midpoint is $(3, 0)$; it is the affine combination $\frac{1}{2}p + \frac{1}{2}q$. □

Exercise 1.42 (Mixture model). A portfolio vector is $0.2v_1 + 0.5v_2 + 0.3v_3$. Why is it both affine and convex?

Hint. Check the coefficient sum and signs. □

Solution. The coefficients sum to 1 and are nonnegative, so the combination is convex and hence affine. □

Challenges

Exercise 1.43 (Restricted scalars). Regard $\mathbb{R}$ as a vector space over $\mathbb{Q}$. Explain why $\{1, \sqrt{2}\}$ is independent.

Hint. A rational relation $a + b\sqrt{2} = 0$ with $b \neq 0$ would make $\sqrt{2}$ rational. □

Solution. Thus $b = 0$, then $a = 0$; the list is $\mathbb{Q}$ -independent. □

Exercise 1.44 (Operation redesign). On $\mathbb{R}$, define $x \oplus y = x + y - 1$ and $a \odot x = ax + (1 - a)$. Identify the zero vector for these operations.

Hint. Solve $x \oplus z = x$. □

Solution. $z = 1$. Indeed $x \oplus 1 = x$, and the formulas arise by translating the usual vector-space structure by 1. □

Chapter summary

Linear algebra begins with two operations: adding vectors and scaling them by elements of a field. The point of the abstract definition is not to hide geometry but to identify exactly which arguments depend only on those operations. This chapter moves back and forth between concrete coordinate vectors, functions and polynomials so that linear combinations become a working language rather than a formal slogan. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 2

Subspaces, Span and Linear Independence

Subspaces are the linear subsets of a vector space, span is the smallest subspace generated by specified vectors, and linear independence is the condition that a generating description has no redundancy. These three ideas form the basic grammar of finite-dimensional reasoning.

Learning goals

After this chapter, the reader should be able to:

- determine whether a subset is a subspace using both the three-condition and linear-combination tests;
- compute sums and intersections of explicitly presented subspaces;
- find all linear relations among a finite list;
- extract a minimal generating list from a redundant spanning list;
- construct a counterexample showing that a union of subspaces need not be a subspace;
- translate between dependence relations and solutions of homogeneous systems;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

2.1 Subspaces

A subset $W \subseteq V$ is a subspace when it contains 0 and is closed under addition and scalar multiplication. Equivalently, $au + bv \in W$ for all $u, v \in W$ and $a, b \in \mathbb{F}$.

Example 2.1 (Testing a constraint subspace). Let

$$W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}.$$

For $u, v \in W$ and scalars a, b ,

$$(au + bv)_1 - 2(au + bv)_2 + (au + bv)_3 = a(u_1 - 2u_2 + u_3) + b(v_1 - 2v_2 + v_3) = 0.$$

Thus W is a subspace. Solving the equation gives $x = 2s - t$, so

$$W = \text{span}\{(2, 1, 0), (-1, 0, 1)\}.$$

The defining equation proves closure; the parametrization exhibits a spanning set. These are two complementary ways to understand the same subspace.

2.2 Intersections and sums

Arbitrary intersections of subspaces are subspaces. The union of two subspaces need not be a subspace, but the sum $U + W = \{u + w\}$ always is.

2.3 Span

The span of $S \subseteq V$ is the set of all finite linear combinations of elements of S . It is the smallest subspace containing S .

Example 2.2 (Removing a redundant generator). Consider

$$v_1 = (1, 0, 1), \quad v_2 = (0, 1, 1), \quad v_3 = (1, 1, 2).$$

Since $v_3 = v_1 + v_2$,

$$\text{span}(v_1, v_2, v_3) = \text{span}(v_1, v_2).$$

Moreover v_1, v_2 are independent: the first two coordinates of $av_1 + bv_2 = 0$ force $a = b = 0$.

The calculation shows how span and independence work together: the third vector adds no direction, and deleting it preserves the generated subspace.

2.4 Linear dependence

A finite list $v_1, \dots, v_r$ is linearly dependent when some nontrivial scalar relation $a_1v_1 + \dots + a_rv_r = 0$ exists. Otherwise the list is linearly independent.

2.5 Redundancy and exchange

If a list is dependent, at least one vector is in the span of the others. This simple observation drives the exchange arguments used to prove uniqueness of dimension.

2.6 Homogeneous systems

Columns of a matrix are linearly dependent exactly when the corresponding homogeneous system has a nonzero solution. Row reduction converts an abstract dependence question into an algorithm.

Example 2.3 (Dependence as a null-space computation). Put the vectors

$$v_1 = (1, 1, 0), \quad v_2 = (0, 1, 1), \quad v_3 = (1, 2, 1)$$

as columns of A . A relation $c_1v_1 + c_2v_2 + c_3v_3 = 0$ is exactly a solution of $Ac = 0$. Since $v_3 = v_1 + v_2$,

$$(-1, -1, 1)^T \in \ker A.$$

Row reduction gives one free variable, so every relation is a scalar multiple of this one.

This is the computational bridge: column dependence is not a new problem; it is a homogeneous linear system.

2.7 Core structural results

Theorem 2.4 (Subspace criterion). *A nonempty subset $W \subseteq V$ is a subspace if and only if $au + bv \in W$ for all $u, v \in W$ and all scalars a, b .*

Proof. Taking $a = b = 0$ gives $0 \in W$; special choices recover closure under addition and scalar multiplication. The converse is immediate. $\square$

Theorem 2.5 (Dependence lemma). *A list $v_1, \dots, v_r$ is dependent if and only if one vector lies in the span of the preceding vectors after a suitable reordering.*

Proof. Solve a nontrivial relation for a vector whose coefficient is nonzero. Conversely, any span relation rearranges to a nontrivial dependence relation. $\square$

2.8 Worked examples

Example 2.6 (Solution spaces). The solution set of a homogeneous linear system $Ax = 0$ is a subspace of $\mathbb{F}^n$ because matrix multiplication is linear.

Example 2.7 (Polynomial constraints). Polynomials of degree at most 4 satisfying $p(1) = 0$ form a subspace: evaluation at 1 respects addition and scalar multiplication.

Example 2.8 (A nonexample). The set $\{(x, y) : xy = 0\}$ is the union of the coordinate axes and is not a subspace because $(1, 0) + (0, 1) = (1, 1)$ leaves the set.

2.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

2.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 2.9 (Intersection and sum of planes). Let $U = \{(x, y, z) : z = 0\}$ and $W = \{(x, y, z) : y = 0\}$. Find bases for $U \cap W$ and $U + W$.

Solution. The intersection requires $y = z = 0$, so $U \cap W = \text{span}\{(1, 0, 0)\}$. Since $(1, 0, 0), (0, 1, 0) \in U$ and $(0, 0, 1) \in W$, the sum contains the standard basis and hence equals $\mathbb{R}^3$. $\square$

Problem 2.10 (A union that fails). Give an explicit failure of closure showing that the union of the x - and y -axes in $\mathbb{R}^2$ is not a subspace.

Solution. Both $(1, 0)$ and $(0, 1)$ lie in the union, but their sum $(1, 1)$ lies on neither axis. Therefore the union is not closed under addition. $\square$

Problem 2.11 (Span from row reduction). Determine whether $(2, 1, 3)$ lies in the span of $(1, 0, 1)$ and $(0, 1, 1)$.

Solution. Solve $a(1, 0, 1) + b(0, 1, 1) = (2, 1, 3)$. The first two coordinates force $a = 2, b = 1$, and then the third gives $a + b = 3$, which holds. Hence the vector lies in the span. $\square$

Problem 2.12 (Redundant generator). Show that $\text{span}(u, v, u+2v) = \text{span}(u, v)$ for arbitrary vectors u, v .

Solution. The third vector already lies in $\text{span}(u, v)$, so adding it cannot enlarge the span. Conversely u and v occur in the larger generating list, so the smaller span is contained in the larger one. $\square$

Problem 2.13 (Subspace defined by a functional). Let $W = \{p \in \mathcal{P}_3 : p(1) = 0\}$. Prove W is a subspace and find a convenient spanning set.

Solution. Evaluation at 1 is linear, so W is its kernel and hence a subspace. Every p with $p(1) = 0$ is divisible by $x-1$, so $W = (x-1)\mathcal{P}_2$ and is spanned by $(x-1), x(x-1), x^2(x-1)$. $\square$

Problem 2.14 (Dependence with a parameter). For which a are $(1, a)$ and $(a, 1)$ linearly dependent in $\mathbb{R}^2$?

Solution. Two planar vectors are dependent exactly when the determinant $\begin{vmatrix} 1 & a \\ a & 1 \end{vmatrix} = 1 - a^2$ vanishes. Thus $a = \pm 1$. $\square$

Problem 2.15 (Minimal spanning extraction). From the list $(1, 0, 0), (0, 1, 0), (1, 1, 0), (0, 0, 1)$ extract a basis of its span.

Solution. The third vector is the sum of the first two, so it is redundant. The remaining vectors are the standard basis of $\mathbb{R}^3$, hence a basis of the span. $\square$

Problem 2.16 (A nonlinear constraint). Why is $\{(x, y) : xy = 0\}$ not a subspace even though it contains 0 and is closed under scalar multiplication?

Solution. It is not closed under addition: $(1, 0)$ and $(0, 1)$ satisfy the constraint, but their sum $(1, 1)$ does not. All subspace conditions are necessary. $\square$

Problem 2.17 (Independence of monomials). Prove $1, x, \dots, x^n$ are linearly independent in $\mathcal{P}_n$ without invoking dimension.

Solution. If $\sum_{k=0}^n a_k x^k$ is the zero polynomial, then every coefficient of that polynomial is zero. Hence all $a_k = 0$, which is exactly linear independence. $\square$

Problem 2.18 (Sum not direct). Let $U = \text{span}(e_1, e_2)$ and $W = \text{span}(e_2, e_3)$ in $\mathbb{R}^3$. Explain why $U + W = \mathbb{R}^3$ but the sum is not direct.

Solution. The sum contains e_1, e_2, e_3 , so it is all of $\mathbb{R}^3$. But e_2 is a nonzero vector in both U and W , so $U \cap W \neq \{0\}$; therefore the representation $u + w$ is not unique. $\square$

Problem 2.19 (Solving a dependence relation). Find a nontrivial relation among $(1, 1, 0), (0, 1, 1), (1, 0, -1)$.

Solution. Adding the second and third gives $(1, 1, 0)$, so $-(1, 1, 0) + (0, 1, 1) + (1, 0, -1) = 0$. Thus the list is dependent. $\square$

Problem 2.20 (Subspace synthesis). Prove that the solution set of any homogeneous system $Ax = 0$ is a subspace, and explain why the analogous set $Ax = b$ with $b \neq 0$ generally is not.

Solution. If $Ax = Ay = 0$, then $A(\alpha x + \beta y) = \alpha Ax + \beta Ay = 0$, so the homogeneous solution set is a subspace. If $Ax = b$ with $b \neq 0$, then $x = 0$ is not a solution, so the solution set cannot be a vector subspace; it is typically an affine translate of the kernel. $\square$

2.11 Exercises with complete solutions

Exercise 2.21. Show that the intersection of any family of subspaces is a subspace.

Hint. Take two arbitrary vectors in the intersection and test an arbitrary linear combination inside every defining subspace. Take two arbitrary vectors from the intersection and test an arbitrary linear combination in every member of the family. $\square$

Solution. The zero vector lies in every member. If two vectors lie in the intersection, every linear combination lies in each member and hence in the intersection. $\square$

Exercise 2.22. Show that $U \cup W$ is a subspace only if $U \subseteq W$ or $W \subseteq U$.

Hint. Assume neither subspace contains the other; choose $u \in U \setminus W$ and $w \in W \setminus U$, then test $u + w$. $\square$

Solution. If neither contains the other, choose $u \in U \setminus W$ and $w \in W \setminus U$. Then $u + w$ belongs to neither subspace, contradicting closure. $\square$

Exercise 2.23. Find the span of $(1, 0, 1)$ and $(0, 1, 1)$ in $\mathbb{R}^3$.

Hint. Write a general combination of the generators and eliminate the free coefficients to obtain equations for the span. $\square$

Solution. It is $\{(a, b, a + b) : a, b \in \mathbb{R}\}$, equivalently the plane $z = x + y$. $\square$

Exercise 2.24. Are $1 + x$, $x + x^2$, and $1 - x^2$ independent?

Hint. Set up one coefficient equation for each polynomial coefficient; a nonzero solution is exactly a dependence relation. $\square$

Solution. No. The first minus the second minus the third is zero: $(1 + x) - (x + x^2) - (1 - x^2) = 0$. $\square$

Exercise 2.25. Prove that a list containing the zero vector is dependent.

Hint. Give the zero vector coefficient 1 and every other vector coefficient 0. $\square$

Solution. Give the zero vector coefficient 1 and all other vectors coefficient 0. $\square$

Exercise 2.26. Determine whether $(1, 2)$ and $(2, 4)$ are independent.

Hint. Check whether one vector is a scalar multiple of the other before doing any row reduction. $\square$

Solution. They are dependent because $(2, 4) = 2(1, 2)$. $\square$

Exercise 2.27. Show that the span of a set equals the intersection of all subspaces containing it.

Hint. Take two arbitrary vectors in the intersection and test an arbitrary linear combination inside every defining subspace. Use the minimality property of span twice: first show the span lies in every containing subspace, then use that the span itself is one of them. $\square$

Solution. The span is itself a subspace containing the set, and every subspace containing the set contains all its finite linear combinations. $\square$

Exercise 2.28. Relate pivots in a matrix to independence of its columns.

Hint. Translate the statement into $Ax = 0$. Pivot columns and free variables tell you whether a nonzero coefficient vector exists. Translate column independence into the homogeneous system $Ax = 0$; free variables are the obstruction. $\square$

Solution. A homogeneous system $Ax = 0$ has only the zero solution exactly when every column variable is a pivot variable; this is equivalent to column independence. $\square$

2.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 2.29 (Plane span). Parametrize $\text{span}\{(1, 0, 1), (0, 1, 1)\}$.

Hint. Use coefficients a, b . $\square$

Solution. The span is $\{(a, b, a + b) : a, b \in \mathbb{R}\}$, the plane $z = x + y$. $\square$

Exercise 2.30 (Intersection). Find $U \cap W$ for $U = \{z = 0\}$ and $W = \{y = 0\}$ in $\mathbb{R}^3$.

Hint. Impose both equations. $\square$

Solution. $U \cap W = \{(x, 0, 0)\} = \text{span}(e_1)$. $\square$

Exercise 2.31 (Sum). For the same U, W , compute $U + W$.

Hint. Check whether the standard basis lies in the sum. $\square$

Solution. $e_1, e_2 \in U$ and $e_3 \in W$, so $U + W = \mathbb{R}^3$. $\square$

Exercise 2.32 (Parameter dependence). For which a are $(1, a)$ and $(a, 1)$ dependent?

Hint. Set the 2×2 determinant to zero. $\square$

Solution. $1 - a^2 = 0$, hence $a = \pm 1$. $\square$

Exercise 2.33 (Relation space). Find all relations among $(1, 0), (0, 1), (1, 1)$.

Hint. Solve $a(1, 0) + b(0, 1) + c(1, 1) = 0$. $\square$

Solution. $a + c = 0, b + c = 0$, so $(a, b, c) = t(-1, -1, 1)$. $\square$

Proofs

Exercise 2.34 (Intersection theorem). Prove an arbitrary intersection of subspaces is a subspace.

Hint. Test a general linear combination in every member of the family. □

Solution. Every member contains zero and is closed under linear combinations, so the intersection has the same properties. □

Exercise 2.35 (Union criterion). Prove $U \cup W$ is a subspace only when $U \subseteq W$ or $W \subseteq U$.

Hint. If neither contains the other, choose $u \in U \setminus W$ and $w \in W \setminus U$. □

Solution. Then $u + w$ lies in neither subspace; otherwise subtraction would put one chosen vector in the other subspace. □

Exercise 2.36 (Smallest subspace). Prove $\text{span } S$ is the intersection of all subspaces containing S .

Hint. Use both containments and the defining minimality of span . □

Solution. Every containing subspace contains all finite combinations, while $\text{span } S$ itself is one of the containing subspaces. □

Exercise 2.37 (Redundancy lemma). Show that if $v_r \in \text{span}(v_1, \dots, v_{r-1})$, deleting v_r does not change the span.

Hint. One containment is automatic; substitute the span expression for v_r for the other. □

Solution. Every combination using v_r can be rewritten using only $v_1, \dots, v_{r-1}$. □

Counterexamples

Exercise 2.38 (Union of axes). Show the union of the coordinate axes in $\mathbb{R}^2$ is not a subspace.

Hint. Add one vector from each axis. □

Solution. $(1, 0)$ and $(0, 1)$ lie in the union but $(1, 1)$ does not. □

Exercise 2.39 (Nonhomogeneous solution set). Show $\{x : Ax = b\}$ with $b \neq 0$ need not be a subspace.

Hint. Check whether 0 can solve the equation. □

Solution. $A0 = 0 \neq b$, so the zero vector is absent whenever the system is consistent with nonzero b . □

Exercise 2.40 (Scalar-closed is not enough). Give a set containing 0 and closed under scalar multiplication but not under addition.

Hint. Use the union of two distinct lines through the origin. □

Solution. The union of the coordinate axes works; it is scalar-closed but $(1, 0) + (0, 1)$ leaves it. □

Applications

Exercise 2.41 (Conservation constraint). Show vectors (x, y, z) with $x + y + z = 0$ form a subspace and give a basis.

Hint. Solve $z = -x - y$. □

Solution. The set is $\text{span}\{(1, 0, -1), (0, 1, -1)\}$. □

Exercise 2.42 (Polynomial boundary condition). Find a basis of $W = \{p \in \mathcal{P}_3 : p(1) = 0\}$.

Hint. Factor $x - 1$. □

Solution. $W = (x - 1)\mathcal{P}_2$, with basis $x - 1, x(x - 1), x^2(x - 1)$. □

Challenges

Exercise 2.43 (Direct-sum test). For $U = \text{span}(e_1, e_2)$, $W = \text{span}(e_2, e_3)$, decide whether $U + W$ is direct.

Hint. A sum is direct iff the intersection is zero. □

Solution. $U \cap W = \text{span}(e_2) \neq 0$, so the sum is not direct. □

Exercise 2.44 (Dependence without coordinates). If u, v are independent, prove $u + v, u - v$ are independent over a field of characteristic not 2.

Hint. Add and subtract a relation $a(u + v) + b(u - v) = 0$. □

Solution. The coefficients of u, v are $a + b, a - b$; independence gives $a + b = a - b = 0$, hence $2a = 2b = 0$, so $a = b = 0$. □

Chapter summary

Subspaces are the linear subsets of a vector space, span is the smallest subspace generated by specified vectors, and linear independence is the condition that a generating description has no redundancy. These three ideas form the basic grammar of finite-dimensional reasoning. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 3

Bases and Dimension

A basis is simultaneously a generating system and a coordinate system with no redundancy. Finite-dimensional vector spaces become computationally manageable because every vector then has a unique finite coordinate list. Dimension records the size of any basis and is independent of the particular basis chosen.

Learning goals

After this chapter, the reader should be able to:

- test whether a finite list is a basis by checking both spanning and independence;
- extend an independent list to a basis and reduce a spanning list to a basis;
- compute the dimension of a subspace from an explicit spanning set or homogeneous system;
- prove that two bases of a finite-dimensional vector space have the same number of vectors;
- apply the dimension formula to compute $\dim(U + W)$ or $\dim(U \cap W)$;
- decide whether an injective or surjective linear map can exist from dimension data alone;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

3.1 Bases

A basis of V is a linearly independent list that spans V . The two conditions guarantee existence and uniqueness of coordinates.

3.2 Coordinates relative to a basis

If $B = (v_1, \dots, v_n)$ is a basis, every v has a unique representation $v = \sum a_i v_i$. The coordinate column is written $[v]_B$.

Example 3.1 (Coordinates in a nonstandard basis). Let $B = ((1, 1), (1, -1))$. To find $[(5, 1)]_B$, solve

$$a(1, 1) + b(1, -1) = (5, 1).$$

Then $a + b = 5$ and $a - b = 1$, so $a = 3$, $b = 2$. Therefore

$$[(5, 1)]_B = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

Checking gives $3(1, 1) + 2(1, -1) = (5, 1)$.

The check is important: coordinates are coefficients relative to the chosen basis, not the ambient coordinates of the vector.

3.3 Extension and reduction

An independent list in a finite-dimensional space can be extended to a basis, while a spanning list can be reduced to a basis by deleting redundant vectors.

Example 3.2 (Extending an independent list). In $\mathbb{R}^3$, start with

$$u = (1, 1, 0), \quad v = (0, 1, 1).$$

They are independent. Test $e_1 = (1, 0, 0)$: if $e_1 = au + bv$, then the first coordinate gives $a = 1$, the third gives $b = 0$, but the second becomes $1 \neq 0$. Hence $e_1 \notin \text{span}(u, v)$. Therefore u, v, e_1 is a basis.

The method is systematic: extend an independent list by adding any vector outside its current span.

3.4 Dimension

All bases of a finite-dimensional vector space have the same number of vectors. That common number is $\dim V$.

3.5 Dimension of subspaces

If $W \subseteq V$ and V is finite-dimensional, then $\dim W \leq \dim V$, with equality only when $W = V$.

3.6 Dimension formula

For finite-dimensional subspaces $U, W \subseteq V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Example 3.3 (Using the dimension formula as a consistency test). Suppose $U, W \subseteq V$ satisfy

$$\dim U = 5, \quad \dim W = 4, \quad \dim(U \cap W) = 2.$$

Then

$$\dim(U + W) = 5 + 4 - 2 = 7.$$

If V had dimension 6, these data would be impossible because $U + W \subseteq V$. Thus dimension formulas do more than compute numbers: they detect inconsistent geometric claims before any coordinates are chosen.

3.7 Core structural results

Theorem 3.4 (Basis uniqueness of coordinates). *A list is a basis if and only if every vector has a unique representation as a linear combination of the list.*

Proof. Spanning gives existence. If two representations exist, subtract them to obtain a dependence relation; independence forces equal coefficients. $\square$

Theorem 3.5 (Dimension formula for sums). *For finite-dimensional U, W , one has $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.*

Proof. Choose a basis of $U \cap W$, extend it separately to bases of U and W , and check that the union of the extensions spans $U + W$ and is independent. $\square$

3.8 Worked examples

Example 3.6 (Polynomial dimension). The space $\mathcal{P}_n$ of polynomials of degree at most n has basis $1, x, \dots, x^n$ and dimension $n + 1$.

Example 3.7 (Symmetric matrices). The real symmetric $n \times n$ matrices have dimension $n(n + 1)/2$, one free entry for each diagonal or upper-triangular position.

Example 3.8 (Trace-zero matrices). The trace-zero subspace of $M_n(\mathbb{F})$ has dimension $n^2 - 1$.

3.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

3.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 3.9 (Basis test by uniqueness). Show that $B = ((1, 1), (1, -1))$ is a basis of $\mathbb{R}^2$ and find the coordinates of (a, b) .

Solution. Solve $c_1(1, 1) + c_2(1, -1) = (a, b)$. Then $c_1 + c_2 = a$ and $c_1 - c_2 = b$, giving $c_1 = (a + b)/2$, $c_2 = (a - b)/2$. Every vector has a unique representation, so B is a basis. $\square$

Problem 3.10 (Dimension of a constrained polynomial space). Find the dimension of $W = \{p \in \mathcal{P}_4 : p(0) = p(1) = 0\}$.

Solution. The conditions imply $p(x) = x(x - 1)q(x)$ with $q \in \mathcal{P}_2$. Multiplication by $x(x - 1)$ is injective, so W is isomorphic to $\mathcal{P}_2$ and has dimension 3. $\square$

Problem 3.11 (Extend an independent list). Extend $(1, 1, 0)$ and $(0, 1, 1)$ to a basis of $\mathbb{R}^3$.

Solution. The two vectors are independent. Choose, for instance, $e_1 = (1, 0, 0)$. It is not in their span because solving $a(1, 1, 0) + b(0, 1, 1) = e_1$ forces $a = 1$, $b = 0$, but then the second coordinate is 1. Thus the three vectors form a basis. $\square$

Problem 3.12 (Reduce a spanning list). Reduce $(1, 0, 0), (0, 1, 0), (1, 1, 0), (0, 0, 1)$ to a basis without changing its span.

Solution. Delete $(1, 1, 0)$ because it is the sum of the first two. The remaining three standard vectors are independent and span $\mathbb{R}^3$. $\square$

Problem 3.13 (Dimension of symmetric matrices). Compute $\dim \text{Sym}_n(\mathbb{R})$.

Solution. A symmetric matrix is determined by its n diagonal entries and its $n(n-1)/2$ entries above the diagonal. Thus the dimension is $n + n(n-1)/2 = n(n+1)/2$. $\square$

Problem 3.14 (Dimension formula in action). If $\dim U = 5$, $\dim W = 4$, and $\dim(U+W) = 7$, find $\dim(U \cap W)$.

Solution. The dimension formula gives $7 = 5 + 4 - \dim(U \cap W)$, so the intersection has dimension 2. $\square$

Problem 3.15 (Full-dimension subspace). Prove that a subspace W of a finite-dimensional space V with $\dim W = \dim V$ must equal V .

Solution. A basis of W is an independent list in V with exactly $\dim V$ vectors. Any such list is a basis of V , so it spans V . Since it lies in W , we obtain $V \subseteq W$, hence equality. $\square$

Problem 3.16 (Trace-zero matrices). Why does the space of trace-zero $n \times n$ matrices have dimension $n^2 - 1$?

Solution. The trace map $M_n(\mathbb{F}) \rightarrow \mathbb{F}$ is a nonzero linear functional and hence surjective. Its kernel is the trace-zero subspace. Rank-nullity gives dimension $n^2 - 1$. $\square$

Problem 3.17 (Coordinate uniqueness). Suppose $B = (v_1, \dots, v_n)$ is a basis and $\sum a_i v_i = \sum b_i v_i$. Prove $a_i = b_i$ for every i .

Solution. Subtract the two representations to obtain $\sum (a_i - b_i) v_i = 0$. Basis vectors are independent, so every coefficient $a_i - b_i$ is zero. $\square$

Problem 3.18 (Finite versus infinite dimension). Explain why the extension theorem for a finite independent list does not by itself prove every vector space has a basis.

Solution. In finite-dimensional spaces the extension process terminates because there is a finite spanning list. For arbitrary vector spaces, one needs a maximality principle such as Zorn's lemma to guarantee a maximal independent set; the elementary finite algorithm is insufficient. $\square$

Problem 3.19 (Direct-sum dimension). If $V = U \oplus W$ with finite dimensions, prove $\dim V = \dim U + \dim W$.

Solution. Take bases of U and W . Their union spans V because every vector decomposes as $u + w$. If a linear combination of the union is zero, the U -part equals minus the W -part and hence lies in $U \cap W = \{0\}$, forcing all coefficients to vanish. $\square$

Problem 3.20 (Basis synthesis). A list of n vectors in an n -dimensional space spans. Why is it automatically a basis?

Solution. If the list were dependent, one vector could be removed without changing the span, producing a spanning list of fewer than n vectors, impossible in an n -dimensional space. Hence the list is independent and therefore a basis. $\square$

3.11 Exercises with complete solutions

Exercise 3.21. Show that $(1, 0), (1, 1)$ is a basis of $\mathbb{R}^2$.

Hint. Start with the spanning condition and the independence condition separately; a basis requires both. $\square$

Solution. The two vectors are not scalar multiples, so they are independent; two independent vectors in a two-dimensional space form a basis. $\square$

Exercise 3.22. Find the coordinates of $(3, 2)$ in the basis $((1, 0), (1, 1))$.

Hint. Write each new basis vector in the old coordinates and use those coordinate columns to build the conversion matrix. Use the exchange lemma to remove one redundant vector without changing the span, then repeat. $\square$

Solution. Solve $a(1, 0) + b(1, 1) = (3, 2)$. Then $b = 2$ and $a = 1$, so the coordinate column is $(1, 2)^T$. $\square$

Exercise 3.23. Find a basis for polynomials in $\mathcal{P}_3$ satisfying $p(0) = 0$.

Hint. Row-reduce a spanning matrix and count pivot columns rather than counting the original generators. $\square$

Solution. The condition kills the constant term, so x, x^2, x^3 is a basis. $\square$

Exercise 3.24. Why can no four vectors in $\mathbb{R}^3$ be independent?

Hint. Begin with a basis of the smaller subspace and extend it to a basis of the larger one. $\square$

Solution. Any independent list has length at most the dimension, which is 3. $\square$

Exercise 3.25. If $U \subseteq V$ and $\dim U = \dim V < \infty$, prove $U = V$.

Hint. For the dimension formula, begin with a basis of $U \cap W$, then extend it separately inside U and W . $\square$

Solution. A basis of U is an independent list in V of full dimension, hence already a basis of V . $\square$

Exercise 3.26. Compute the dimension of skew-symmetric $n \times n$ matrices over a field of characteristic not 2.

Hint. Compare the size of an independent list with the size of a spanning list in the same finite-dimensional space. $\square$

Solution. Diagonal entries are zero and entries below the diagonal are determined by those above it, giving $n(n-1)/2$. $\square$

Exercise 3.27. Suppose $\dim U = 3$, $\dim W = 4$, and $\dim(U \cap W) = 2$. Find $\dim(U + W)$.

Hint. To test whether a list can be a basis, compare its length with the dimension before doing any computation. $\square$

Solution. The dimension formula gives $3 + 4 - 2 = 5$. $\square$

Exercise 3.28. Explain why every finite spanning set contains a basis.

Hint. Use rank-nullity as a dimension count: injectivity kills the nullity and surjectivity forces full rank. $\square$

Solution. Repeatedly delete any vector lying in the span of the others. The process terminates and preserves span; the remaining list is independent. $\square$

3.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 3.29 (Basis coordinates). For $B = ((1, 0), (1, 1))$, find $[(4, 3)]_B$.

Hint. Solve $a(1, 0) + b(1, 1) = (4, 3)$. $\square$

Solution. $b = 3, a = 1$, so the coordinates are $(1, 3)^T$. $\square$

Exercise 3.30 (Polynomial subspace dimension). Find the dimension of $\{p \in \mathcal{P}_5 : p(0) = 0\}$.

Hint. The constant coefficient is forced to zero. $\square$

Solution. A basis is x, x^2, x^3, x^4, x^5 , so the dimension is 5. $\square$

Exercise 3.31 (Symmetric matrices). Compute $\dim \text{Sym}_4(\mathbb{R})$.

Hint. Count diagonal and above-diagonal entries. $\square$

Solution. $4 + 6 = 10$. $\square$

Exercise 3.32 (Dimension formula). If $\dim U = 6, \dim W = 5, \dim(U \cap W) = 3$, find $\dim(U + W)$.

Hint. Use inclusion–exclusion for dimensions. $\square$

Solution. $6 + 5 - 3 = 8$. $\square$

Exercise 3.33 (Constraint count). Find $\dim\{(x_1, \dots, x_5) : x_1 + \dots + x_5 = 0\}$.

Hint. One independent homogeneous equation removes one degree of freedom. $\square$

Solution. The dimension is 4. $\square$

Proofs

Exercise 3.34 (Full-dimension subspace). Prove $U \subseteq V$, $\dim U = \dim V < \infty$ implies $U = V$.

Hint. A basis of U already has the maximum possible length in V . $\square$

Solution. That basis is a basis of V , hence spans $V \subseteq U$. $\square$

Exercise 3.35 (Spanning list of length n). Prove n spanning vectors in an n -dimensional space form a basis.

Hint. If dependent, delete a redundant vector. $\square$

Solution. Dependence would leave fewer than n spanning vectors, contradicting dimension. $\square$

Exercise 3.36 (Independent list of length n). Prove n independent vectors in an n -dimensional space span.

Hint. Extend the independent list to a basis. $\square$

Solution. There is no room to add another basis vector, so the list is already a basis. $\square$

Exercise 3.37 (Direct-sum dimension). Prove $\dim(U \oplus W) = \dim U + \dim W$.

Hint. Take bases of U, W and prove their union is a basis. $\square$

Solution. The union spans by decomposition and is independent because $U \cap W = 0$. $\square$

Counterexamples

Exercise 3.38 (Infinite-dimensional warning). Give an independent list in an infinite-dimensional space that is not a basis.

Hint. Use $1, x$ inside $\mathbb{R}[x]$. $\square$

Solution. $\{1, x\}$ is independent but does not span x^2 , so it is not a basis. $\square$

Exercise 3.39 (Spanning without independence). Give three spanning vectors of $\mathbb{R}^2$ that are not a basis.

Hint. Add a redundant vector to the standard basis. $\square$

Solution. $(1, 0), (0, 1), (1, 1)$ span but are dependent. $\square$

Exercise 3.40 (Independent without spanning). Give two independent vectors in $\mathbb{R}^3$ that do not span.

Hint. Use e_1, e_2 . $\square$

Solution. They are independent but miss e_3 . $\square$

Applications

Exercise 3.41 (Data coordinates). A three-feature model is restricted by one independent linear constraint. What is the dimension of its feasible subspace?

Hint. One independent homogeneous constraint reduces dimension by one. $\square$

Solution. The dimension is 2. $\square$

Exercise 3.42 (Matrix parameter count). How many parameters specify a 5×5 skew-symmetric real matrix?

Hint. Diagonal entries vanish and lower entries are determined by upper entries. $\square$

Solution. $\binom{5}{2} = 10$. $\square$

Challenges

Exercise 3.43 (Two subspaces in $\mathbb{R}^8$). In $\mathbb{R}^8$, can two 5-dimensional subspaces have zero intersection?

Hint. Use $\dim(U + W) \leq 8$. □

Solution. No. The dimension formula gives $\dim(U \cap W) \geq 5 + 5 - 8 = 2$. □

Exercise 3.44 (Basis replacement). If $B = (v_1, \dots, v_n)$ is a basis and $w = \sum a_i v_i$ with $a_k \neq 0$, prove replacing v_k by w still gives a basis.

Hint. Solve the equation for v_k in terms of w and the other v_i . □

Solution. The new list spans; it has n vectors in an n -dimensional space, hence is a basis. □

Chapter summary

A basis is simultaneously a generating system and a coordinate system with no redundancy. Finite-dimensional vector spaces become computationally manageable because every vector then has a unique finite coordinate list. Dimension records the size of any basis and is independent of the particular basis chosen. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 4

Coordinates and Change of Basis

Coordinates turn an abstract vector space into a copy of $\mathbb{F}^n$, but the copy depends on a chosen basis. Change-of-basis matrices explain precisely how two coordinate descriptions of the same vector are related. This is the first systematic appearance of conjugation and covariance.

Learning goals

After this chapter, the reader should be able to:

- compute the coordinate vector of an abstract vector relative to a specified ordered basis;
- construct a change-of-basis matrix from one ordered basis to another;
- convert coordinate vectors between two bases using the change-of-basis matrix and its inverse;
- derive the similarity relation between matrices of the same operator in different bases;
- choose a basis adapted to a subspace or decomposition in order to simplify coordinates;
- detect and correct an invalid change-of-basis computation by checking what the columns represent;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

4.1 Coordinate isomorphisms

For an ordered basis $B = (v_1, \dots, v_n)$, the coordinate map $\kappa_B : V \rightarrow \mathbb{F}^n$, $v \mapsto [v]_B$, is a linear isomorphism.

Example 4.1 (A coordinate isomorphism in practice). For $B = ((1, 1), (1, -1))$, the coordinate map sends

$$(x, y) \mapsto [(x, y)]_B = \begin{pmatrix} (x+y)/2 \\ (x-y)/2 \end{pmatrix}.$$

It is linear, and reconstructing gives

$$\frac{x+y}{2}(1,1) + \frac{x-y}{2}(1,-1) = (x,y).$$

Thus the coordinate map is visibly invertible.

This example makes the abstraction concrete: choosing a basis is choosing a linear isomorphism from the vector space to a coordinate space.

4.2 Transition matrices

Given bases B, C , the matrix $P_{C \leftarrow B}$ is defined by $[v]_C = P_{C \leftarrow B}[v]_B$. Its columns are the C -coordinates of the vectors of B .

Example 4.2 (Building and checking a transition matrix). Let $B = ((1,1), (1,-1))$ and E be the standard basis. Since the columns of $P_{E \leftarrow B}$ are the E -coordinates of the B -vectors,

$$P_{E \leftarrow B} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

For $v = 3b_1 + 2b_2$,

$$[v]_E = P_{E \leftarrow B} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Direct reconstruction gives the same vector, which checks both the columns and the direction of the arrow in the notation.

4.3 Inverse transitions

Changing from B to C and back gives $P_{B \leftarrow C} = P_{C \leftarrow B}^{-1}$.

4.4 Coordinate covariants

Vectors and linear maps do not change when the basis changes; their coordinate arrays do. Good notation separates the object from its representation.

4.5 Dual coordinates

Linear functionals transform by the inverse transpose of the vector change-of-basis matrix. This anticipates the distinction between vectors and covectors.

4.6 Conditioning of bases

A mathematically valid basis can be numerically poor if its vectors are nearly dependent. Orthogonal bases mitigate this instability.

Example 4.3 (Near dependence and numerical instability). Take $B_\varepsilon = ((1,0), (1,\varepsilon))$. Its basis matrix is

$$P_\varepsilon = \begin{pmatrix} 1 & 1 \\ 0 & \varepsilon \end{pmatrix}, \quad P_\varepsilon^{-1} = \begin{pmatrix} 1 & -1/\varepsilon \\ 0 & 1/\varepsilon \end{pmatrix}.$$

For every nonzero ε this is a valid basis, but when $|\varepsilon|$ is small, the inverse has very large entries.

So algebraic invertibility and numerical reliability are different questions. The determinant confirms validity; the size of the inverse warns about sensitivity.

4.7 Core structural results

Theorem 4.4 (Transition matrix rule). *The columns of $P_{C \leftarrow B}$ are $[v_1]_C, \dots, [v_n]_C$ when $B = (v_1, \dots, v_n)$.*

Proof. Apply the transition identity to each basis vector. Its B -coordinate is the corresponding standard vector, so the resulting C -coordinate is the corresponding column. $\square$

Theorem 4.5 (Cocycle identity). *For bases B, C, D , $P_{D \leftarrow B} = P_{D \leftarrow C} P_{C \leftarrow B}$.*

Proof. Apply the two coordinate changes successively to an arbitrary coordinate vector and use uniqueness of the representing matrix. $\square$

4.8 Worked examples

Example 4.6 (Two planar bases). For $B = ((1, 0), (1, 1))$ and standard E , $P_{E \leftarrow B} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Example 4.7 (Polynomial bases). The monomial basis and shifted basis $1, (x - 1), (x - 1)^2$ of $\mathcal{P}_2$ are related by expanding powers of $x - 1$.

Example 4.8 (Scaling a basis). Replacing v_i by $c_i v_i$ multiplies the corresponding transition column by c_i .

4.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

4.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 4.9 (Transition matrix from columns). For $B = ((1, 1), (1, -1))$ and the standard basis E , compute $P_{E \leftarrow B}$ and its inverse.

Solution. The columns are the E -coordinates of the B -vectors, so $P_{E \leftarrow B} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Its inverse is $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, which is $P_{B \leftarrow E}$. $\square$

Problem 4.10 (Coordinates of a polynomial). Find the coordinates of $x^2 + 1$ in $B = (1, x - 1, (x - 1)^2)$.

Solution. Write $x = (x - 1) + 1$. Then $x^2 + 1 = (x - 1)^2 + 2(x - 1) + 2$, so the coordinate vector is $(2, 2, 1)^T$. $\square$

Problem 4.11 (Cocycle identity). Given bases B, C, D , prove $P_{D \leftarrow B} = P_{D \leftarrow C} P_{C \leftarrow B}$.

Solution. For every vector v , first change B -coordinates to C -coordinates and then to D -coordinates. Thus $[v]_D = P_{D \leftarrow C} P_{C \leftarrow B} [v]_B$. Uniqueness of the matrix realizing this coordinate transformation gives the identity. $\square$

Problem 4.12 (A badly conditioned basis). Why is $B_\varepsilon = ((1, 0), (1, \varepsilon))$ a basis for $\varepsilon \neq 0$ but numerically poor when $|\varepsilon|$ is small?

Solution. The determinant of the basis matrix is ε , so it is invertible for $\varepsilon \neq 0$. Its inverse contains entries of size $1/\varepsilon$, so small perturbations in coordinates can be greatly amplified as $\varepsilon \rightarrow 0$. $\square$

Problem 4.13 (Permutation of basis vectors). If $C = (v_2, v_1, v_3)$ is obtained from $B = (v_1, v_2, v_3)$ by swapping the first two vectors, describe $P_{C \leftarrow B}$.

Solution. It is the permutation matrix that swaps the first two coordinates: $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. The same matrix is its inverse. $\square$

Problem 4.14 (Scaling a basis). Let $C = (2v_1, 3v_2)$ when $B = (v_1, v_2)$. Find $P_{C \leftarrow B}$.

Solution. If $v = av_1 + bv_2 = c(2v_1) + d(3v_2)$, then $c = a/2$ and $d = b/3$. Thus $P_{C \leftarrow B} = \text{diag}(1/2, 1/3)$. $\square$

Problem 4.15 (Coordinates versus objects). Two coordinate columns for the same vector are different. Explain why this does not contradict uniqueness of coordinates.

Solution. Coordinates are unique only after the basis is fixed. Different bases define different coordinate isomorphisms $V \rightarrow \mathbb{F}^n$, so the same abstract vector can have different coordinate columns without ambiguity. $\square$

Problem 4.16 (Dual transformation). Suppose vector coordinates change by $x_C = P x_B$. How must a row covector ℓ change so that the scalar $\ell(x)$ is invariant?

Solution. We require $\ell_C x_C = \ell_B x_B$ for all x_B . Since $x_C = P x_B$, this gives $\ell_C P = \ell_B$, hence $\ell_C = \ell_B P^{-1}$. In column-covector convention this becomes the inverse transpose. $\square$

Problem 4.17 (Coordinate map isomorphism). Prove the coordinate map $\kappa_B : V \rightarrow \mathbb{F}^n$ is linear and bijective.

Solution. Linearity follows by combining coefficients in the basis expansion. Every coordinate vector gives a linear combination of the basis vectors, so the map is onto. Uniqueness of basis coordinates gives injectivity. $\square$

Problem 4.18 (Three-basis computation). If $P_{C \leftarrow B} = A$ and $P_{D \leftarrow C} = B$, what is $P_{B \leftarrow D}$?

Solution. First $P_{D \leftarrow B} = BA$. Therefore the reverse transition is $(BA)^{-1} = A^{-1}B^{-1}$. $\square$

Problem 4.19 (Basis from columns). Why does every invertible matrix P determine a new ordered basis relative to a fixed basis E ?

Solution. Take as new basis vectors the vectors whose E -coordinate columns are the columns of P . Invertibility means those columns are independent and span the whole space, so they form a basis. $\square$

Problem 4.20 (Coordinate synthesis). Explain why changing basis is not the same operation as applying a linear transformation to vectors.

Solution. A basis change changes only the description of the same vector: the geometric object is fixed and coordinates are transformed by a transition matrix. Applying an operator changes the vector itself. Confusing these roles is the source of many incorrect conjugation formulas. $\square$

4.11 Exercises with complete solutions

Exercise 4.21. Compute $P_{E \leftarrow B}$ for $B = ((1, 1), (1, -1))$.

Hint. Write each new basis vector in the old coordinates and use those coordinate columns to build the conversion matrix. Solve $v = a_1b_1 + \cdots + a_nb_n$; the coefficients you obtain are the coordinate vector. $\square$

Solution. Its columns are the basis vectors in standard coordinates: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. $\square$

Exercise 4.22. Find $P_{B \leftarrow E}$ for the preceding basis.

Hint. Build the change-of-basis matrix column by column from the new basis vectors written in the old coordinates. $\square$

Solution. Invert the matrix to get $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. $\square$

Exercise 4.23. Verify the cocycle identity for three arbitrary bases.

Hint. Check the direction of the change-of-basis matrix by applying it to a coordinate vector whose answer you know. $\square$

Solution. For every v , substitute $[v]_C = P_{C \leftarrow B}[v]_B$ into $[v]_D = P_{D \leftarrow C}[v]_C$; equality for all coordinate vectors gives the matrix identity. $\square$

Exercise 4.24. Why must every transition matrix be invertible?

Hint. The inverse change-of-basis matrix reverses the coordinate conversion; verify by composing the two conversions. $\square$

Solution. It represents the composition of two coordinate isomorphisms, hence an isomorphism. $\square$

Exercise 4.25. Compute coordinates of $p(x) = x^2$ in $B = (1, x - 1, (x - 1)^2)$.

Hint. For an operator matrix, insert identity changes of coordinates on the domain and codomain before simplifying. $\square$

Solution. Since $x = (x - 1) + 1$, $x^2 = (x - 1)^2 + 2(x - 1) + 1$, so the coordinates are $(1, 2, 1)^T$. $\square$

Exercise 4.26. How do coordinates change if the basis order is permuted?

Hint. Write each new basis vector in the old coordinates and use those coordinate columns to build the conversion matrix. Similarity should appear as $P^{-1}AP$; determine which side corresponds to changing input coordinates first. $\square$

Solution. The coordinate column is permuted inversely to the basis order; the transition is represented by a permutation matrix. $\square$

Exercise 4.27. Explain why the vector itself is basis-independent.

Hint. Write each new basis vector in the old coordinates and use those coordinate columns to build the conversion matrix. An adapted basis should begin with a basis of the distinguished subspace and then be extended to the whole space. $\square$

Solution. A basis selects a coordinate isomorphism but does not alter the element $v \in V$; only its numerical representative changes. $\square$

Exercise 4.28. Why are nearly parallel planar basis vectors numerically problematic?

Hint. Test a proposed formula on each basis vector; a matrix representation is determined by those columns. $\square$

Solution. The transition matrix is close to singular, so its inverse can have large entries and amplify small data errors. $\square$

4.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 4.29 (Transition columns). For $B = ((1, 1), (2, 1))$, compute $P_{E \leftarrow B}$.

Hint. Use the basis vectors as columns in standard coordinates. $\square$

Solution. $\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$. $\square$

Exercise 4.30 (Reverse transition). Find $P_{B \leftarrow E}$ for the preceding basis.

Hint. Invert $P_{E \leftarrow B}$. $\square$

Solution. The determinant is -1 , so $P_{B \leftarrow E} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$. $\square$

Exercise 4.31 (Coordinate conversion). If $[v]_B = (3, -1)^T$ and $P_{E \leftarrow B} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$, find $[v]_E$.

Hint. Multiply the transition matrix by the coordinate column. $\square$

Solution. $[v]_E = (1, 2)^T$. $\square$

Exercise 4.32 (Shifted polynomial basis). Find $[x^2]_{(1, x-1, (x-1)^2)}$.

Hint. Write $x = (x - 1) + 1$ and expand. $\square$

Solution. $x^2 = 1 + 2(x - 1) + (x - 1)^2$, so coordinates are $(1, 2, 1)^T$. $\square$

Exercise 4.33 (Cocycle). If $P_{C \leftarrow B} = A$, $P_{D \leftarrow C} = B_0$, express $P_{D \leftarrow B}$.

Hint. Apply the two coordinate changes in succession. $\square$

Solution. $P_{D \leftarrow B} = B_0 A$. $\square$

Proofs

Exercise 4.34 (Transition invertibility). Prove every transition matrix is invertible.

Hint. The reverse coordinate change provides an inverse map. □

Solution. $P_{B \leftarrow C} P_{C \leftarrow B} = I$, so the matrices are inverses. □

Exercise 4.35 (Cocycle identity). Prove $P_{D \leftarrow B} = P_{D \leftarrow C} P_{C \leftarrow B}$.

Hint. Apply both sides to an arbitrary B -coordinate vector. □

Solution. Both products produce the unique D -coordinate column of the same vector. □

Exercise 4.36 (Coordinate map linearity). Prove $v \mapsto [v]_B$ is linear.

Hint. Compare basis coefficients of $av + bw$. □

Solution. Coefficients combine componentwise, so $[av + bw]_B = a[v]_B + b[w]_B$. □

Exercise 4.37 (Similarity derivation). Derive $[T]_C = P^{-1}[T]_B P$ for $P = P_{B \leftarrow C}$.

Hint. Follow a C -coordinate vector into B , apply T , then return. □

Solution. The three steps are P , $[T]_B$, P^{-1} , giving the conjugation formula. □

Counterexamples

Exercise 4.38 (Wrong direction). Explain why using $P_{C \leftarrow B}$ where $P_{B \leftarrow C}$ is required can give a plausible but wrong result.

Hint. Test the matrix on a basis vector whose converted coordinates are known. □

Solution. The two matrices are inverses, generally unequal; a basis-vector test exposes the reversed direction. □

Exercise 4.39 (Singular columns). Why can a singular matrix not be a transition matrix?

Hint. A transition matrix represents an isomorphism between coordinate spaces. □

Solution. A singular matrix is not invertible, so it cannot encode reversible basis conversion. □

Exercise 4.40 (Coordinates are not vectors). Give an example where the same vector has different coordinate columns in two bases.

Hint. Use $v = (1, 1)$, standard basis, and $B = ((1, 1), (1, -1))$. □

Solution. $[v]_E = (1, 1)^T$ but $[v]_B = (1, 0)^T$. □

Applications

Exercise 4.41 (Adapted basis). For $U = \text{span}(1, 1) \subset \mathbb{R}^2$, choose a basis adapted to U .

Hint. Begin with a basis of U and add any vector outside it. □

Solution. $((1, 1), (1, -1))$ is adapted; U is the first coordinate axis in these coordinates. □

Exercise 4.42 (Numerical conditioning). Why is $B_\varepsilon = ((1, 0), (1, \varepsilon))$ unstable for small ε ?

Hint. Inspect the inverse of its basis matrix. □

Solution. The inverse contains $1/\varepsilon$, so small input perturbations can cause large coordinate changes. □

Challenges

Exercise 4.43 (Dual coordinate law). If vector columns transform by $x_C = Px_B$, derive the row-covector law preserving $\ell(x)$.

Hint. Require $\ell_C x_C = \ell_B x_B$ for all x_B . □

Solution. $\ell_C P = \ell_B$, so $\ell_C = \ell_B P^{-1}$; in column convention this is inverse transpose. □

Exercise 4.44 (Three basis reversal). If $P_{C \leftarrow B} = A$, $P_{D \leftarrow C} = B_0$, find $P_{B \leftarrow D}$.

Hint. First compose forward, then invert. □

Solution. $P_{D \leftarrow B} = B_0 A$, hence $P_{B \leftarrow D} = A^{-1} B_0^{-1}$. □

Chapter summary

Coordinates turn an abstract vector space into a copy of $\mathbb{F}^n$, but the copy depends on a chosen basis. Change-of-basis matrices explain precisely how two coordinate descriptions of the same vector are related. This is the first systematic appearance of conjugation and covariance. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 5

Linear Transformations

Linear transformations are the morphisms of vector spaces: they preserve exactly the operations that define linear structure. Once linearity is recognized, a map is determined by its action on a basis, compositions remain linear, and apparently different formulas become instances of the same operator calculus.

Learning goals

After this chapter, the reader should be able to:

- verify or refute linearity by testing additivity and homogeneity;
- determine a linear transformation uniquely from its values on a basis;
- construct a linear map satisfying prescribed images of basis vectors;
- compute compositions of linear transformations and identify when two maps commute;
- decide whether a proposed inverse map is linear and actually inverts the original transformation;
- translate geometric actions such as projections, reflections, and coordinate permutations into linear maps;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

5.1 Definition and first consequences

A map $T : V \rightarrow W$ is linear if $T(av + bw) = aT(v) + bT(w)$. Then $T(0) = 0$ and all finite linear combinations pass through T .

Example 5.1 (A fast linearity diagnosis). Consider

$$F(x, y) = (x + y, xy).$$

Although the first coordinate is linear, the second is not. Indeed

$$F(1, 0) + F(0, 1) = (1, 0) + (1, 0) = (2, 0),$$

while

$$F(1, 1) = (2, 1).$$

Thus additivity fails.

A useful diagnostic order is: first test $F(0) = 0$; then test additivity or homogeneity on simple vectors. One failed identity settles nonlinearity, so there is no reason to expand a general formula unnecessarily.

5.2 Determination by a basis

A linear map is uniquely determined by its values on a basis of the domain. Conversely, arbitrary images of basis vectors extend uniquely to a linear map.

Example 5.2 (Constructing a map from basis data). Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfy

$$T(e_1) = (1, 0, 1), \quad T(e_2) = (0, 1, 1).$$

Since $(x, y) = xe_1 + ye_2$, linearity forces

$$T(x, y) = xT(e_1) + yT(e_2) = (x, y, x + y).$$

Substituting e_1, e_2 recovers the prescribed values, so the formula is checked.

This is the basic construction principle: arbitrary images may be assigned to basis vectors, and there is exactly one linear extension.

5.3 Algebra of operators

For fixed V, W , the set $\mathcal{L}(V, W)$ is a vector space under pointwise addition and scalar multiplication. Endomorphisms $\mathcal{L}(V)$ also compose.

5.4 Composition

The composite of linear maps is linear. Composition is associative and distributes over addition but is generally not commutative.

Example 5.3 (Composition order matters). Let $P(x, y) = (x, 0)$ and $R(x, y) = (-y, x)$. Then

$$(RP)(x, y) = R(x, 0) = (0, x),$$

whereas

$$(PR)(x, y) = P(-y, x) = (-y, 0).$$

Thus $RP \neq PR$.

Geometrically, projecting and then rotating is not the same as rotating and then projecting. The calculation explains why operator multiplication is noncommutative even though vector addition is commutative.

5.5 Inverse maps

A bijective linear map has a linear inverse. Such maps are isomorphisms and identify vector spaces as having the same linear structure.

5.6 Important operators

Differentiation on polynomials, evaluation functionals, coordinate projections, inclusion maps, shifts, and matrix multiplication are canonical examples.

5.7 Core structural results

Theorem 5.4 (Extension from a basis). *If $B = (v_1, \dots, v_n)$ is a basis of V and $w_i \in W$ are arbitrary, there is a unique linear $T : V \rightarrow W$ with $T(v_i) = w_i$.*

Proof. Write each v uniquely as $\sum a_i v_i$ and define $T(v) = \sum a_i w_i$. Uniqueness follows because any linear map must use this formula. $\square$

Theorem 5.5 (Inverse of an isomorphism). *If $T : V \rightarrow W$ is linear and bijective, then $T^{-1} : W \rightarrow V$ is linear.*

Proof. Write $y_j = T(x_j)$. Then $T^{-1}(ay_1 + by_2) = T^{-1}(T(ax_1 + bx_2)) = ax_1 + bx_2$. $\square$

5.8 Worked examples

Example 5.6 (Differentiation). $D : \mathcal{P}_n \rightarrow \mathcal{P}_{n-1}$, $D(p) = p'$, is linear because differentiation respects sums and scalar multiples.

Example 5.7 (Evaluation). For fixed a , $E_a(p) = p(a)$ is a linear functional on a polynomial space.

Example 5.8 (Shear). $T(x, y) = (x + y, y)$ is linear and invertible, with inverse $(u, v) \mapsto (u - v, v)$.

5.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

5.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 5.9 (Basis images determine a map). Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfy $T(e_1) = (1, 0, 1)$ and $T(e_2) = (0, 1, 1)$. Compute $T(x, y)$.

Solution. Linearity gives $T(x, y) = xT(e_1) + yT(e_2) = (x, y, x + y)$. No other linear map can have the prescribed basis images. $\square$

Problem 5.10 (Detecting nonlinearity quickly). Is $F(x, y) = (x + y, xy)$ linear?

Solution. No. The second component is quadratic. For instance $F(2, 0) = (2, 0)$ while $2F(1, 0) = (2, 0)$ happens to agree, but $F(1, 1) = (2, 1)$ whereas $F(1, 0) + F(0, 1) = (2, 0)$, so additivity fails. $\square$

Problem 5.11 (Differentiation as an operator). On $\mathcal{P}_3$, determine $\ker D$ and $\operatorname{im} D$ for $D(p) = p'$.

Solution. The derivative is zero exactly for constant polynomials, so $\ker D = \mathcal{P}_0$. Every polynomial of degree at most 2 has an antiderivative in $\mathcal{P}_3$, so $\operatorname{im} D = \mathcal{P}_2$. $\square$

Problem 5.12 (Composition order). Let $P(x, y) = (x, 0)$ and $R(x, y) = (-y, x)$. Compare RP and PR .

Solution. $RP(x, y) = R(x, 0) = (0, x)$, whereas $PR(x, y) = P(-y, x) = (-y, 0)$. Thus composition is not commutative. $\square$

Problem 5.13 (A nilpotent map). Define $N(x, y, z) = (y, z, 0)$. Determine the smallest k with $N^k = 0$.

Solution. $N^2(x, y, z) = (z, 0, 0)$ and $N^3 = 0$, while $N^2 \neq 0$. Thus the nilpotency index is 3. $\square$

Problem 5.14 (Evaluation functional). Show $E_1 : \mathcal{P}_2 \rightarrow \mathbb{R}$, $E_1(p) = p(1)$, is linear and find its values on $1, x, x^2$.

Solution. Evaluation respects sums and scalar multiples. Each of $1, x, x^2$ takes value 1 at $x = 1$, so the functional is determined by the coordinate row $(1, 1, 1)$. $\square$

Problem 5.15 (Inverse linearity). If T is a bijective linear map, prove T^{-1} is linear.

Solution. Write $y_i = T(x_i)$. Then $T^{-1}(ay_1 + by_2) = T^{-1}(T(ax_1 + bx_2)) = ax_1 + bx_2 = aT^{-1}y_1 + bT^{-1}y_2$. $\square$

Problem 5.16 (Zero map versus zero vector). Explain the difference between the zero vector in W and the zero element of $\mathcal{L}(V, W)$.

Solution. The zero vector is a single element 0_W . The zero operator is a function $0 : V \rightarrow W$ sending every v to 0_W . It is the additive identity in the vector space of linear maps. $\square$

Problem 5.17 (A projection decomposition). Let $P(x, y, z) = (x, y, 0)$. Show $P^2 = P$ and describe its fixed vectors and kernel.

Solution. Applying P twice changes nothing after the first application, so $P^2 = P$. Fixed vectors satisfy $z = 0$, the xy -plane; the kernel is the z -axis. $\square$

Problem 5.18 (A linear map from a formula). For $T(A) = A + A^T$ on $M_2(\mathbb{R})$, prove linearity and characterize the image.

Solution. Transpose is linear, so $T(\alpha A + \beta B) = \alpha T(A) + \beta T(B)$. Every output is symmetric. Conversely, if S is symmetric then $T(S/2) = S$, so the image is exactly the symmetric matrices. $\square$

Problem 5.19 (Restriction to a subspace). Let $T(x, y, z) = (x + y, y + z, z)$. Is the plane $U = \{z = 0\}$ invariant?

Solution. If $(x, y, 0) \in U$, then $T(x, y, 0) = (x + y, y, 0) \in U$. Hence the restriction $T|_U$ is a well-defined linear operator on U . $\square$

Problem 5.20 (Operator synthesis). Why does specifying values on a spanning list fail to define a linear map unless all dependence relations are respected?

Solution. If $\sum a_i v_i = 0$ is a relation in the spanning list, linearity requires $\sum a_i T(v_i) = 0$. Arbitrary assigned images may violate this condition, giving contradictory values for the same vector. A basis avoids the problem because it has no nontrivial relations. $\square$

5.11 Exercises with complete solutions

Exercise 5.21. Show that $T(x, y) = (x + 2y, 3x - y)$ is linear.

Hint. Check $T(u+v) = T(u)+T(v)$ and $T(\lambda u) = \lambda T(u)$ separately; one failure is enough. $\square$

Solution. Each output coordinate is a homogeneous linear expression in x, y , so direct substitution verifies preservation of linear combinations. $\square$

Exercise 5.22. Is $T(x, y) = (x + 1, y)$ linear?

Hint. Write each input vector in the chosen basis and use linearity to extend the prescribed basis values. $\square$

Solution. No. A linear map must send 0 to 0, but this map sends $(0, 0)$ to $(1, 0)$. $\square$

Exercise 5.23. Construct the unique linear map with $T(e_1) = (1, 1)$ and $T(e_2) = (2, 0)$.

Hint. To construct the map, choose images of basis vectors first; linearity then forces every other value. $\square$

Solution. For $(x, y) = xe_1 + ye_2$, $T(x, y) = x(1, 1) + y(2, 0) = (x + 2y, x)$. $\square$

Exercise 5.24. Show that composition of linear maps is linear.

Hint. For a composition, compute the inner map first and keep track of which space each intermediate vector lies in. $\square$

Solution. Apply the inner map to a linear combination, then use linearity of the outer map. $\square$

Exercise 5.25. Prove that two linear maps agreeing on a basis agree everywhere.

Hint. To test invertibility, verify both $ST = I$ and $TS = I$; one-sided cancellation is not enough in an arbitrary setting. $\square$

Solution. Every vector is a linear combination of basis vectors, and both maps distribute over that combination. $\square$

Exercise 5.26. Give a nonzero nilpotent operator on $\mathbb{R}^2$.

Hint. Compare the effect of the two maps on a basis; equality on a basis implies equality everywhere. $\square$

Solution. Take $N(x, y) = (y, 0)$. Then $N \neq 0$ but $N^2 = 0$. $\square$

Exercise 5.27. Show that $\mathcal{L}(V, W)$ is a vector space.

Hint. For a geometric transformation, compute what it does to the standard basis before writing a general formula. $\square$

Solution. Verify the vector-space axioms pointwise; the zero vector is the zero map. $\square$

Exercise 5.28. Explain why a bijective nonlinear map need not be an isomorphism of vector spaces.

Hint. If a constant term survives when the input is zero, the map cannot be linear. $\square$

Solution. An isomorphism must preserve addition and scalar multiplication; bijectivity alone preserves only set structure. $\square$

5.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 5.29 (Formula from basis images). If $T(e_1) = (1, 2)$, $T(e_2) = (-1, 3)$, find $T(x, y)$.

Hint. Use $T(xe_1 + ye_2) = xT(e_1) + yT(e_2)$. $\square$

Solution. $T(x, y) = (x - y, 2x + 3y)$. $\square$

Exercise 5.30 (Composition). For $T(x, y) = (x + y, y)$, $S(u, v) = (u - v, v)$, compute $S \circ T$.

Hint. Substitute $T(x, y)$ into S . $\square$

Solution. $(S \circ T)(x, y) = (x, y)$, so $S = T^{-1}$. $\square$

Exercise 5.31 (Nilpotent power). For $N(x, y, z) = (y, z, 0)$, compute N^2, N^3 .

Hint. Apply the same formula repeatedly. $\square$

Solution. $N^2(x, y, z) = (z, 0, 0)$ and $N^3 = 0$. $\square$

Exercise 5.32 (Projection). For $P(x, y, z) = (x, y, 0)$, find all fixed vectors.

Hint. Solve $P(v) = v$. $\square$

Solution. The fixed vectors are exactly $z = 0$, the xy -plane. $\square$

Exercise 5.33 (Evaluation). On $\mathcal{P}_2$, compute $E_2(1 - 3x + x^2)$.

Hint. Substitute $x = 2$. $\square$

Solution. $1 - 6 + 4 = -1$. $\square$

Proofs

Exercise 5.34 (Basis determination). Prove two linear maps agreeing on a basis agree everywhere.

Hint. Expand an arbitrary vector in the basis. □

Solution. Linearity makes both maps the same linear combination of identical basis images. □

Exercise 5.35 (Composition linear). Prove the composite of two linear maps is linear.

Hint. Pass a general linear combination through the inner map and then the outer map. □

Solution. $(ST)(av + bw) = S(aTv + bTw) = aSTv + bSTw$. □

Exercise 5.36 (Inverse linear). Prove the inverse of a bijective linear map is linear.

Hint. Write $y_i = T(x_i)$ and pull a combination back through T . □

Solution. $T^{-1}(ay_1 + by_2) = T^{-1}T(ax_1 + bx_2) = ax_1 + bx_2$. □

Exercise 5.37 (Operator space). Prove $\mathcal{L}(V, W)$ is closed under linear combinations.

Hint. Check $(aS + bT)(cv + dw)$. □

Solution. Pointwise operations preserve the linearity identity, so $aS + bT$ is linear. □

Counterexamples

Exercise 5.38 (Translation). Show $F(x, y) = (x + 1, y)$ is not linear.

Hint. Test the zero vector. □

Solution. $F(0, 0) = (1, 0) \neq 0$. □

Exercise 5.39 (Quadratic component). Show $F(x, y) = (x, y^2)$ is not linear.

Hint. Test homogeneity with scalar 2. □

Solution. $F(2x, 2y) = (2x, 4y^2) \neq 2F(x, y)$ when $y \neq 0$. □

Exercise 5.40 (Noncommuting maps). Give two linear maps on $\mathbb{R}^2$ that do not commute.

Hint. Use a projection and a quarter-turn. □

Solution. With $P(x, y) = (x, 0)$, $R(x, y) = (-y, x)$, $RP(1, 1) = (0, 1)$ but $PR(1, 1) = (-1, 0)$. □

Applications

Exercise 5.41 (Reflection). Write the reflection across the x -axis as a linear map and describe its square.

Hint. Flip the sign of the second coordinate. □

Solution. $R(x, y) = (x, -y)$, and $R^2 = I$. □

Exercise 5.42 (Differentiation). Why is differentiation a linear map $\mathcal{P}_3 \rightarrow \mathcal{P}_2$?

Hint. Use $(ap + bq)' = ap' + bq'$. □

Solution. Differentiation preserves all linear combinations and lowers degree by one. □

Challenges

Exercise 5.43 (Images on dependent generators). Can one define a linear map by prescribing arbitrary images of $v_1, v_2, v_1 + v_2$?

Hint. The dependence relation must be preserved. □

Solution. Only if the third prescribed image equals the sum of the first two; otherwise linearity gives a contradiction. □

Exercise 5.44 (Idempotent decomposition). If $P^2 = P$, prove every v decomposes as $v = Pv + (v - Pv)$ with first term in $\text{im } P$ and second in $\ker P$.

Hint. Apply P to $v - Pv$. □

Solution. $P(v - Pv) = Pv - P^2v = 0$, while $Pv \in \text{im } P$; the displayed equality is immediate. □

Chapter summary

Linear transformations are the morphisms of vector spaces: they preserve exactly the operations that define linear structure. Once linearity is recognized, a map is determined by its action on a basis, compositions remain linear, and apparently different formulas become instances of the same operator calculus. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 6

Kernels, Images and Isomorphisms

Kernel and image measure exactly how a linear transformation fails to be injective or surjective. The rank–nullity theorem balances those two defects and turns structural questions into dimension counts. Quotient spaces then package the first isomorphism theorem in its natural form.

Learning goals

After this chapter, the reader should be able to:

- compute bases for the kernel and image of a linear transformation;
- use rank-nullity to determine missing dimensions and to test consistency of computed bases;
- decide injectivity from the kernel and surjectivity from the image;
- prove that a finite-dimensional map is an isomorphism from equivalent injective, surjective, or rank conditions;
- construct an explicit isomorphism between two vector spaces of the same finite dimension;
- relate solutions of $Tx = b$ to one particular solution plus the kernel of T ;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

6.1 Kernel

$\ker T = \{v : T(v) = 0\}$ is a subspace of the domain. A linear map is injective exactly when its kernel is zero.

Example 6.1 (Kernel and image from one formula). For

$$T(x, y, z) = (x + y, y + z),$$

the kernel satisfies $x + y = 0$, $y + z = 0$, hence

$$\ker T = \text{span}\{(-1, 1, -1)\}.$$

The images of the standard basis are $(1, 0)$, $(1, 1)$, $(0, 1)$, which span $\mathbb{R}^2$, so $\text{im } T = \mathbb{R}^2$.

The dimensions $1 + 2 = 3$ check rank–nullity. Computing both sides and then checking the dimension balance is a reliable way to catch algebra mistakes.

6.2 Image and rank

$\text{im } T = \{T(v) : v \in V\}$ is a subspace of the codomain. Its dimension is the rank.

Example 6.2 (Rank–nullity as a prediction tool). Suppose $T : \mathbb{R}^7 \rightarrow \mathbb{R}^4$ is surjective. Then

$$\text{rank } T = 4$$

without any row reduction. Rank–nullity immediately gives

$$\dim \ker T = 7 - 4 = 3.$$

Thus every equation $Tx = b$ has a three-parameter family of solutions once one particular solution is known.

Dimension information can therefore predict the size of a solution family before individual solutions are computed.

6.3 Rank–nullity

For finite-dimensional V , $\dim V = \dim \ker T + \dim \text{im } T$.

6.4 Injective versus surjective

For a linear endomorphism of a finite-dimensional space, injective, surjective and bijective are equivalent.

6.5 Quotients

For a subspace $U \subseteq V$, the quotient V/U consists of cosets $v + U$. It has dimension $\dim V - \dim U$ in finite dimensions.

6.6 First isomorphism theorem

The map $V/\ker T \rightarrow \text{im } T$, $v + \ker T \mapsto T(v)$, is a well-defined linear isomorphism.

Example 6.3 (Quotienting away exactly the kernel). For $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (x, y)$, the kernel is the z -axis. Two vectors have the same image exactly when they differ by a kernel vector:

$$T(v) = T(w) \iff v - w \in \ker T.$$

Hence a coset $v + \ker T$ records precisely the information that survives under T . The map

$$\mathbb{R}^3/\ker T \longrightarrow \mathbb{R}^2, \quad v + \ker T \mapsto T(v)$$

is therefore well defined and bijective. The quotient has removed exactly the redundancy of the original map.

6.7 Core structural results

Theorem 6.4 (Rank–nullity theorem). *If $T : V \rightarrow W$ and V is finite-dimensional, then $\dim V = \dim \ker T + \dim \operatorname{im} T$.*

Proof. Choose a basis of the kernel and extend it to a basis of V . The images of the added basis vectors form a basis of the image. $\square$

Theorem 6.5 (First isomorphism theorem). *Every linear map induces an isomorphism $V/\ker T \cong \operatorname{im} T$.*

Proof. The induced map is well-defined because changing a representative by a kernel vector does not change its image; injectivity and surjectivity are immediate. $\square$

6.8 Worked examples

Example 6.6 (Differentiation). On $\mathcal{P}_n$, differentiation has kernel the constants and image $\mathcal{P}_{n-1}$, so nullity 1 plus rank n equals $n + 1$.

Example 6.7 (Projection). The projection $\mathbb{R}^3 \rightarrow \mathbb{R}^2$, $(x, y, z) \mapsto (x, y)$, has one-dimensional kernel and rank 2.

Example 6.8 (Trace functional). $\operatorname{tr} : M_n \rightarrow \mathbb{F}$ is surjective; its kernel has dimension $n^2 - 1$.

6.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

6.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 6.9 (Rank–nullity from equations). For $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$, $T(x_1, x_2, x_3, x_4) = (x_1 + x_3, x_2 + x_4)$, find rank and nullity.

Solution. The map is onto because $(a, b) = T(a, b, 0, 0)$, so rank is 2. The kernel is given by $x_3 = -x_1$, $x_4 = -x_2$, hence has dimension 2. Rank plus nullity is 4. $\square$

Problem 6.10 (Kernel detects injectivity). Prove that a linear map is injective iff its kernel is $\{0\}$.

Solution. If $T(v) = T(w)$, then $T(v - w) = 0$; trivial kernel forces $v = w$. Conversely, if T is injective and $T(v) = 0 = T(0)$, then $v = 0$. $\square$

Problem 6.11 (First isomorphism theorem). For $T(x, y, z) = (x, y)$, identify $\mathbb{R}^3/\ker T$ with $\mathbb{R}^2$.

Solution. The kernel is the z -axis. Map the coset $(x, y, z) + \ker T$ to (x, y) . This is well-defined, linear, injective, and onto; vertical displacement is exactly what the quotient forgets. $\square$

Problem 6.12 (Trace kernel). Compute the dimension of $\{A \in M_3(\mathbb{R}) : \operatorname{tr} A = 0\}$.

Solution. Trace is a nonzero linear functional from a 9-dimensional space to $\mathbb{R}$, hence has rank 1. Rank-nullity gives kernel dimension 8. $\square$

Problem 6.13 (Image of differentiation). For $D : \mathcal{P}_4 \rightarrow \mathcal{P}_3$, determine rank and nullity.

Solution. The kernel consists of constants, dimension 1. Every cubic has an antiderivative of degree at most 4, so D is onto $\mathcal{P}_3$, which has dimension 4. Thus rank 4, nullity 1. $\square$

Problem 6.14 (Injective endomorphism). Why is an injective linear endomorphism of a finite-dimensional space automatically surjective?

Solution. If $\dim V = n$ and the kernel is zero, rank-nullity gives rank n . The image is an n -dimensional subspace of the n -dimensional codomain V , so it equals V . $\square$

Problem 6.15 (A quotient dimension). If $\dim V = 10$ and $\dim U = 4$, compute $\dim(V/U)$.

Solution. Choose a basis of U and extend it to a basis of V . The six cosets of the added basis vectors form a basis of V/U , so the quotient dimension is $10 - 4 = 6$. $\square$

Problem 6.16 (Rank under composition). Show $\operatorname{rank}(ST) \leq \min(\operatorname{rank} S, \operatorname{rank} T)$.

Solution. The image of ST is $S(\operatorname{im} T)$, so its dimension cannot exceed $\dim \operatorname{im} T$. It is also a subspace of $\operatorname{im} S$, so it cannot exceed $\operatorname{rank} S$. $\square$

Problem 6.17 (Same kernel, different maps). Give two nonzero linear maps $\mathbb{R}^2 \rightarrow \mathbb{R}$ with the same kernel.

Solution. Take $T(x, y) = x$ and $S(x, y) = 2x$. Both have kernel the y -axis, but they are different maps. The kernel alone does not determine a linear map. $\square$

Problem 6.18 (Surjective map splitting). Let $T : V \rightarrow W$ be surjective with finite dimensions. Show there is a subspace $U \subset V$ such that $T|_U : U \rightarrow W$ is an isomorphism.

Solution. Choose a basis of $\ker T$ and extend it to a basis of V . Let U be the span of the added vectors. Rank-nullity shows their number is $\dim W$, and their images form a basis of W , so the restriction is bijective. $\square$

Problem 6.19 (Kernel of a matrix operator). Let $L_A : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ be $L_A(X) = AX$ with invertible A . Find its kernel and rank.

Solution. If $AX = 0$, multiply by A^{-1} to get $X = 0$, so the kernel is trivial. The domain has dimension 4, hence rank is 4; in fact L_A is an isomorphism. $\square$

Problem 6.20 (Structural synthesis). Explain why rank-nullity is a quantitative form of the first isomorphism theorem.

Solution. The first isomorphism theorem says $V/\ker T \cong \operatorname{im} T$. Taking finite dimensions gives $\dim V - \dim \ker T = \dim \operatorname{im} T$, which rearranges exactly to rank-nullity. $\square$

6.11 Exercises with complete solutions

Exercise 6.21. Find the kernel of $T(x, y, z) = (x + y, y + z)$.

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. Set $T(v) = 0$ and solve; a basis of the solution space is a basis of the kernel. $\square$

Solution. Solve $x + y = 0$ and $y + z = 0$: $(x, y, z) = t(-1, 1, -1)$, so the kernel is one-dimensional. $\square$

Exercise 6.22. Find the rank of the preceding map.

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. Generate the image from the images of a basis of the domain, then remove dependent vectors. $\square$

Solution. The domain has dimension 3 and the kernel dimension is 1, so rank–nullity gives rank 2. $\square$

Exercise 6.23. Prove that zero kernel is equivalent to injectivity.

Hint. Set the output equal to zero and solve the resulting homogeneous equations; the solution vectors generate the kernel. Use rank–nullity after computing either rank or nullity to check the other dimension. $\square$

Solution. If $T(v) = T(w)$ then $T(v - w) = 0$. Zero kernel forces $v = w$; the converse follows by comparing $T(v) = T(0)$. $\square$

Exercise 6.24. Why does surjectivity imply rank equals codomain dimension?

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. Injectivity is equivalent to $\ker T = \{0\}$; try to prove or refute exactly that statement. $\square$

Solution. The image equals the codomain, so their dimensions are equal. $\square$

Exercise 6.25. Show $\dim(V/U) = \dim V - \dim U$ in finite dimensions.

Hint. Surjectivity asks whether the image spans the codomain; compare its dimension with $\dim W$. $\square$

Solution. Extend a basis of U to a basis of V ; the cosets of the added basis vectors form a basis of the quotient. $\square$

Exercise 6.26. Compute the nullity of the trace map on $M_3(\mathbb{R})$.

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. In equal finite dimensions, show that injective $\Rightarrow$ full rank $\Rightarrow$ surjective. $\square$

Solution. The domain has dimension 9, the trace map has rank 1, so nullity is 8. $\square$

Exercise 6.27. For an endomorphism of an n -dimensional space, show injective implies surjective.

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. To build an isomorphism, send one basis bijectively to another basis and extend linearly. $\square$

Solution. Injectivity gives nullity 0, so rank–nullity gives rank n , which equals the codomain dimension. $\square$

Exercise 6.28. Explain geometrically what quotienting $\mathbb{R}^3$ by the z -axis does.

Hint. For $Tx = b$, subtract two solutions; their difference lies in the kernel. $\square$

Solution. Vectors differing by a vertical displacement are identified; the quotient is two-dimensional and can be identified with the xy -plane. $\square$

6.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 6.29 (Kernel basis). Find a basis of $\ker T$ for $T(x, y, z) = (x + y, y + z)$.

Hint. Solve the homogeneous equations. $\square$

Solution. $(x, y, z) = t(-1, 1, -1)$, so the basis is $\{(-1, 1, -1)\}$. $\square$

Exercise 6.30 (Image basis). For the same map, find a basis of the image.

Hint. Use images of the standard basis and remove redundancy. $\square$

Solution. The images include $(1, 0)$ and $(0, 1)$, so the image is $\mathbb{R}^2$ with the standard basis. $\square$

Exercise 6.31 (Rank-nullity). A map $T : \mathbb{R}^9 \rightarrow \mathbb{R}^5$ has rank 4. Find nullity.

Hint. Use domain dimension = rank + nullity. $\square$

Solution. Nullity = $9 - 4 = 5$. $\square$

Exercise 6.32 (Quotient dimension). If $\dim V = 12$, $\dim U = 7$, find $\dim(V/U)$.

Hint. Use $\dim(V/U) = \dim V - \dim U$. $\square$

Solution. The quotient dimension is 5. $\square$

Exercise 6.33 (Affine solution set). If $Tx = b$ has one solution x_0 and $\ker T = \text{span}\{k_1, k_2\}$, describe all solutions.

Hint. Subtract x_0 from an arbitrary solution. $\square$

Solution. All solutions are $x_0 + ak_1 + bk_2$. $\square$

Proofs

Exercise 6.34 (Kernel criterion). Prove T is injective iff $\ker T = 0$.

Hint. Use $T(v) = T(w) \iff T(v - w) = 0$. □

Solution. Trivial kernel forces $v = w$; injectivity forces any kernel vector to equal zero. □

Exercise 6.35 (Surjective rank criterion). Prove $T : V \rightarrow W$ is surjective iff $\text{rank } T = \dim W$ in finite dimensions.

Hint. The image is a subspace of W . □

Solution. A subspace equals W exactly when it has full dimension. □

Exercise 6.36 (Equal-dimension equivalence). For $\dim V = \dim W < \infty$, prove injective iff surjective.

Hint. Use rank-nullity and compare rank with the common dimension. □

Solution. Zero nullity gives full rank and surjectivity; full rank gives zero nullity and injectivity. □

Exercise 6.37 (First isomorphism well-definedness). Why is $v + \ker T \mapsto T(v)$ well defined?

Hint. Compare two representatives of the same coset. □

Solution. If $v - w \in \ker T$, then $T(v - w) = 0$, so $T(v) = T(w)$. □

Counterexamples

Exercise 6.38 (Kernel not enough). Give two distinct nonzero maps $\mathbb{R}^2 \rightarrow \mathbb{R}$ with the same kernel.

Hint. Scale one functional by a nonzero scalar. □

Solution. $T(x, y) = x$ and $S(x, y) = 2x$ both have kernel the y -axis. □

Exercise 6.39 (Infinite-dimensional warning). Give an injective linear map on an infinite-dimensional space that is not surjective.

Hint. Use the shift on polynomial coefficients or sequences. □

Solution. On $\mathbb{R}[x]$, $T(p) = xp$ is injective but no polynomial with nonzero constant term lies in the image. □

Exercise 6.40 (Surjective not injective). Give a surjective linear map $\mathbb{R}^3 \rightarrow \mathbb{R}^2$ that is not injective.

Hint. Use coordinate projection. □

Solution. $(x, y, z) \mapsto (x, y)$ is onto and has the nonzero z -axis as kernel. □

Applications

Exercise 6.41 (Redundant measurements). A measurement map $\mathbb{R}^6 \rightarrow \mathbb{R}^4$ has rank 3. How many invisible directions are there?

Hint. Invisible directions form the kernel. □

Solution. Nullity = $6 - 3 = 3$. □

Exercise 6.42 (Trace constraint). Find the dimension of trace-zero 4×4 matrices.

Hint. Trace is a surjective functional of rank one. □

Solution. $16 - 1 = 15$. □

Challenges

Exercise 6.43 (Rank of composition). Prove $\text{rank}(ST) \leq \min(\text{rank } S, \text{rank } T)$.

Hint. Compare $\text{im}(ST)$ with $\text{im } S$ and with $S(\text{im } T)$. □

Solution. $\text{im}(ST) \subseteq \text{im } S$, and its dimension cannot exceed the dimension of $\text{im } T$. □

Exercise 6.44 (Splitting a surjection). If $T : V \rightarrow W$ is surjective in finite dimensions, prove there is $U \subseteq V$ with $T|_U : U \rightarrow W$ an isomorphism.

Hint. Extend a basis of $\ker T$ to a basis of V . □

Solution. Let U span the added basis vectors; their images form a basis of W by the rank-nullity proof. □

Chapter summary

Kernel and image measure exactly how a linear transformation fails to be injective or surjective. The rank–nullity theorem balances those two defects and turns structural questions into dimension counts. Quotient spaces then package the first isomorphism theorem in its natural form. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Part II

Matrices and Operators

Chapter 7

Matrix Representation of Linear Maps

Matrices are coordinate representations of linear maps. The matrix is not the transformation itself: it depends on chosen bases in the domain and codomain. Once this distinction is kept visible, matrix multiplication, change of basis, rank and similarity all acquire transparent conceptual meanings.

Learning goals

After this chapter, the reader should be able to:

- construct the matrix of a linear map from its action on a basis;
- recover the action of a linear map from its matrix and coordinate conventions;
- compute the matrix of a composition and explain why matrix multiplication appears in that order;
- change the matrix of an operator when the basis changes;
- identify the column space and null space of a matrix as image and kernel data of the associated map;
- check a matrix representation by applying it to basis vectors and comparing the resulting coordinates;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

7.1 Matrix of a linear map

For bases B of V and C of W , the matrix $[T]_{C \leftarrow B}$ is defined by $[T(v)]_C = [T]_{C \leftarrow B}[v]_B$.

Example 7.1 (Building a matrix column by column). Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be $T(x, y) = (x - y, 2x, y)$. The standard matrix is obtained from basis images:

$$T(e_1) = (1, 2, 0), \quad T(e_2) = (-1, 0, 1),$$

so

$$[T] = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Multiplying this matrix by $(x, y)^T$ reproduces $(x - y, 2x, y)^T$, which checks the construction.

The columns are not arbitrary bookkeeping: each is the coordinate image of one domain basis vector.

7.2 Column rule

The j th column of $[T]_{C \leftarrow B}$ is $[T(v_j)]_C$.

7.3 Composition and multiplication

Coordinates turn composition into matrix multiplication: $[S \circ T] = [S][T]$ with compatible bases.

Example 7.2 (Matrix multiplication from composition). Let $T(x, y) = (x + y, y)$ and $S(u, v) = (u, 2v)$. Their standard matrices are

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Since $S \circ T$ means apply T first, its matrix is

$$BA = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

Directly, $(S \circ T)(x, y) = (x + y, 2y)$, agreeing with the product.

This calculation fixes the order convention: composition order is reversed in the written matrix product.

7.4 Identity and inverses

The identity has the identity matrix in the same basis. An isomorphism is represented by an invertible matrix.

7.5 Change of basis for operators

For an endomorphism, changing from basis B to C gives $[T]_C = P_{C \leftarrow B}[T]_B P_{B \leftarrow C}$. Thus representations of the same operator are similar.

Example 7.3 (Same operator, different basis matrix). Suppose an operator has matrix A in basis B , and $P = P_{B \leftarrow C}$ converts C -coordinates to B -coordinates. Starting with $[v]_C$, first obtain $P[v]_C$, apply A , and finally return using P^{-1} . Thus

$$[T]_C = P^{-1}AP.$$

The formula is best remembered by following one vector through the coordinate changes rather than memorizing a conjugation pattern.

7.6 Rank and row reduction

The rank of a matrix equals the dimension of the image of its associated linear map. Row operations preserve rank.

7.7 Core structural results

Theorem 7.4 (Matrix composition rule). *With compatible bases, $[S \circ T] = [S][T]$.*

Proof. Apply both sides to an arbitrary coordinate vector and use the defining representation identity twice. $\square$

Theorem 7.5 (Similarity under basis change). *If $A = [T]_B$ and $P = P_{C \leftarrow B}$, then $[T]_C = PAP^{-1}$.*

Proof. Use $[v]_C = P[v]_B$ and express the coordinate identity for $T(v)$ in the two bases. $\square$

7.8 Worked examples

Example 7.6 (Derivative matrix). On $\mathcal{P}_2$ in the monomial basis, differentiation is represented

by $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

Example 7.7 (Rotation). A planar rotation through angle θ has standard matrix $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

Example 7.8 (Projection). Projection onto the first coordinate axis has matrix $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

7.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

7.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 7.9 (Matrix from basis images). Find the standard matrix of $T(x, y, z) = (x + z, 2y, x - z)$.

Solution. The columns are $T(e_1) = (1, 0, 1)$, $T(e_2) = (0, 2, 0)$, and $T(e_3) = (1, 0, -1)$, giving $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix}$. $\square$

Problem 7.10 (Composition matrix). Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Which matrix represents first applying A and then B ?

Solution. Column vectors are acted on from the left, so first A then B is represented by BA . Matrix multiplication encodes composition in reverse verbal order. $\square$

Problem 7.11 (Similarity from basis change). If $[T]_B = A$ and $P = P_{C \leftarrow B}$, derive $[T]_C = PAP^{-1}$.

Solution. For $x_C = Px_B$, we have $[T(v)]_C = PAx_B = PAP^{-1}x_C$. Since this holds for every coordinate column, the C -matrix is PAP^{-1} . $\square$

Problem 7.12 (Derivative matrix). Write the matrix of $D : \mathcal{P}_3 \rightarrow \mathcal{P}_2$ in monomial bases.

Solution. $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. Thus the columns are $(0, 0, 0)^T$, $(1, 0, 0)^T$, $(0, 2, 0)^T$, $(0, 0, 3)^T$. $\square$

Problem 7.13 (Rank from columns). For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$ find the rank.

Solution. Every column is a scalar multiple of $(1, 2)^T$, and the matrix is nonzero. Hence the column space is one-dimensional and rank is 1. $\square$

Problem 7.14 (Elementary matrix meaning). What linear map does left multiplication by the elementary matrix that swaps rows 1 and 2 represent?

Solution. It composes a matrix-valued coordinate representation with the coordinate permutation swapping the first two output coordinates. Algebraically it performs the corresponding row operation. $\square$

Problem 7.15 (Inverse matrix as inverse map). Why does A^{-1} represent T^{-1} when A represents an isomorphism T in fixed bases?

Solution. The composition $T^{-1}T = I$ gives $[T^{-1}]A = I$ by the composition rule, so $[T^{-1}] = A^{-1}$. $\square$

Problem 7.16 (Rectangular matrix). What does a 3×2 matrix represent geometrically?

Solution. It represents a linear map from a two-dimensional coordinate space to a three-dimensional coordinate space. Its two columns are the images of the two domain basis vectors expressed in the codomain basis. $\square$

Problem 7.17 (Same operator, different matrices). Can two different matrices represent the same linear operator?

Solution. Yes, if different bases are used. The matrices are related by similarity for an endomorphism. With the basis fixed, however, the representing matrix is unique. $\square$

Problem 7.18 (Projection matrix). Find the standard matrix for projection onto $\text{span}(e_1, e_3)$ in $\mathbb{R}^3$.

Solution. The map sends (x, y, z) to $(x, 0, z)$, so the matrix is $\text{diag}(1, 0, 1)$. $\square$

Problem 7.19 (Column space and image). Explain why the column space of a matrix is the coordinate image of its linear map.

Solution. Every output is $Ax = \sum_j x_j a_j$, a linear combination of the columns a_j . Conversely every such combination occurs for the coefficient vector x , so the image equals the column span. $\square$

Problem 7.20 (Matrix synthesis). Why is similarity an equivalence relation on square matrices?

Solution. Reflexivity uses I , symmetry uses P^{-1} , and transitivity follows because $B = PAP^{-1}$ and $C = QBQ^{-1}$ imply $C = (QP)A(QP)^{-1}$. It therefore captures 'same operator in different bases'. $\square$

7.11 Exercises with complete solutions

Exercise 7.21. Find the matrix of $T(x, y) = (x + 2y, 3x - y)$ in the standard basis.

Hint. The j -th matrix column is the coordinate vector of $T(b_j)$ in the codomain basis. $\square$

Solution. The columns are $T(e_1) = (1, 3)$ and $T(e_2) = (2, -1)$, so $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$. $\square$

Exercise 7.22. Why do columns record images of basis vectors?

Hint. Apply the map to a basis of the domain and extract an independent spanning list from the resulting vectors. Multiply the matrix by a coordinate vector only after confirming which basis those coordinates use. $\square$

Solution. The coordinate vector of the j th basis vector is e_j , and multiplying by a matrix selects its j th column. $\square$

Exercise 7.23. Prove that the matrix of an inverse map is the inverse matrix.

Hint. For composition, apply the rightmost matrix first; this mirrors the order of the functions. $\square$

Solution. From $T^{-1}T = I$, the composition rule gives $[T^{-1}][T] = I$. $\square$

Exercise 7.24. Compute the rank of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

Hint. Check matrix dimensions before multiplying; they encode the domain and codomain compatibility. $\square$

Solution. The second row and second column are scalar multiples of the first, while the matrix is nonzero, so rank is 1. $\square$

Exercise 7.25. Show that similar matrices have the same rank.

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. To change basis, convert input coordinates, apply the old matrix, then convert the output coordinates. $\square$

Solution. Left and right multiplication by invertible matrices correspond to isomorphisms and do not change image dimension. $\square$

Exercise 7.26. Write the standard matrix of reflection across the x -axis.

Hint. The null space and column space are just the kernel and image of the associated linear map. □

Solution. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. □

Exercise 7.27. Explain why matrix multiplication is generally noncommutative.

Hint. Recover the map by applying the matrix to each standard basis vector and reading off the columns. □

Solution. Composition of transformations is order-sensitive; for example, a projection followed by a rotation differs from the reverse order. □

Exercise 7.28. Relate elementary row operations to invertible matrices.

Hint. Test a proposed representation on basis vectors; if one column is wrong, the whole matrix is wrong. □

Solution. Each elementary row operation is left multiplication by an elementary invertible matrix. □

7.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 7.29 (Matrix from a formula). Find the standard matrix of $T(x, y) = (2x - y, x + 3y)$.

Hint. Use $T(e_1), T(e_2)$ as columns. □

Solution. $\begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$. □

Exercise 7.30 (Apply a representation). For $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$, compute $A(3, 1)^T$.

Hint. Multiply row by column. □

Solution. $(5, -1)^T$. □

Exercise 7.31 (Composition matrix). If $[S] = B$, $[T] = A$, what is $[S \circ T]$?

Hint. The inner map acts first. □

Solution. $[S \circ T] = BA$. □

Exercise 7.32 (Identity matrix). What matrix represents the identity in any ordered basis?

Hint. Each basis vector maps to itself. □

Solution. The identity matrix. □

Exercise 7.33 (Matrix of inverse). If T is invertible with matrix A , what is the matrix of T^{-1} in the corresponding bases?

Hint. Compose the two maps. □

Solution. A^{-1} . □

Proofs

Exercise 7.34 (Column theorem). Prove the j -th column of $[T]_{C \leftarrow B}$ is $[T(b_j)]_C$.

Hint. Apply the matrix to the coordinate vector e_j . □

Solution. Since $[b_j]_B = e_j$, the product selects column j , which must be $[T(b_j)]_C$. □

Exercise 7.35 (Representation uniqueness). Prove a linear map has a unique matrix once domain and codomain bases are fixed.

Hint. A matrix is determined by its columns. □

Solution. The basis images determine T , and their coordinate columns determine the matrix uniquely. □

Exercise 7.36 (Composition proof). Derive matrix multiplication from composition on basis coordinates.

Hint. Write $[Tv]_C = A[v]_B$ and then apply S . □

Solution. $[STv]_D = B[Tv]_C = BA[v]_B$. □

Exercise 7.37 (Change-of-basis proof). Derive the similarity formula for an endomorphism.

Hint. Insert coordinate changes before and after the operator. □

Solution. If $P = P_{B \leftarrow C}$, then $[T]_C = P^{-1}[T]_B P$. □

Counterexamples

Exercise 7.38 (Counterexample: order). Show AB and BA can differ.

Hint. Choose a projection and shear matrix. □

Solution. For $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $AB \neq BA$. □

Exercise 7.39 (Counterexample: same array). Explain why the same numerical matrix can represent different maps if bases differ.

Hint. The matrix acts on coordinate columns, whose meanings depend on bases. □

Solution. Changing the identification of coordinates changes the underlying basis vectors, so the same array need not describe the same geometric map. □

Exercise 7.40 (Counterexample: rectangular inverse). Why can a 2×3 matrix not have a two-sided inverse?

Hint. A two-sided inverse would make the corresponding spaces isomorphic. □

Solution. $\mathbb{F}^3$ and $\mathbb{F}^2$ have different finite dimensions. □

Applications

Exercise 7.41 (Application: rotation). Write the matrix of a 90° rotation in the standard basis.

Hint. Track e_1, e_2 . □

Solution. $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. □

Exercise 7.42 (Application: derivative). Find the matrix of $D : \mathcal{P}_2 \rightarrow \mathcal{P}_1$ in monomial bases.

Hint. Differentiate $1, x, x^2$. □

Solution. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. □

Challenges

Exercise 7.43 (Challenge: adapted matrix). If U is T -invariant and the basis begins with a basis of U , what block shape must $[T]$ have?

Hint. Images of the first basis vectors remain in U . □

Solution. The lower-left block is zero, so $[T] = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$. □

Exercise 7.44 (Challenge: rank representation). Why is rank independent of the chosen matrix representation?

Hint. Changing bases multiplies the matrix by invertible matrices on left and right. □

Solution. Invertible row/coordinate changes do not change the dimension of the image, so rank is invariant. □

Chapter summary

Matrices are coordinate representations of linear maps. The matrix is not the transformation itself: it depends on chosen bases in the domain and codomain. Once this distinction is kept visible, matrix multiplication, change of basis, rank and similarity all acquire transparent conceptual meanings. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 8

Determinants and Trace

The determinant compresses invertibility, oriented volume scaling and alternating multilinearity into one scalar. Trace measures the sum of diagonal action and is the simplest similarity invariant. Together they are the first examples of basis-independent quantities extracted from a matrix representation.

Learning goals

After this chapter, the reader should be able to:

- compute determinants efficiently using row operations, triangular form, or cofactor expansion;
- track the effect of row swaps, row scalings, and row additions on the determinant;
- use the determinant to decide invertibility and linear independence for square matrices;
- prove and apply multiplicativity $\det(AB) = \det(A)\det(B)$;
- compute traces and use cyclicity $\text{tr}(AB) = \text{tr}(BA)$ in concrete identities;
- interpret determinant and trace as basis-independent invariants of a linear operator;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

8.1 Alternating multilinear characterization

The determinant is the unique alternating multilinear function of the columns with $\det I = 1$.

8.2 Permutation formula

For $A = (a_{ij})$, $\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_i a_{i,\sigma(i)}$.

8.3 Row operations

Swapping rows changes sign, scaling a row scales the determinant, and adding a multiple of one row to another leaves it unchanged.

8.4 Invertibility

A square matrix is invertible exactly when its determinant is nonzero.

Example 8.1 (A determinant hypothesis that matters). If A and B are square matrices, then

$$\det(AB) = \det A \det B.$$

Consequently AB is invertible exactly when both factors are invertible: a zero determinant in either factor forces $\det(AB) = 0$, while two nonzero determinants give a nonzero product.

This gives a scalar certificate for a structural fact about composition of linear isomorphisms.

8.5 Geometric meaning

Over $\mathbb{R}$, $|\det A|$ is the factor by which the associated linear map scales n -dimensional volume; the sign records orientation.

8.6 Trace

$\operatorname{tr}(A) = \sum_i a_{ii}$ is linear and satisfies $\operatorname{tr}(AB) = \operatorname{tr}(BA)$, hence is invariant under similarity.

Example 8.2 (Determinant as an invertibility check). For

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}, \quad \det A = 4 - 3 = 1.$$

Thus A is invertible. Solving $A(x, y)^T = (5, 8)^T$ gives $(x, y) = (2, 1)$, and substitution checks the solution.

The determinant does not produce the solution by itself; it tells us that a unique solution must exist for every right-hand side.

Example 8.3 (Trace survives change of basis). Let $B = P^{-1}AP$. Using cyclicity of trace,

$$\operatorname{tr}(B) = \operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A).$$

Hence trace is attached to the operator, not to one particular matrix representation.

A practical check follows: if two matrices are claimed to be similar but have different traces, the claim is impossible.

8.7 Core structural results

Theorem 8.4 (Multiplicativity of determinant). *For square matrices A, B , $\det(AB) = \det(A) \det(B)$.*

Proof. For fixed B , the map sending the columns of A to $\det(AB)$ is alternating multilinear and has normalization $\det(B)$ at $A = I$; uniqueness gives the formula. $\square$

Theorem 8.5 (Cyclic trace identity). *For compatible square matrices, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.*

Proof. Expand the diagonal entries: $\operatorname{tr}(AB) = \sum_{i,j} a_{ij}b_{ji}$ and interchange the dummy indices. $\square$

8.8 Worked examples

Example 8.6 (Triangular matrices). The determinant of a triangular matrix is the product of its diagonal entries, and its trace is their sum.

Example 8.7 (Area scaling). The matrix $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ scales signed area by 6.

Example 8.8 (Shear). A shear $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ has determinant 1 although it changes angles.

8.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

8.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 8.9 (Determinant by elimination). Compute $\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 5 \end{pmatrix}$ and explain why no expansion is needed.

Solution. The matrix is upper triangular, so the determinant is the product of diagonal entries: $1 \cdot 1 \cdot 5 = 5$. □

Problem 8.10 (Effect of a row replacement). What happens to the determinant when row 2 is replaced by row 2 + 7 row 1?

Solution. The determinant is unchanged. Multilinearity expands the new determinant into the old one plus 7 times a determinant with two equal rows, and the latter is zero by alternation. □

Problem 8.11 (Area scaling). What signed area factor is produced by $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$?

Solution. $\det A = 6 - 1 = 5$. Thus oriented area is multiplied by 5; ordinary area is multiplied by $|\det A| = 5$. □

Problem 8.12 (Invertibility test). Why does $\det A = 0$ imply the columns are dependent?

Solution. The determinant is alternating multilinear and vanishes on dependent columns. Conversely, if the columns were independent they would form a basis and the associated linear map would be invertible, forcing nonzero determinant. □

Problem 8.13 (Trace cyclicity). Verify directly that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Solution. $\text{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji}$. Interchanging the dummy indices gives $\sum_j \sum_i b_{ji} a_{ij} = \text{tr}(BA)$. $\square$

Problem 8.14 (Similarity invariants). Show determinant and trace are invariant under $A \mapsto PAP^{-1}$.

Solution. Multiplicativity gives $\det(PAP^{-1}) = \det P \det A \det P^{-1} = \det A$. Cyclicity gives $\text{tr}(PAP^{-1}) = \text{tr}(AP^{-1}P) = \text{tr} A$. $\square$

Problem 8.15 (Determinant of a reflection). Find the determinant of reflection across a hyperplane in $\mathbb{R}^n$.

Solution. The reflection fixes the $(n-1)$ -dimensional hyperplane and negates one normal direction. In an adapted basis the matrix is $\text{diag}(1, \dots, 1, -1)$, so the determinant is -1 . $\square$

Problem 8.16 (Determinant of a shear). Why does $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ preserve area for every t ?

Solution. Its determinant is 1. Geometrically a shear slides parallel slices without changing base or height, consistent with area preservation. $\square$

Problem 8.17 (Trace of rank one). For column vectors u, v , compute $\text{tr}(uv^T)$.

Solution. The diagonal entries are $u_i v_i$, so the trace is $\sum_i u_i v_i = v^T u$. $\square$

Problem 8.18 (Block triangular determinant). If $M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$ with square diagonal blocks, what is $\det M$?

Solution. The determinant is $\det A \det C$. This follows by expansion or by reducing to triangular block operations; the off-diagonal block B does not affect the product of diagonal block determinants. $\square$

Problem 8.19 (Orientation test). How can determinant distinguish the two connected components of $GL_n(\mathbb{R})$?

Solution. An invertible real matrix has determinant either positive or negative. Continuous paths in GL_n cannot cross determinant 0, so the sign cannot change along a path. Positive and negative determinants correspond to the two orientation classes. $\square$

Problem 8.20 (Determinant synthesis). Why is the determinant better viewed as a property of an endomorphism than of one matrix?

Solution. Changing basis replaces the matrix by a similar one, and determinant is similarity invariant. Thus $\det T$ is well-defined independently of coordinates; a matrix merely computes it. $\square$

8.11 Exercises with complete solutions

Exercise 8.21. Compute $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Hint. Use row operations to reach triangular form, but record which operations change the determinant and how. $\square$

Solution. The permutation formula gives $ad - bc$. $\square$

Exercise 8.22. Show that a matrix with two equal columns has determinant zero.

Hint. Reduce to triangular form while recording how each row operation changes the determinant. Expand along a row or column with the most zeros rather than using a full permutation formula. $\square$

Solution. Alternation changes the determinant sign when the equal columns are swapped, but the matrix is unchanged, so the determinant equals its negative. $\square$

Exercise 8.23. Why is $\det A \neq 0$ equivalent to invertibility?

Hint. Place the vectors as columns and ask when the determinant vanishes; that is exactly the dependence condition in the square case. A square matrix is singular exactly when its determinant vanishes; connect that to a nontrivial kernel. $\square$

Solution. Nonzero determinant implies full column independence; equivalently the associated endomorphism has zero kernel and hence is invertible. $\square$

Exercise 8.24. Compute the determinant of a diagonal matrix.

Hint. Reduce to triangular form while recording how each row operation changes the determinant. For $\det(AB)$, first handle the invertible case or use the effect of elementary matrices. $\square$

Solution. Only the identity permutation contributes, so the determinant is the product of diagonal entries. $\square$

Exercise 8.25. Prove $\operatorname{tr}(PAP^{-1}) = \operatorname{tr}(A)$.

Hint. Write the diagonal entries of AB as a double sum and interchange the summation indices. The determinant of a triangular matrix is the product of its diagonal entries. $\square$

Solution. Use cyclicity: $\operatorname{tr}(PAP^{-1}) = \operatorname{tr}(AP^{-1}P) = \operatorname{tr}(A)$. $\square$

Exercise 8.26. What determinant does a reflection of $\mathbb{R}^n$ across a hyperplane have?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. For trace, write the diagonal entries of AB and interchange the summation indices to compare with BA . $\square$

Solution. It has eigenvalues 1 on the hyperplane and -1 on a normal direction, so determinant -1 . $\square$

Exercise 8.27. Compute the trace of a rank-one matrix uv^T .

Hint. Similarity preserves determinant and trace because the factors P and P^{-1} cancel inside these invariants. $\square$

Solution. The trace is $v^T u = \sum_i u_i v_i$. $\square$

Exercise 8.28. Show that determinant and trace are basis-independent for an endomorphism.

Hint. Place the vectors as columns and ask when the determinant vanishes; that is exactly the dependence condition in the square case. Use determinant as a quick consistency check for an alleged inverse: $\det(A^{-1}) = 1/\det(A)$. $\square$

Solution. Two matrix representations are similar; determinant and trace are invariant under similarity. $\square$

8.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 8.29 (Two-by-two determinant). Compute $\det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

Hint. Use $ad - bc$. □

Solution. $12 - 2 = 10$. □

Exercise 8.30 (Triangular determinant). Compute the determinant of an upper-triangular matrix with diagonal $2, -1, 5$.

Hint. Multiply diagonal entries. □

Solution. -10 . □

Exercise 8.31 (Trace). Compute the trace of $\begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix}$.

Hint. Add diagonal entries. □

Solution. -3 . □

Exercise 8.32 (Scaling determinant). For $A \in M_3$, what is $\det(2A)$?

Hint. Scaling all three rows by 2 multiplies determinant by 2^3 . □

Solution. $\det(2A) = 8 \det A$. □

Exercise 8.33 (Invertibility). Can a matrix with determinant 0 be invertible?

Hint. Use $\det(AA^{-1}) = 1$. □

Solution. No; invertibility would force $\det A \det A^{-1} = 1$. □

Proofs

Exercise 8.34 (Alternating property). Prove a matrix with two equal rows has determinant zero.

Hint. Swap the equal rows. □

Solution. Alternation changes the sign but leaves the matrix unchanged, so $\det A = -\det A$; over characteristic not 2, this gives zero, and the determinant axioms give zero generally. □

Exercise 8.35 (Multiplicativity consequence). Prove if AB is invertible then both square factors are invertible.

Hint. Use determinants. □

Solution. $\det(AB) = \det A \det B \neq 0$, so both determinants are nonzero. □

Exercise 8.36 (Trace cyclicity). Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Hint. Expand diagonal entries as double sums. ☐

Solution. Both equal $\sum_{i,j} a_{ij}b_{ji}$. ☐

Exercise 8.37 (Similarity invariance). Prove determinant is invariant under similarity.

Hint. Compute $\det(P^{-1}AP)$. ☐

Solution. $\det(P^{-1})\det(A)\det(P) = \det A$. ☐

Counterexamples

Exercise 8.38 (Counterexample: trace product). Is $\operatorname{tr}(AB) = \operatorname{tr} A \operatorname{tr} B$ always true?

Hint. Test identity matrices. ☐

Solution. No. For $A = B = I_2$, the left side is 2, the right side 4. ☐

Exercise 8.39 (Counterexample: determinant sum). Is $\det(A+B) = \det A + \det B$ generally true?

Hint. Test $A = B = I_2$. ☐

Solution. No: $\det(2I_2) = 4$ but $\det I_2 + \det I_2 = 2$. ☐

Exercise 8.40 (Counterexample: same trace). Do equal traces imply similarity?

Hint. Compare I_2 and a nontrivial Jordan block with eigenvalue 1. ☐

Solution. No; both have trace 2, but one is diagonalizable and the other is not. ☐

Applications

Exercise 8.41 (Application: area scaling). What oriented area factor does $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ produce?

Hint. Use the determinant. ☐

Solution. The factor is 6. ☐

Exercise 8.42 (Application: orientation). What does a negative determinant say about an invertible real planar map?

Hint. Interpret the sign of determinant. ☐

Solution. It reverses orientation. ☐

Challenges

Exercise 8.43 (Challenge: rank-one update). Let $A = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. Compute determinant and trace for all t .

Hint. Use triangular formulas. ☐

Solution. $\det A = 1$, $\operatorname{tr} A = 2$, independent of t ; these invariants do not detect the shear parameter. ☐

Exercise 8.44 (Challenge: commutator trace). Show $\operatorname{tr}(AB - BA) = 0$.

Hint. Use cyclicity of trace. ☐

Solution. $\operatorname{tr}(AB) - \operatorname{tr}(BA) = 0$. ☐

Chapter summary

The determinant compresses invertibility, oriented volume scaling and alternating multilinearity into one scalar. Trace measures the sum of diagonal action and is the simplest similarity invariant. Together they are the first examples of basis-independent quantities extracted from a matrix representation. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 9

Eigenvalues and Eigenvectors

Eigenvectors are directions preserved by an operator up to scaling, and eigenvalues are the corresponding scale factors. The characteristic polynomial packages all eigenvalues algebraically. This chapter emphasizes both geometric meaning and the limitations of eigenvector methods when a full eigenbasis does not exist.

Learning goals

After this chapter, the reader should be able to:

- compute eigenvalues from the characteristic polynomial and compute a basis for each eigenspace;
- distinguish algebraic multiplicity from geometric multiplicity in explicit examples;
- prove that eigenvectors belonging to distinct eigenvalues are linearly independent;
- decide whether a given vector or subspace is preserved by an operator using eigenvector information;
- construct examples with repeated eigenvalues that are or are not diagonalizable;
- check an eigenpair directly and diagnose arithmetic errors by substituting back into $Tv = \lambda v$;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

9.1 Eigenvectors and eigenvalues

A nonzero v is an eigenvector of T with eigenvalue λ when $T(v) = \lambda v$.

Example 9.1 (A complete eigensystem calculation). For

$$A = \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix},$$

the characteristic equation is $(4 - \lambda)(2 - \lambda) = 0$, so $\lambda = 4, 2$. For $\lambda = 4$, $(A - 4I)v = 0$ gives $y = 0$, hence $E_4 = \text{span}(1, 0)$. For $\lambda = 2$, $2x + y = 0$, so $E_2 = \text{span}(1, -2)$.

Check:

$$A(1, 0)^T = 4(1, 0)^T, \quad A(1, -2)^T = 2(1, -2)^T.$$

The substitution check catches sign errors that the characteristic polynomial alone cannot reveal.

9.2 Eigenspaces

$E_\lambda = \ker(T - \lambda I)$ is a subspace. Distinct eigenspaces meet only at zero.

9.3 Characteristic polynomial

For a matrix A , $\chi_A(t) = \det(tI - A)$. Its roots are precisely the eigenvalues over the chosen field.

Example 9.2 (Field dependence of the spectrum). For the quarter-turn matrix

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

$$\chi_R(t) = t^2 + 1.$$

There are no real eigenvalues, but over $\mathbb{C}$ the roots are $\pm i$. Solving $(R - iI)v = 0$ gives, for example, $v = (1, -i)$.

The operator has not changed; extending the scalar field has supplied roots that were unavailable over $\mathbb{R}$.

9.4 Algebraic and geometric multiplicity

The algebraic multiplicity is the multiplicity of a root of χ_A ; the geometric multiplicity is $\dim E_\lambda$ and never exceeds it.

Example 9.3 (Repeated eigenvalue: diagonalizable or defective?). Compare

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Both have characteristic polynomial $(t - 1)^2$. For I , $E_1 = \mathbb{R}^2$, so geometric multiplicity is 2. For J , $(J - I)v = 0$ forces $y = 0$, so geometric multiplicity is 1.

Thus eigenvalues with algebraic multiplicity do not determine similarity: eigenspace dimension records additional structure.

9.5 Independence of eigenvectors

Eigenvectors associated with distinct eigenvalues are linearly independent.

9.6 Field dependence

A real matrix may have no real eigenvalues but acquire complex eigenvalues after extending scalars to $\mathbb{C}$.

9.7 Core structural results

Theorem 9.4 (Distinct eigenvalues give independence). *Eigenvectors corresponding to distinct eigenvalues are linearly independent.*

Proof. Use induction. Apply $T - \lambda_r I$ to a dependence relation; the last term vanishes and the remaining coefficients are multiplied by nonzero scalars. $\square$

Theorem 9.5 (Geometric multiplicity bound). *For an eigenvalue λ , $1 \leq \dim E_\lambda \leq$ its algebraic multiplicity.*

Proof. The lower bound is the definition of eigenvalue. The upper bound follows after choosing a basis beginning with an eigenspace basis, which forces a block with λ repeated on the diagonal. $\square$

9.8 Worked examples

Example 9.6 (Diagonal matrix). For $D = \text{diag}(d_1, \dots, d_n)$, each standard vector is an eigenvector with eigenvalue d_i .

Example 9.7 (Rotation). A nontrivial planar rotation has no real eigenvectors unless the angle is 0 or π , but over $\mathbb{C}$ it has eigenvalues $e^{\pm i\theta}$.

Example 9.8 (Shear). $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only the eigenvalue 1 and only a one-dimensional eigenspace.

9.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

9.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 9.9 (Defective shear). For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ compute algebraic and geometric multiplicities of $\lambda = 1$.

Solution. The characteristic polynomial is $(t - 1)^2$, so algebraic multiplicity is 2. The eigenspace solves $(A - I)v = 0$, hence $y = 0$, so it is one-dimensional. Geometric multiplicity $1 < 2$, making the matrix defective. $\square$

Problem 9.10 (Rotation and complexification). Find the eigenvalues of the real rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ over $\mathbb{R}$ and $\mathbb{C}$.

Solution. The characteristic polynomial is $t^2 + 1$. It has no real roots, so no real eigenvalues; over $\mathbb{C}$ the eigenvalues are i and $-i$. $\square$

Problem 9.11 (Parameter collision). For $A_t = \begin{pmatrix} t & 1 \\ 0 & 2 \end{pmatrix}$, when are there two distinct eigenvalues?

Solution. The eigenvalues are the diagonal entries t and 2 . They are distinct exactly when $t \neq 2$. At $t = 2$ the matrix becomes a nontrivial Jordan-type block with only one eigendirection. $\square$

Problem 9.12 (Eigenvectors of powers). If $Tv = \lambda v$, what is $T^k v$?

Solution. Induction gives $T^k v = \lambda^k v$. Thus eigenvectors of T remain eigenvectors of every polynomial in T , with eigenvalues transformed by the polynomial. $\square$

Problem 9.13 (Zero eigenvalue and singularity). Show that 0 is an eigenvalue of a square matrix iff the matrix is singular.

Solution. 0 is an eigenvalue iff there is $v \neq 0$ with $Av = 0$, i.e. iff the kernel is nontrivial. A square matrix is invertible exactly when its kernel is zero. $\square$

Problem 9.14 (Distinct eigenvectors). Why are eigenvectors for distinct eigenvalues independent?

Solution. In a dependence relation, apply $T - \lambda_r I$ to eliminate one eigenvector while multiplying the others by nonzero scalars. Induction reduces to a shorter relation, forcing all coefficients to zero. $\square$

Problem 9.15 (Triangular spectrum). Read the eigenvalues of $\begin{pmatrix} 3 & * & * \\ 0 & -1 & * \\ 0 & 0 & 3 \end{pmatrix}$.

Solution. The characteristic polynomial of a triangular matrix is the product $(t-3)(t+1)(t-3)$, so the eigenvalues are 3 with algebraic multiplicity 2 and -1 with multiplicity 1 . $\square$

Problem 9.16 (Eigenspace as kernel). Why is E_λ always a subspace?

Solution. $E_\lambda = \ker(T - \lambda I)$. Kernels of linear maps are subspaces, so closure and the zero vector follow automatically. $\square$

Problem 9.17 (Eigenvalue of an inverse). If T is invertible and $Tv = \lambda v$, what eigenvalue does v have for T^{-1} ?

Solution. Invertibility implies $\lambda \neq 0$. Applying T^{-1} gives $v = \lambda T^{-1}v$, so $T^{-1}v = \lambda^{-1}v$. $\square$

Problem 9.18 (Trace and eigenvalues). For a triangularizable operator, relate trace to eigenvalues counted with algebraic multiplicity.

Solution. In a triangular representation the diagonal entries are the eigenvalues with multiplicity, and trace is the sum of diagonal entries. Since trace is basis-invariant, it equals the eigenvalue sum. $\square$

Problem 9.19 (Determinant and eigenvalues). For a triangularizable operator, relate determinant to eigenvalues.

Solution. In triangular form the determinant is the product of diagonal entries, which are the eigenvalues with algebraic multiplicity. Therefore $\det T$ equals their product. $\square$

Problem 9.20 (Spectral synthesis). Why can knowing all eigenvalues still be insufficient to determine an operator up to similarity?

Solution. Eigenvalues record only the roots of the characteristic polynomial. Operators with the same eigenvalues can have different eigenspace dimensions or Jordan block structures, such as the identity and a nontrivial shear with eigenvalue 1. $\square$

9.11 Exercises with complete solutions

Exercise 9.21. Find the eigenvalues of $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Form $\det(A - \lambda I)$ first; only after finding λ should you solve $(A - \lambda I)v = 0$. $\square$

Solution. They are the diagonal entries 2 and 3. $\square$

Exercise 9.22. Find the eigenspace for eigenvalue 1 of $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Substitute the proposed pair back into $Av = \lambda v$ before doing any further theory. $\square$

Solution. Solve $(A - I)v = 0$, giving $y = 0$; the eigenspace is the span of $(1, 0)$. $\square$

Exercise 9.23. Why is the zero vector excluded from the definition of eigenvector?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. For a repeated eigenvalue, compute the eigenspace dimension; multiplicity in the polynomial alone does not decide diagonalizability. $\square$

Solution. It satisfies $T(0) = \lambda 0$ for every scalar, so including it would carry no information about the operator. $\square$

Exercise 9.24. Prove eigenspaces are subspaces.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. To prove independence, apply a polynomial in T that kills all but one eigencomponent. $\square$

Solution. They are kernels of the linear maps $T - \lambda I$. $\square$

Exercise 9.25. Show that eigenvalues depend on the scalar field.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. An eigenspace is a kernel: $E_\lambda = \ker(T - \lambda I)$. Use kernel methods to find a basis. $\square$

Solution. The real rotation by 90° has characteristic polynomial $t^2 + 1$, with no real roots but roots $\pm i$ over $\mathbb{C}$. $\square$

Exercise 9.26. If T is invertible, can 0 be an eigenvalue?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Distinct eigenvalues give distinct invariant directions; try a linear relation and apply T . $\square$

Solution. No. A zero eigenvalue would give a nonzero vector in the kernel. $\square$

Exercise 9.27. What are the eigenvalues of a triangular matrix?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Compare the sum of eigenspace dimensions with the dimension of the whole space. $\square$

Solution. They are its diagonal entries, because the characteristic polynomial is the product of $t - a_{ii}$. $\square$

Exercise 9.28. Explain the difference between algebraic and geometric multiplicity for a shear.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. For triangular matrices, read candidate eigenvalues from the diagonal before computing eigenvectors. $\square$

Solution. For the 2×2 shear, eigenvalue 1 has algebraic multiplicity 2 but geometric multiplicity 1. $\square$

9.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 9.29 (Check eigenpair). Check whether $v = (1, 2)$ is an eigenvector of $A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$.

Hint. Compute Av . $\square$

Solution. $Av = (3, 6) = 3v$, so yes, with eigenvalue 3. $\square$

Exercise 9.30 (Characteristic roots). Find eigenvalues of $\begin{pmatrix} 2 & 1 \\ 0 & 5 \end{pmatrix}$.

Hint. Use triangular form. $\square$

Solution. 2 and 5. $\square$

Exercise 9.31 (Eigenspace). Find E_2 for $A = \begin{pmatrix} 2 & 1 \\ 0 & 5 \end{pmatrix}$.

Hint. Solve $(A - 2I)v = 0$. $\square$

Solution. $y = 0$, so $E_2 = \text{span}(1, 0)$. $\square$

Exercise 9.32 (Power action). If $Tv = -2v$, compute T^5v .

Hint. Iterate the eigenvalue. $\square$

Solution. $T^5v = (-2)^5v = -32v$. $\square$

Exercise 9.33 (Inverse eigenvalue). If T is invertible and $Tv = 4v$, what does T^{-1} do to v ?

Hint. Apply T^{-1} . $\square$

Solution. $T^{-1}v = \frac{1}{4}v$. $\square$

Proofs

Exercise 9.34 (Distinct-eigenvalue proof). Prove eigenvectors for distinct eigenvalues are independent in the two-vector case.

Hint. Apply $T - \lambda_2 I$ to $av_1 + bv_2 = 0$. □

Solution. $a(\lambda_1 - \lambda_2)v_1 = 0$, so $a = 0$, then $b = 0$. □

Exercise 9.35 (Eigenspace subspace). Prove E_λ is a subspace.

Hint. Identify it as a kernel. □

Solution. $E_\lambda = \ker(T - \lambda I)$, hence is a subspace. □

Exercise 9.36 (Zero eigenvalue). Prove 0 is an eigenvalue iff a square matrix is singular.

Hint. Translate $Av = 0$ with $v \neq 0$. □

Solution. A nonzero kernel vector exists exactly when the matrix is not invertible. □

Exercise 9.37 (Polynomial action). If $Tv = \lambda v$, prove $p(T)v = p(\lambda)v$.

Hint. Use $T^k v = \lambda^k v$. □

Solution. Expand p as a sum of powers and apply linearity. □

Counterexamples

Exercise 9.38 (Counterexample: zero vector). Why is the zero vector excluded from the definition of eigenvector?

Hint. Test $T0 = \lambda 0$. □

Solution. The equation holds for every λ , so including zero would destroy the meaning of eigenvalue. □

Exercise 9.39 (Counterexample: repeated root). Give a matrix with characteristic polynomial $(t - 1)^2$ that is not diagonalizable.

Hint. Use a shear Jordan block. □

Solution. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has one-dimensional eigenspace. □

Exercise 9.40 (Counterexample: field dependence). Give a real matrix with no real eigenvalues.

Hint. Use a quarter-turn. □

Solution. $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $t^2 + 1$. □

Applications

Exercise 9.41 (Application: recurrence mode). If a state starts in an eigenvector direction with eigenvalue 0.9, what happens after n iterations?

Hint. Use $T^n v = \lambda^n v$. □

Solution. The state is scaled by 0.9^n and decays to zero. □

Exercise 9.42 (Application: unstable mode). What does an eigenvalue 1.2 imply for repeated iteration along its eigenvector?

Hint. Compute 1.2^n . □

Solution. The magnitude grows exponentially like 1.2^n . □

Challenges

Exercise 9.43 (Challenge: collision parameter). For $A_t = \begin{pmatrix} t & 1 \\ 0 & 2 \end{pmatrix}$, classify when it has two distinct eigenvalues.

Hint. Read diagonal entries. □

Solution. Distinct when $t \neq 2$; at $t = 2$ only eigenvalue 2 remains. □

Exercise 9.44 (Challenge: trace/determinant check). A 2×2 matrix has eigenvalues 3, -2 . What trace and determinant must it have?

Hint. For triangularizable matrices, use sum and product. □

Solution. Trace 1, determinant -6 . □

Chapter summary

Eigenvectors are directions preserved by an operator up to scaling, and eigenvalues are the corresponding scale factors. The characteristic polynomial packages all eigenvalues algebraically. This chapter emphasizes both geometric meaning and the limitations of eigenvector methods when a full eigenbasis does not exist. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 10

Invariant Subspaces and Triangularization

Invariant subspaces let a large operator be studied through smaller pieces. Triangularization is the systematic expression of this idea: once an invariant flag is chosen, the matrix becomes triangular and its diagonal reveals the eigenvalues. Over an algebraically closed field, every operator on a nonzero finite-dimensional space admits such a form.

Learning goals

After this chapter, the reader should be able to:

- test whether a subspace is invariant under a given operator;
- construct a basis adapted to an invariant subspace and derive the resulting block or triangular matrix form;
- read eigenvalues from the diagonal of a triangular matrix;
- prove that an operator is triangularizable when its characteristic polynomial splits over the scalar field;
- build an invariant flag one subspace at a time from eigenvectors and quotient-space reasoning;
- distinguish triangularizability from diagonalizability by giving explicit examples;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

10.1 Invariant subspaces

A subspace U is T -invariant when $T(U) \subseteq U$. The restriction $T|_U$ is then an operator on U .

Example 10.1 (Testing invariance by a spanning set). Let $T(x, y, z) = (x + y, y + z, z)$ and $U = \{(x, y, 0)\}$. It is enough to test a basis of U :

$$T(e_1) = e_1, \quad T(e_2) = e_1 + e_2.$$

Both images lie in U , so $T(U) \subseteq U$.

The spanning-set test is efficient because linearity then sends every linear combination in U back into U .

10.2 Quotient operators

If U is invariant, T induces an operator on V/U by $v + U \mapsto T(v) + U$.

10.3 Invariant flags

A chain $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ with $\dim V_k = k$ and each V_k invariant is a complete invariant flag.

Example 10.2 (Reading invariant subspaces from triangular form). For an upper-triangular matrix in basis $v_1, \dots, v_n$, each subspace

$$V_k = \text{span}(v_1, \dots, v_k)$$

is invariant: the first k columns have no entries below row k . Thus triangular form is equivalent to preserving a complete flag of subspaces.

The matrix pattern and the geometric filtration are two descriptions of the same structure.

10.4 Triangular matrices and flags

An operator is upper triangular in a basis exactly when the spans of its initial basis segments form an invariant flag.

Example 10.3 (An invariant line starts triangularization). Suppose $Tv_1 = \lambda v_1$. Extend v_1 to a basis $v_1, \dots, v_n$. Since the first basis vector spans an invariant line, the first column of the matrix of T has zeros below the first entry:

$$[T] = \begin{pmatrix} \lambda & * \\ 0 & * \end{pmatrix}.$$

Repeating the argument on an invariant quotient or a nested invariant sequence produces triangular form.

This explains geometrically why eigenvectors are the first step toward triangularization.

10.5 Triangularization theorem

Over an algebraically closed field, every operator on a finite-dimensional nonzero vector space is triangularizable.

10.6 Block triangular form

Even without a complete flag, a single invariant subspace produces a block upper-triangular matrix, separating the restriction and quotient actions.

10.7 Core structural results

Theorem 10.4 (Flag criterion for triangularity). *A basis $v_1, \dots, v_n$ gives an upper-triangular matrix for T if and only if $\text{span}(v_1, \dots, v_k)$ is invariant for every k .*

Proof. Upper triangularity says $T(v_k)$ lies in the span of the first k basis vectors. This is exactly the invariant-flag condition. $\square$

Theorem 10.5 (Triangularization over an algebraically closed field). *Every operator on a finite-dimensional complex vector space has an upper-triangular matrix in some basis.*

Proof. Choose an eigenvector, use its span as the first invariant subspace, pass to the quotient, and induct on dimension. $\square$

10.8 Worked examples

Example 10.6 (Nilpotent shift). The map $N(e_1) = 0$, $N(e_k) = e_{k-1}$ preserves the standard flag and is strictly upper triangular in the reversed standard convention.

Example 10.7 (Polynomial operator). Differentiation preserves the nested polynomial spaces $\mathcal{P}_0 \subset \mathcal{P}_1 \subset \dots$.

Example 10.8 (Real rotation). A planar rotation through 90° has no invariant real line, so it cannot be triangularized over $\mathbb{R}$.

10.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

10.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 10.9 (Invariant line). For $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, show $\text{span}(e_1)$ is invariant.

Solution. $Ae_1 = 2e_1$, so every scalar multiple of e_1 is mapped back into the same line. It is in fact an eigenspace. $\square$

Problem 10.10 (Block form from invariance). If U is invariant under T , why does a basis beginning with a basis of U give a block upper-triangular matrix?

Solution. Images of basis vectors in U stay in U , so their coordinate columns have zero components in the complementary basis directions. That forces the lower-left block to vanish. $\square$

Problem 10.11 (Quotient operator). Define the induced operator on V/U when U is T -invariant and prove it is well-defined.

Solution. Set $\overline{T}(v + U) = T(v) + U$. If $v - w \in U$, invariance gives $T(v - w) \in U$, so $T(v) + U = T(w) + U$. Thus the definition is independent of representative. $\square$

Problem 10.12 (Flag criterion). Explain why an upper-triangular matrix preserves the coordinate flag $V_k = \text{span}(e_1, \dots, e_k)$.

Solution. The first k columns of an upper-triangular matrix have zeros below row k , so the image of any vector supported in the first k coordinates remains supported there. $\square$

Problem 10.13 (Triangularization over $\mathbb{C}$). Outline the induction proving every complex operator is triangularizable.

Solution. Choose an eigenvector and its invariant line. Pass to the quotient, which has one lower dimension, triangularize the induced operator there by induction, then lift the quotient basis and prepend the eigenvector. $\square$

Problem 10.14 (Failure over $\mathbb{R}$). Why is rotation by 90° not triangularizable over $\mathbb{R}$?

Solution. A real upper-triangular matrix has real diagonal entries and hence real eigenvalues. The rotation has characteristic polynomial $t^2 + 1$ and no real eigenvalue, so no real triangular form exists. $\square$

Problem 10.15 (Polynomial invariant subspaces). Show $\mathcal{P}_k$ is invariant under differentiation on $\mathcal{P}_n$.

Solution. If $\deg p \leq k$, then $\deg p' \leq k - 1 \leq k$ (or $p' = 0$). Hence differentiation maps $\mathcal{P}_k$ into itself. $\square$

Problem 10.16 (Invariant complement need not exist). Give a 2×2 example with an invariant line but no invariant complementary line.

Solution. The shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ preserves $\text{span}(e_1)$. Its only eigenspace is that line, so no distinct invariant line can complement it. $\square$

Problem 10.17 (Restriction spectrum). If U is invariant, why must every eigenvalue of $T|_U$ also be an eigenvalue of T ?

Solution. An eigenvector of the restriction is a nonzero vector $u \in U$ with $T(u) = \lambda u$, which is also an eigenvector of T on the whole space. $\square$

Problem 10.18 (Nested kernels). For a nilpotent N , show $\ker N \subseteq \ker N^2 \subseteq \dots$ are invariant.

Solution. If $N^k v = 0$, then $N^k(Nv) = N^{k+1}v = 0$, so $Nv \in \ker N^k$ after adjusting indices appropriately; each kernel is invariant under N . $\square$

Problem 10.19 (Diagonal form as strongest triangular form). How does diagonalization strengthen triangularization?

Solution. Triangularization produces a complete invariant flag but can retain couplings above the diagonal. Diagonalization decomposes the space into one-dimensional invariant eigenspaces, eliminating all off-diagonal couplings. $\square$

Problem 10.20 (Invariant-subspace synthesis). Why are invariant subspaces the natural mechanism for induction in operator theory?

Solution. They let one split the problem into a restriction on a smaller subspace and an induced operator on a lower-dimensional quotient. Structural information can then be assembled from these two simpler pieces. $\square$

10.11 Exercises with complete solutions

Exercise 10.21. Show every eigenspace is invariant.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Test invariance by applying T to a basis of the subspace; every image must remain in the subspace. $\square$

Solution. If $v \in E_\lambda$, then $T(v) = \lambda v \in E_\lambda$. $\square$

Exercise 10.22. If U is invariant, prove the quotient operator is well-defined.

Hint. Apply the operator to a basis of the proposed subspace; invariance means every resulting vector stays in its span. Choose a basis of the invariant subspace first and extend it to a basis of the whole space. $\square$

Solution. If $v + U = w + U$, then $v - w \in U$, so $T(v - w) \in U$ and $T(v) + U = T(w) + U$. $\square$

Exercise 10.23. Why does an upper-triangular matrix preserve the coordinate flag?

Hint. Choose a basis adapted to the invariant flag; the vanishing entries below the diagonal encode preservation of the successive subspaces. With an adapted basis, entries below the relevant block vanish because vectors in the invariant subspace cannot leave it. $\square$

Solution. The first k columns have zeros below row k , so vectors supported in the first k coordinates map back into that subspace. $\square$

Exercise 10.24. Give an invariant subspace of a diagonal operator.

Hint. Apply the operator to a basis of the proposed subspace; invariance means every resulting vector stays in its span. For a triangular matrix, the characteristic polynomial is the product of $(\lambda - a_{ii})$. $\square$

Solution. The span of any subset of eigenbasis vectors is invariant. $\square$

Exercise 10.25. Why may triangularization fail over $\mathbb{R}$?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Build the invariant flag one eigenvector at a time; then pass to the induced map on a quotient. $\square$

Solution. The induction starts by choosing an eigenvector; a real operator can have no real eigenvalue. $\square$

Exercise 10.26. Show the characteristic polynomial of a triangular matrix is the product of $t - a_{ii}$.

Hint. Reduce to triangular form while recording how each row operation changes the determinant. Triangularizable does not mean diagonalizable: look for a nonzero Jordan off-diagonal entry. $\square$

Solution. The matrix $tI - A$ is triangular, so its determinant is the product of its diagonal entries. $\square$

Exercise 10.27. Describe the block form associated with an invariant subspace U .

Hint. Apply the operator to a basis of the proposed subspace; invariance means every resulting vector stays in its span. Over a field where the characteristic polynomial splits, start with one eigenvector and induct on dimension. $\square$

Solution. Choose a basis of U and extend it to V . The lower-left block is zero because $T(U) \subseteq U$. $\square$

Exercise 10.28. Explain why triangularization already reveals all eigenvalues.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Check whether the proposed subspace is generated by eigenvectors; such a span is automatically invariant. $\square$

Solution. The characteristic polynomial is the product of t minus the diagonal entries, so the diagonal entries are precisely the eigenvalues with multiplicity. $\square$

10.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 10.29 (Invariant line). Is $\text{span}(1, 0)$ invariant under $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$?

Hint. Apply A to the spanning vector. $\square$

Solution. Yes: $A(1, 0) = 2(1, 0)$. $\square$

Exercise 10.30 (Noninvariant line). Is $\text{span}(0, 1)$ invariant for the same matrix?

Hint. Apply $A(0, 1)$. $\square$

Solution. No: $A(0, 1) = (1, 3)$ is not a multiple of $(0, 1)$. $\square$

Exercise 10.31 (Invariant kernel). If $ST = TS$, show $\ker T$ is S -invariant.

Hint. For $v \in \ker T$, compute $T(Sv)$. $\square$

Solution. $T(Sv) = S(Tv) = 0$, so $Sv \in \ker T$. $\square$

Exercise 10.32 (Triangular spectrum). Read the eigenvalues of an upper-triangular matrix with diagonal $1, 1, 4$.

Hint. Use the diagonal entries. $\square$

Solution. 1 with algebraic multiplicity 2 , and 4 . $\square$

Exercise 10.33 (Flag). For an upper-triangular matrix, which coordinate subspaces form an invariant flag?

Hint. Use the spans of the first basis vectors. $\square$

Solution. $\text{span}(e_1) \subset \text{span}(e_1, e_2) \subset \cdots$. $\square$

Proofs

Exercise 10.34 (Invariant-span proof). If U is spanned by vectors whose images lie in U , prove U is invariant.

Hint. Apply linearity to a general spanning combination. □

Solution. Every $u = \sum a_i v_i$ maps to $\sum a_i T v_i \in U$. □

Exercise 10.35 (Eigenspace invariance). Prove every eigenspace of T is T -invariant.

Hint. If $v \in E_\lambda$, compute Tv . □

Solution. $Tv = \lambda v \in E_\lambda$. □

Exercise 10.36 (Polynomial invariance). If U is T -invariant, prove it is invariant under every polynomial $p(T)$.

Hint. First prove invariance under powers. □

Solution. Repeated application stays in U , and linear combinations of the powers also stay in U . □

Exercise 10.37 (Triangularization criterion). Explain why a basis adapted to a complete invariant flag gives an upper-triangular matrix.

Hint. Track where each basis vector may map in the flag. □

Solution. The j -th vector maps into the span of the first j vectors, so entries below the diagonal vanish. □

Counterexamples

Exercise 10.38 (Counterexample: subspace). Give a subspace that is not invariant under a rotation by 90° .

Hint. Use the x -axis. □

Solution. The x -axis maps to the y -axis, so it is not invariant. □

Exercise 10.39 (Counterexample: sum decomposition). If U, W are invariant, must every complement of U be invariant?

Hint. Use a diagonal matrix and a slanted complement. □

Solution. No. For $A = \text{diag}(1, 2)$, the complement $\text{span}(1, 1)$ of the x -axis is not invariant. □

Exercise 10.40 (Counterexample: real triangularization). Give a real operator that cannot be triangularized over $\mathbb{R}$.

Hint. A real triangular matrix has real diagonal eigenvalues. □

Solution. A 90° rotation has no real eigenvalues, so it cannot have real triangular form. □

Applications

Exercise 10.41 (Application: block structure). If $V = U \oplus W$ with both subspaces invariant, what matrix form arises in an adapted basis?

Hint. No component crosses between the invariant summands. □

Solution. The matrix is block diagonal $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$. □

Exercise 10.42 (Application: quotient). If U is invariant, why does T induce an operator on V/U ?

Hint. Check independence of coset representative. □

Solution. Define $\bar{T}(v + U) = Tv + U$; if $v - w \in U$, invariance gives $T(v - w) \in U$. □

Challenges

Exercise 10.43 (Challenge: commuting family). If S, T commute and $E_\lambda(T)$ is one-dimensional, what can be said about its nonzero vector for S ?

Hint. Commutation makes the eigenspace S -invariant. □

Solution. A linear map on a one-dimensional invariant space is scalar multiplication, so the vector is also an eigenvector of S . □

Exercise 10.44 (Challenge: triangular determinant). Use triangularization to explain why determinant equals the product of eigenvalues when an operator triangularizes.

Hint. Diagonal entries of triangular form are eigenvalues. □

Solution. Determinant is the product of diagonal entries, hence of eigenvalues with algebraic multiplicity. □

Chapter summary

Invariant subspaces let a large operator be studied through smaller pieces. Triangularization is the systematic expression of this idea: once an invariant flag is chosen, the matrix becomes triangular and its diagonal reveals the eigenvalues. Over an algebraically closed field, every operator on a nonzero finite-dimensional space admits such a form. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 11

Diagonalization and Minimal Polynomials

Diagonalization asks whether an operator can be decomposed into independent one-dimensional modes. The minimal polynomial gives the sharp algebraic criterion: an operator is diagonalizable exactly when its minimal polynomial splits into distinct linear factors. This viewpoint unifies eigenvalue counting with polynomial identities satisfied by the operator.

Learning goals

After this chapter, the reader should be able to:

- decide diagonalizability from eigenspace dimensions and from the minimal polynomial;
- construct a diagonalizing basis when enough independent eigenvectors exist;
- compute the minimal polynomial from annihilating relations or a known canonical form;
- use the minimal polynomial to reduce high powers or polynomial expressions in an operator;
- prove that a split operator is diagonalizable exactly when its minimal polynomial has no repeated root;
- compare characteristic and minimal polynomials without assuming that they are equal;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

11.1 Diagonalizable operators

An operator is diagonalizable when it has a basis of eigenvectors, equivalently when some matrix representation is diagonal.

Example 11.1 (Diagonalization by an eigenbasis). For

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix},$$

eigenvectors may be chosen $v_2 = (1, 0)$ for eigenvalue 2 and $v_3 = (1, 1)$ for eigenvalue 3. With

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

one checks $AP = PD$, hence $P^{-1}AP = D$.

The equation $AP = PD$ is often the fastest verification because its columns say exactly $Av_i = \lambda_i v_i$.

11.2 Direct sums of eigenspaces

If the characteristic polynomial splits, diagonalizability is equivalent to V being the direct sum of its eigenspaces.

11.3 Minimal polynomial

The minimal polynomial m_T is the monic polynomial of least positive degree with $m_T(T) = 0$.

Example 11.2 (Finding a minimal polynomial). For

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

we have $J - I \neq 0$ but

$$(J - I)^2 = 0.$$

Therefore the minimal polynomial is

$$m_J(t) = (t - 1)^2.$$

The characteristic polynomial is also $(t - 1)^2$, but the key information is the repeated factor in the minimal polynomial: it certifies that J is not diagonalizable.

11.4 Divisibility property

Every polynomial p with $p(T) = 0$ is divisible by m_T . In particular, m_T divides the characteristic polynomial.

11.5 Square-free criterion

If m_T splits, then T is diagonalizable exactly when m_T has no repeated root.

Example 11.3 (Square-free minimal polynomial criterion). Suppose

$$m_T(t) = (t - 2)(t + 1)$$

over the scalar field. The factors are distinct, so the minimal polynomial is square-free and splits. Hence T is diagonalizable.

By contrast, the polynomial $(t - 2)^2(t + 1)$ does not by itself prove nondiagonalizability, but if it is the actual minimal polynomial then the repeated factor rules diagonalization out. The criterion concerns the minimal polynomial, not merely the characteristic polynomial.

11.6 Spectral projectors by interpolation

For distinct eigenvalues, Lagrange interpolation polynomials in T produce projections onto eigenspaces.

11.7 Core structural results

Theorem 11.4 (Dimension criterion). *If χ_T splits, then T is diagonalizable if and only if the sum of eigenspace dimensions equals $\dim V$.*

Proof. Distinct eigenspaces form a direct sum. Their union of bases is an eigenbasis exactly when the direct sum has full dimension. $\square$

Theorem 11.5 (Minimal-polynomial criterion). *An operator is diagonalizable if and only if its minimal polynomial is a product of distinct linear factors.*

Proof. For a diagonal operator, a polynomial annihilates it exactly when it vanishes at every diagonal eigenvalue. Conversely, use Lagrange polynomials to construct complementary eigenspace projectors. $\square$

11.8 Worked examples

Example 11.6 (Projection). A projection $P^2 = P$ has minimal polynomial dividing $t(t-1)$, which has distinct roots, so every projection is diagonalizable.

Example 11.7 (Involution). If $T^2 = I$ over characteristic not 2, then m_T divides $(t-1)(t+1)$, so T is diagonalizable.

Example 11.8 (Nontrivial shear). The shear has minimal polynomial $(t-1)^2$, so it is not diagonalizable.

11.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

11.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 11.9 (Projection minimal polynomial). If $P^2 = P$ and $P \neq 0, I$, determine the minimal polynomial.

Solution. P satisfies $t(t-1)$, and neither t nor $t-1$ alone annihilates it because P is neither zero nor identity. Thus $m_P(t) = t(t-1)$, which has distinct roots, so P is diagonalizable. $\square$

Problem 11.10 (Involution). If $T^2 = I$ over a field of characteristic not 2, prove T is diagonalizable.

Solution. The minimal polynomial divides $(t - 1)(t + 1)$, a product of distinct linear factors. Therefore the minimal polynomial is square-free and split, which is the diagonalizability criterion. $\square$

Problem 11.11 (Shear obstruction). Why is a nontrivial shear not diagonalizable?

Solution. For $A = I + N$ with $N \neq 0$ and $N^2 = 0$, the minimal polynomial is $(t - 1)^2$. The repeated factor violates the square-free minimal-polynomial criterion. $\square$

Problem 11.12 (Eigenspace dimension test). An operator on a 5-dimensional space has eigenspaces of dimensions 2, 1, 2 for three distinct eigenvalues. Is it diagonalizable?

Solution. Distinct eigenspaces form a direct sum, and their dimensions sum to 5. Their combined bases therefore form a basis of eigenvectors, so the operator is diagonalizable. $\square$

Problem 11.13 (Minimal polynomial from diagonal data). For $D = \text{diag}(1, 1, 3, 5, 5)$ find m_D .

Solution. The minimal polynomial must vanish at each distinct diagonal value and needs no repeated factors: $m_D = (t - 1)(t - 3)(t - 5)$. $\square$

Problem 11.14 (Annihilating polynomial divisibility). Why does the minimal polynomial divide every polynomial p with $p(T) = 0$?

Solution. Divide $p = qm_T + r$ with $\deg r < \deg m_T$. Evaluating at T gives $0 = r(T)$. Minimality of m_T forces $r = 0$, so $m_T \mid p$. $\square$

Problem 11.15 (Spectral projectors). For distinct eigenvalues λ_1, λ_2 , write the projector onto E_{λ_1} as a polynomial in T .

Solution. $P_1 = (T - \lambda_2 I)/(\lambda_1 - \lambda_2)$. It acts as 1 on E_{λ_1} and 0 on E_{λ_2} . $\square$

Problem 11.16 (Repeated characteristic root). Why does a repeated root of the characteristic polynomial not imply non-diagonalizability?

Solution. The identity has characteristic polynomial $(t - 1)^n$ but minimal polynomial $t - 1$. Diagonalizability is controlled by repeated roots in the minimal polynomial, not merely in the characteristic polynomial. $\square$

Problem 11.17 (Direct eigenspace sum). Prove eigenspaces for distinct eigenvalues have zero intersection.

Solution. If v lies in both E_λ and E_μ with $\lambda \neq \mu$, then $\lambda v = Tv = \mu v$, so $(\lambda - \mu)v = 0$ and hence $v = 0$. $\square$

Problem 11.18 (Diagonalization by eigenbasis). Given n independent eigenvectors in an n -dimensional space, how do you diagonalize the operator?

Solution. Use the eigenvectors as an ordered basis. The image of each basis vector is its eigenvalue times itself, so the representing matrix is diagonal with those eigenvalues. $\square$

Problem 11.19 (Polynomial functional test). If T is diagonalizable, why is every polynomial $p(T)$ diagonalizable in the same basis?

Solution. In an eigenbasis, T is diagonal. Applying a polynomial entrywise to the diagonal gives another diagonal matrix, so $p(T)$ is diagonal in the same basis. $\square$

Problem 11.20 (Diagonalization synthesis). Compare the roles of characteristic and minimal polynomials in diagonalization.

Solution. The characteristic polynomial tells which eigenvalues are available and their algebraic multiplicities. The minimal polynomial records the largest Jordan-type obstruction. Splitting of the characteristic polynomial is necessary, while square-freeness of the split minimal polynomial is the decisive criterion. $\square$

11.11 Exercises with complete solutions

Exercise 11.21. Why is every operator with distinct eigenvalues equal in number to the dimension diagonalizable?

Hint. Start with a shortest linear relation among the eigenvectors and apply $T - \lambda I$ for one of the eigenvalues. Compute the eigenspace dimension for each eigenvalue and compare the total with $\dim V$. $\square$

Solution. Eigenvectors for distinct eigenvalues are independent, yielding a basis. $\square$

Exercise 11.22. Find the minimal polynomial of a diagonal matrix with diagonal entries 1, 1, 3.

Hint. Factor the minimal polynomial. Diagonalizability requires the split factors to be distinct. A split operator is diagonalizable exactly when the minimal polynomial has distinct linear factors. $\square$

Solution. It is $(t - 1)(t - 3)$; repeated diagonal occurrences do not create repeated factors in the minimal polynomial. $\square$

Exercise 11.23. Find the minimal polynomial of a nontrivial 2×2 shear.

Hint. To find the minimal polynomial, start from the characteristic polynomial and test lower-degree divisors that still annihilate the operator. $\square$

Solution. It is $(t - 1)^2$: $A - I$ is nonzero but $(A - I)^2 = 0$. $\square$

Exercise 11.24. Show every projection is diagonalizable without computing a determinant.

Hint. Set the output equal to zero and solve the resulting homogeneous equations; the solution vectors generate the kernel. Reduce high powers of T modulo the minimal polynomial instead of multiplying matrices repeatedly. $\square$

Solution. Decompose $V = \operatorname{im} P \oplus \ker P$. The operator acts as 1 on the image and 0 on the kernel. $\square$

Exercise 11.25. If $T^2 = I$, what are the possible eigenvalues?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. If $T = PDP^{-1}$, powers and polynomials of T are obtained by applying the same function to D . $\square$

Solution. Any eigenvalue λ satisfies $\lambda^2 = 1$, so $\lambda = \pm 1$. $\square$

Exercise 11.26. Why must m_T divide every annihilating polynomial?

Hint. Repeated roots in the characteristic polynomial are harmless if the minimal polynomial has no repeated factor. $\square$

Solution. Divide p by m_T ; the remainder has smaller degree and also annihilates T , so minimality forces the remainder to vanish. $\square$

Exercise 11.27. Construct the spectral projector onto E_{λ_i} for distinct eigenvalues.

Hint. Start with a shortest linear relation among the eigenvectors and apply $T - \lambda I$ for one of the eigenvalues. Use the smallest annihilating polynomial; the characteristic polynomial always annihilates by Cayley–Hamilton but need not be minimal. $\square$

Solution. Use $P_i = \prod_{j \neq i} (T - \lambda_j I) / (\lambda_i - \lambda_j)$; it equals 1 on E_{λ_i} and 0 on the other eigenspaces. $\square$

Exercise 11.28. Explain why repeated roots in the characteristic polynomial do not by themselves prevent diagonalization.

Hint. Factor the minimal polynomial. Diagonalizability requires the split factors to be distinct. To refute diagonalizability, exhibit either too few eigenvectors or a repeated factor in the minimal polynomial. $\square$

Solution. The identity matrix has characteristic polynomial $(t - 1)^n$ but is diagonal; what matters is the minimal polynomial, which is $t - 1$. $\square$

11.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 11.29 (Diagonalize 2x2). Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$.

Hint. Use eigenvectors for 2 and 3. $\square$

Solution. Take $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, then $P^{-1}AP = \text{diag}(2, 3)$. $\square$

Exercise 11.30 (Minimal polynomial identity). What is the minimal polynomial of I ?

Hint. Find the monic polynomial of least degree annihilating I . $\square$

Solution. $m_I(t) = t - 1$. $\square$

Exercise 11.31 (Minimal polynomial nilpotent). If $N^3 = 0$ but $N^2 \neq 0$, what is m_N ?

Hint. Nilpotence forces a power of t . $\square$

Solution. $m_N(t) = t^3$. $\square$

Exercise 11.32 (Jordan shear). Find the minimal polynomial of $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Hint. Check $A - I$ and $(A - I)^2$. □

Solution. $(t - 1)^2$. □

Exercise 11.33 (Polynomial reduction). If $m_T(t) = t^2 - 1$, simplify T^5 .

Hint. Use $T^2 = I$. □

Solution. $T^5 = T$. □

Proofs

Exercise 11.34 (Minimal divides annihilator). Prove the minimal polynomial divides every polynomial p with $p(T) = 0$.

Hint. Divide p by m_T with remainder. □

Solution. If $p = qm + r$, then $0 = p(T) = r(T)$; minimality forces $r = 0$. □

Exercise 11.35 (Characteristic annihilation). State how Cayley–Hamilton bounds the minimal polynomial.

Hint. The characteristic polynomial annihilates T . □

Solution. Since $\chi_T(T) = 0$, the minimal polynomial divides χ_T . □

Exercise 11.36 (Distinct eigenvalues criterion). Prove an operator with n distinct eigenvalues on an n -dimensional space is diagonalizable.

Hint. Distinct-eigenvalue eigenvectors are independent. □

Solution. Choose one eigenvector per eigenvalue; n independent vectors form a basis. □

Exercise 11.37 (Square-free criterion). State the split square-free minimal-polynomial criterion for diagonalizability.

Hint. Diagonal form is annihilated by the product of distinct linear factors. □

Solution. T is diagonalizable iff m_T splits into distinct linear factors. □

Counterexamples

Exercise 11.38 (Counterexample: characteristic split). Show a split characteristic polynomial does not guarantee diagonalizability.

Hint. Use a nontrivial Jordan block. □

Solution. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has split characteristic polynomial but is not diagonalizable. □

Exercise 11.39 (Counterexample: repeated char root). Does a repeated root of the characteristic polynomial force nondiagonalizability?

Hint. Test the identity matrix. □

Solution. No. I_2 has $(t - 1)^2$ but is already diagonal. □

Exercise 11.40 (Counterexample: field). Give an operator diagonalizable over $\mathbb{C}$ but not over $\mathbb{R}$.

Hint. Use a planar rotation. □

Solution. A 90° rotation has eigenvalues $\pm i$ and diagonalizes over $\mathbb{C}$, not over $\mathbb{R}$. □

Applications

Exercise 11.41 (Application: powers). If $A = PDP^{-1}$, explain how to compute A^{100} .

Hint. Powers telescope around the diagonal matrix. □

Solution. $A^{100} = PD^{100}P^{-1}$, and D^{100} is obtained entrywise. □

Exercise 11.42 (Application: differential equation). Why does diagonalization simplify $x'(t) = Ax(t)$?

Hint. Change coordinates with an eigenbasis. □

Solution. In eigenbasis coordinates the system decouples into scalar equations $y'_i = \lambda_i y_i$. □

Challenges

Exercise 11.43 (Challenge: same char different minimal). Give two 2×2 matrices with the same characteristic polynomial but different minimal polynomials.

Hint. Use I_2 and a Jordan block. □

Solution. Both have $(t - 1)^2$; I_2 has minimal $t - 1$, the Jordan block has $(t - 1)^2$. □

Exercise 11.44 (Challenge: projections). If $P^2 = P$, determine the possible roots of its minimal polynomial and conclude diagonalizability.

Hint. P is annihilated by $t(t - 1)$. □

Solution. $m_P \mid t(t - 1)$, a product of distinct linear factors, so P is diagonalizable. □

Chapter summary

Diagonalization asks whether an operator can be decomposed into independent one-dimensional modes. The minimal polynomial gives the sharp algebraic criterion: an operator is diagonalizable exactly when its minimal polynomial splits into distinct linear factors. This viewpoint unifies eigenvalue counting with polynomial identities satisfied by the operator. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 12

Canonical Forms

When diagonalization fails, canonical forms measure the failure rather than hiding it. Jordan form over an algebraically closed field records nilpotent chains attached to each eigenvalue, while rational canonical form works over arbitrary fields using invariant factors. Both arise from viewing a vector space as a module over the polynomial ring via $p \cdot v = p(T)v$.

Learning goals

After this chapter, the reader should be able to:

- compute generalized eigenspaces using kernels of powers of $T - \lambda I$;
- construct Jordan chains and assemble them into a Jordan basis when the scalar field permits;
- recover Jordan-block sizes from the growth of $\dim \ker(T - \lambda I)^k$;
- write the Jordan form of a nilpotent operator from its chain lengths;
- use canonical form to compute powers, exponentials, or polynomial functions of an operator;
- explain which data are invariant under similarity and therefore belong to the operator rather than a chosen basis;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

12.1 Generalized eigenvectors

A vector lies in the generalized eigenspace for λ if $(T - \lambda I)^k v = 0$ for some k .

12.2 Nilpotent operators

Nilpotent operators organize into chains $v, Nv, \dots$. Their Jordan blocks encode chain lengths.

12.3 Jordan blocks

A Jordan block $J_r(\lambda)$ has λ on the diagonal and ones on the superdiagonal. Its minimal polynomial is $(t - \lambda)^r$.

Example 12.1 (Recovering Jordan blocks from kernels). Let N be nilpotent on a four-dimensional space with

$$\dim \ker N = 2, \quad \dim \ker N^2 = 3, \quad \dim \ker N^3 = 4.$$

There are two Jordan blocks because $\dim \ker N = 2$. The growth from 2 to 3 to 4 shows one chain has length 3, while the other has length 1. Hence the nilpotent Jordan type is $3 + 1$.

Kernel growth therefore reconstructs block sizes without first finding a Jordan basis.

12.4 Jordan canonical form

Over an algebraically closed field, every operator is similar to a direct sum of Jordan blocks, unique up to block order.

12.5 Primary decomposition

If the minimal polynomial factors into pairwise coprime powers, the space decomposes into invariant primary components.

12.6 Rational canonical form

Over any field, invariant factors determine a block diagonal companion-matrix form. Jordan form is the split refinement of this module-theoretic structure.

Example 12.2 (A companion matrix realizes a polynomial). For

$$p(t) = t^3 + 2t + 1,$$

one companion convention gives

$$C = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & 0 \end{pmatrix}.$$

Its characteristic polynomial is $p(t)$. The matrix packages multiplication by t on the cyclic module $\mathbb{F}[t]/(p)$.

This is why companion blocks are natural rather than ad hoc: they turn a polynomial relation into a linear operator.

Example 12.3 (Why rational form does not require eigenvalues). A real rotation by 90° has characteristic polynomial $t^2 + 1$, which does not split over $\mathbb{R}$. Jordan form over $\mathbb{R}$ is therefore not available in the usual eigenvalue sense. Rational canonical form, however, uses invariant factors over the original field and needs no roots.

Canonical-form choice is field-sensitive: Jordan form emphasizes eigenvalues after splitting; rational form remains intrinsic over the base field.

12.7 Core structural results

Theorem 12.4 (Generalized eigenspace decomposition). *If $m_T(t) = \prod_i (t - \lambda_i)^{r_i}$ with distinct λ_i , then $V = \bigoplus_i \ker(T - \lambda_i I)^{r_i}$.*

Proof. The factors are pairwise coprime. Bézout identities in $\mathbb{F}[t]$ produce polynomial projectors whose images are the primary components. $\square$

Theorem 12.5 (Jordan block invariants). *For a nilpotent operator, the numbers $\dim \ker N^k$ determine the Jordan block sizes.*

Proof. The increment $\dim \ker N^k - \dim \ker N^{k-1}$ counts blocks of size at least k ; successive differences recover the exact sizes. $\square$

12.8 Worked examples

Example 12.6 (Single Jordan block). $J_3(0)$ is nilpotent of index 3; kernel dimensions are 1, 2, 3.

Example 12.7 (Two blocks). $J_2(0) \oplus J_1(0)$ has kernel dimensions 2, 3, revealing two blocks, one of which has length at least 2.

Example 12.8 (Companion matrix). The companion matrix of a monic polynomial has that polynomial as both characteristic and minimal polynomial when it represents a cyclic module.

12.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

12.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 12.9 (Nilpotent chain). For $N = J_3(0)$ compute $\dim \ker N$, $\dim \ker N^2$, and $\dim \ker N^3$.

Solution. The standard Jordan chain has lengths three. The kernels have dimensions 1, 2, 3 respectively. The increments 1, 1, 1 indicate one block of size at least 1, 2, 3. $\square$

Problem 12.10 (Two nilpotent blocks). If a nilpotent operator has Jordan blocks of sizes 3 and 1, compute $\dim \ker N^k$ for $k = 1, 2, 3$.

Solution. At $k = 1$, each block contributes one kernel dimension, giving 2. At $k = 2$, the size-3 block contributes 2 and the size-1 block 1, giving 3. At $k = 3$ the total is 4. $\square$

Problem 12.11 (Geometric multiplicity in Jordan form). How is the geometric multiplicity of λ read from Jordan form?

Solution. Each Jordan block for λ contributes exactly one eigenvector direction, so the geometric multiplicity is the number of Jordan blocks carrying λ . $\square$

Problem 12.12 (Minimal polynomial from blocks). If the largest Jordan block for eigenvalue 2 has size 4 and for eigenvalue -1 has size 2, find the minimal polynomial.

Solution. The exponents are the largest block sizes, so $m_T(t) = (t - 2)^4(t + 1)^2$. $\square$

Problem 12.13 (Diagonalizability from blocks). When is a Jordan form diagonal?

Solution. Exactly when every Jordan block has size 1. Equivalently every eigenvalue's generalized eigenspace is already its ordinary eigenspace. $\square$

Problem 12.14 (Generalized eigenspace). Show $\ker(T - \lambda I)^k$ is T -invariant.

Solution. T commutes with every polynomial in T , so $(T - \lambda I)^k T(v) = T(T - \lambda I)^k v$. If the latter is zero, Tv lies in the same kernel. $\square$

Problem 12.15 (Primary decomposition idea). Why do coprime factors of the minimal polynomial lead to a direct-sum decomposition?

Solution. Bézout identities between the coprime polynomial factors produce polynomial projectors in T whose images are the corresponding primary kernels and whose sum is the identity. $\square$

Problem 12.16 (Real versus complex canonical form). Why may Jordan form fail over $\mathbb{R}$?

Solution. Jordan form requires the minimal polynomial to split into linear factors. A real operator can have irreducible quadratic factors, such as a planar rotation. Rational canonical form does not require splitting. $\square$

Problem 12.17 (Companion matrix). What is the conceptual role of a companion matrix?

Solution. It represents multiplication by t on the cyclic module $\mathbb{F}[t]/(p)$. Its characteristic and minimal polynomial are both the defining monic polynomial p when the module is cyclic. $\square$

Problem 12.18 (Kernel increments). How do the increments $d_k = \dim \ker N^k - \dim \ker N^{k-1}$ encode block sizes?

Solution. d_k counts the number of Jordan blocks of size at least k . Therefore $d_k - d_{k+1}$ counts blocks of size exactly k . $\square$

Problem 12.19 (Similarity invariants). Name data preserved by similarity that are visible in Jordan form.

Solution. Eigenvalues, algebraic and geometric multiplicities, minimal polynomial, and the multiset of Jordan block sizes are all similarity invariants. Jordan form packages them completely over an algebraically closed field. $\square$

Problem 12.20 (Canonical-form synthesis). Why is the module viewpoint $p \cdot v = p(T)v$ natural?

Solution. It turns invariant subspaces into submodules and polynomial identities into annihilators. The structure theorem for finitely generated modules over $\mathbb{F}[t]$ then yields rational canonical form, with Jordan form as the split primary refinement. $\square$

12.11 Exercises with complete solutions

Exercise 12.21. Compute the minimal polynomial of $J_4(\lambda)$.

Hint. Compute $\ker(T - \lambda I)^k$ for successive k ; the growth records the generalized eigenvectors. □

Solution. It is $(t - \lambda)^4$ because the nilpotent part has index 4. □

Exercise 12.22. What is the geometric multiplicity of an eigenvalue in Jordan form?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. A Jordan chain satisfies $(T - \lambda I)v_1 = 0$ and $(T - \lambda I)v_{j+1} = v_j$. □

Solution. It equals the number of Jordan blocks carrying that eigenvalue. □

Exercise 12.23. How can kernel dimensions reveal nilpotent block sizes?

Hint. Compute whichever of rank or nullity is easier, then use $\dim V = \text{rank } T + \text{nullity } T$ as the dimension check. For a nilpotent operator, the largest Jordan block size is the nilpotency index. □

Solution. The increment from $\ker N^{k-1}$ to $\ker N^k$ counts blocks of size at least k . □

Exercise 12.24. When is a Jordan matrix diagonal?

Hint. Compute kernels of successive powers of $T - \lambda I$; their dimension growth reveals the generalized-eigenvector chains. Use differences of successive kernel dimensions to count how many Jordan chains have length at least k . □

Solution. Exactly when every Jordan block has size 1. □

Exercise 12.25. Why is Jordan form not always available over $\mathbb{R}$?

Hint. Compute kernels of successive powers of $T - \lambda I$; their dimension growth reveals the generalized-eigenvector chains. After finding chain generators, order each chain from eigenvector to highest generalized eigenvector before forming the basis. □

Solution. The characteristic/minimal polynomial may not split into linear factors over $\mathbb{R}$. □

Exercise 12.26. State the advantage of rational canonical form.

Hint. To compute a matrix function on a Jordan block, separate the scalar part λI from the nilpotent part. □

Solution. It exists over every field and does not require the characteristic polynomial to split. □

Exercise 12.27. Show that generalized eigenspaces are invariant.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Similarity invariants include the eigenvalues and Jordan-block sizes, not the individual basis vectors in the chains. □

Solution. If $(T - \lambda I)^k v = 0$, then $(T - \lambda I)^k T(v) = T(T - \lambda I)^k v = 0$. □

Exercise 12.28. Relate the largest Jordan block for λ to the minimal polynomial.

Hint. Compute kernels of successive powers of $T - \lambda I$; their dimension growth reveals the generalized-eigenvector chains. Check the total sizes of all blocks against $\dim V$; this catches missing generalized eigenvectors. $\square$

Solution. Its size is the exponent of $(t - \lambda)$ in the minimal polynomial. $\square$

12.12 Graded supplementary exercises

The following exercises extend the original exercise set without replacing it. They are grouped by mathematical role and increase in synthesis.

Standard computations

Exercise 12.29 (Jordan nilpotent block). What are $\dim \ker N$ and $\dim \ker N^2$ for one 3×3 nilpotent Jordan block?

Hint. Follow the Jordan chain. $\square$

Solution. $\dim \ker N = 1$, $\dim \ker N^2 = 2$. $\square$

Exercise 12.30 (Jordan block count). For a nilpotent operator, what does $\dim \ker N$ count?

Hint. Each Jordan block contributes one terminal kernel vector. $\square$

Solution. It equals the number of nilpotent Jordan blocks. $\square$

Exercise 12.31 (Largest block). If $N^4 = 0$ but $N^3 \neq 0$, what is the largest nilpotent Jordan block size?

Hint. Use nilpotency index. $\square$

Solution. The largest block has size 4. $\square$

Exercise 12.32 (Companion polynomial). Write a companion matrix for $t^2 + at + b$ using the convention with ones below the diagonal.

Hint. Place negatives of coefficients in the last column. $\square$

Solution. $\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$. $\square$

Exercise 12.33 (Invariant factors). What divisibility relation is required among invariant factors?

Hint. Recall the rational canonical chain. $\square$

Solution. $f_1 \mid f_2 \mid \cdots \mid f_r$. $\square$

Proofs

Exercise 12.34 (Jordan uniqueness). Explain what data determine Jordan form up to block permutation.

Hint. For each eigenvalue, record all block sizes. □

Solution. The multiset of Jordan block sizes for each eigenvalue determines the form. □

Exercise 12.35 (Kernel-growth proof idea). Why does $\dim \ker N^k - \dim \ker N^{k-1}$ count blocks of size at least k ?

Hint. Inspect one Jordan block at a time. □

Solution. A block of size s contributes one new kernel dimension at stage k exactly when $s \geq k$. □

Exercise 12.36 (Minimal polynomial from Jordan). How is the exponent of $t - \lambda$ in the minimal polynomial read from Jordan form?

Hint. Look at the longest λ -Jordan chain. □

Solution. It is the size of the largest Jordan block for eigenvalue λ . □

Exercise 12.37 (Characteristic polynomial from blocks). Why does the characteristic polynomial multiply the block polynomials?

Hint. Jordan or companion canonical forms are block diagonal. □

Solution. The determinant of a block diagonal $tI - A$ is the product of block determinants. □

Counterexamples

Exercise 12.38 (Counterexample: eigenvalues). Why do eigenvalues alone not determine Jordan form?

Hint. Compare I_2 and a Jordan shear. □

Solution. They have the same eigenvalue multiset but different block structures. □

Exercise 12.39 (Counterexample: Jordan over base field). Why may real Jordan form fail for a real matrix?

Hint. Jordan blocks require eigenvalues in the field. □

Solution. A 90° rotation has eigenvalues $\pm i$, not real. □

Exercise 12.40 (Counterexample: characteristic only). Can the characteristic polynomial alone determine rational canonical form?

Hint. Compare operators with same characteristic polynomial but different minimal polynomials. □

Solution. No; invariant factors depend on finer module structure than the characteristic polynomial. □

Applications

Exercise 12.41 (Application: nilpotent dynamics). How does Jordan block size affect powers of a nilpotent operator?

Hint. A size- s nilpotent block dies exactly at power s . □

Solution. The longest block determines the nilpotency index and how many iterates can remain nonzero. □

Exercise 12.42 (Application: cyclic vector). If V has a cyclic vector for T , what shape does rational canonical form take?

Hint. A cyclic module needs only one invariant factor. □

Solution. The rational canonical form is a single companion block for the minimal/characteristic polynomial. □

Challenges

Exercise 12.43 (Challenge: reconstruct partition). If a nilpotent operator on dimension 6 has kernel dimensions 3, 5, 6 for powers 1, 2, 3, find the block sizes.

Hint. Successive increments 3, 2, 1 count blocks of size at least 1, 2, 3. □

Solution. There are three blocks, two of size at least 2, one of size at least 3; the sizes are 3, 2, 1. □

Exercise 12.44 (Challenge: primary decomposition). If $m_T = (t-1)^2(t+2)^3$, what invariant primary subspaces are suggested?

Hint. Use coprime factors. □

Solution. The space decomposes into the generalized 1-eigenspace $\ker(T - I)^2$ and generalized -2 -eigenspace $\ker(T + 2I)^3$ under the usual primary-decomposition hypotheses. □

Chapter summary

When diagonalization fails, canonical forms measure the failure rather than hiding it. Jordan form over an algebraically closed field records nilpotent chains attached to each eigenvalue, while rational canonical form works over arbitrary fields using invariant factors. Both arise from viewing a vector space as a module over the polynomial ring via $p \cdot v = p(T)v$. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Part III

Euclidean and Hilbert-Space Geometry

Chapter 13

Inner Products and Orthogonality

Inner products add geometry to linear algebra. Length, angle, orthogonality and best approximation all come from a positive definite sesquilinear form. The complex case forces careful placement of conjugation, but the resulting theory unifies Euclidean geometry with finite-dimensional Hilbert-space methods.

Learning goals

After this chapter, the reader should be able to:

- verify that a proposed formula defines an inner product;
- compute norms, distances, and angles from an inner product;
- prove orthogonality statements using bilinearity or sesquilinearity and conjugate symmetry;
- compute orthogonal complements of explicitly described subspaces;
- apply Cauchy–Schwarz and the triangle inequality with equality conditions;
- decompose a vector relative to an orthogonal direct sum and verify the Pythagorean identity;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

13.1 Inner products

A real inner product is symmetric bilinear and positive definite; a complex inner product is conjugate symmetric and sesquilinear.

13.2 Norm and distance

The norm is $\|v\| = \sqrt{\langle v, v \rangle}$ and the induced distance is $\|v - w\|$.

13.3 Cauchy–Schwarz

$|\langle u, v \rangle| \leq \|u\| \|v\|$, with equality precisely when the vectors are linearly dependent.

Example 13.1 (Cauchy–Schwarz with equality check). Take $u = (1, 2, -1)$ and $v = (2, 4, -2)$. Their inner product is

$$\langle u, v \rangle = 2 + 8 + 2 = 12,$$

while

$$\|u\| = \sqrt{6}, \quad \|v\| = \sqrt{24} = 2\sqrt{6}.$$

Hence

$$|\langle u, v \rangle| = 12 = \|u\| \|v\|.$$

Equality is expected because $v = 2u$. This is a useful check on both the arithmetic and the equality condition: Cauchy–Schwarz becomes an equality exactly along one-dimensional directions.

13.4 Orthogonality

Vectors are orthogonal when their inner product is zero. Orthogonal nonzero lists are automatically independent.

13.5 Orthogonal complements

For a subspace U , $U^\perp = \{v : \langle v, u \rangle = 0 \ \forall u \in U\}$. In finite dimensions, $V = U \oplus U^\perp$.

Example 13.2 (Computing an orthogonal complement). Let

$$U = \text{span}\{(1, 1, 0), (0, 1, 1)\} \subset \mathbb{R}^3.$$

A vector $x = (a, b, c)$ lies in $U^\perp$ exactly when

$$a + b = 0, \quad b + c = 0.$$

Thus $a = -b$ and $c = -b$, so

$$U^\perp = \text{span}\{(-1, 1, -1)\}.$$

The dimension check confirms the result: $\dim U = 2$, $\dim U^\perp = 1$, and the dimensions add to 3.

13.6 Parallelogram and polarization

The inner product determines the norm, and conversely a norm comes from an inner product exactly when it satisfies the parallelogram law.

Example 13.3 (Polarization recovers hidden geometry). Suppose a real inner-product norm is known but the inner product is not. For

$$u = (2, 1), \quad v = (1, -1),$$

the polarization identity gives

$$\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2).$$

Here $u + v = (3, 0)$ and $u - v = (1, 2)$, so

$$\langle u, v \rangle = \frac{1}{4}(9 - 5) = 1.$$

Direct computation gives $2 \cdot 1 + 1(-1) = 1$, providing an independent check. The norm therefore encodes the inner-product geometry when the parallelogram law holds.

13.7 Core structural results

Theorem 13.4 (Cauchy–Schwarz inequality). *For all u, v in an inner-product space, $|\langle u, v \rangle| \leq \|u\| \|v\|$.*

Proof. If $v \neq 0$, apply positivity to $u - (\langle u, v \rangle / \|v\|^2)v$ and expand. The case $v = 0$ is immediate. $\square$

Theorem 13.5 (Pythagorean theorem). *If $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.*

Proof. Expand $\langle u + v, u + v \rangle$ and use conjugate symmetry together with the vanishing cross terms. $\square$

13.8 Worked examples

Example 13.6 (Euclidean product). On $\mathbb{R}^n$, $\langle x, y \rangle = x^T y$.

Example 13.7 (Weighted product). For positive weights w_i , $\langle x, y \rangle = \sum_i w_i x_i y_i$ is an inner product.

Example 13.8 (Function product). On continuous functions, $\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt$ is an inner product.

13.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

13.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 13.9 (Cauchy–Schwarz equality). When does equality hold in Cauchy–Schwarz?

Solution. Equality holds exactly when one vector is a scalar multiple of the other. In the standard proof, equality means the orthogonal residual $u - (\langle u, v \rangle / \|v\|^2)v$ has norm zero. $\square$

Problem 13.10 (Orthogonal independence). Prove nonzero pairwise orthogonal vectors are linearly independent.

Solution. Take $\sum a_i v_i = 0$ and inner-product with v_j . All cross terms vanish, leaving $a_j \|v_j\|^2 = 0$. Since $v_j \neq 0$, $a_j = 0$ for every j . $\square$

Problem 13.11 (Weighted inner product). Show $\langle x, y \rangle = 2x_1y_1 + 5x_2y_2$ is an inner product on $\mathbb{R}^2$.

Solution. Bilinearity and symmetry are immediate. Positive definiteness follows because $\langle x, x \rangle = 2x_1^2 + 5x_2^2 > 0$ for every nonzero x . $\square$

Problem 13.12 (Orthogonal complement). Find the orthogonal complement of $\text{span}(1, 1, 1)$ in $\mathbb{R}^3$.

Solution. It is the plane $\{(x, y, z) : x + y + z = 0\}$, since orthogonality to $(1, 1, 1)$ is exactly the displayed equation. $\square$

Problem 13.13 (Pythagoras). If $u \perp v$, prove $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Solution. Expand $\langle u + v, u + v \rangle$. The cross terms vanish by orthogonality, leaving the sum of the squared norms. $\square$

Problem 13.14 (Triangle inequality). Derive the triangle inequality from Cauchy–Schwarz.

Solution. $\|u + v\|^2 = \|u\|^2 + 2\text{Re}\langle u, v \rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2$. Taking nonnegative square roots yields the result. $\square$

Problem 13.15 (Complex conjugation placement). Why must the standard complex inner product include conjugation?

Solution. Without conjugation, $\sum z_i^2$ can be nonreal or negative-looking; for example $i^2 = -1$. With conjugation, $\langle z, z \rangle = \sum |z_i|^2 \geq 0$ and is zero only for $z = 0$. $\square$

Problem 13.16 (Distance to a subspace). If $V = U \oplus U^\perp$ and $v = u + w$, which component controls the distance from v to U ?

Solution. For any $u' \in U$, $v - u' = (u - u') + w$ with orthogonal terms, so $\|v - u'\|^2 = \|u - u'\|^2 + \|w\|^2$. The minimal distance is $\|w\|$. $\square$

Problem 13.17 (Polarization). Recover a real inner product from its norm.

Solution. The real polarization identity is $\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2)$. Expanding the two norms verifies the formula. $\square$

Problem 13.18 (Angle computation). Find the angle between $(1, 1, 0)$ and $(1, 0, 1)$.

Solution. The dot product is 1, each norm is $\sqrt{2}$, so $\cos \theta = 1/2$ and $\theta = \pi/3$. $\square$

Problem 13.19 (Orthogonal decomposition dimension). If U is a subspace of finite-dimensional V , prove $\dim U + \dim U^\perp = \dim V$.

Solution. Choose an orthonormal basis of U and extend it to an orthonormal basis of V . The added basis vectors form a basis of $U^\perp$, so the dimensions add. $\square$

Problem 13.20 (Inner-product synthesis). Why is an inner product more structure than a vector-space norm?

Solution. An inner product determines a norm, but not every norm comes from an inner product. The parallelogram identity characterizes inner-product norms. Thus orthogonality and angle require extra geometric structure beyond linear operations. $\square$

13.11 Exercises with complete solutions

Exercise 13.21. Show nonzero orthogonal vectors are independent.

Hint. Write the relevant inner products explicitly and use bilinearity or sesquilinearity before simplifying norms or angles. Verify positivity, conjugate symmetry, and linearity in the correct argument; do not assume symmetry in the complex case. $\square$

Solution. Take a dependence relation and inner-product with one vector; all cross terms vanish, leaving its coefficient times a positive norm squared. $\square$

Exercise 13.22. Compute the angle between $(1, 0)$ and $(1, 1)$ in $\mathbb{R}^2$.

Hint. Compute the norm from $\|v\|^2 = \langle v, v \rangle$ before using an angle formula. $\square$

Solution. The cosine is $1/(1 \cdot \sqrt{2}) = 1/\sqrt{2}$, so the angle is $\pi/4$. $\square$

Exercise 13.23. Find the orthogonal complement of the x -axis in $\mathbb{R}^2$.

Hint. Use the orthogonal decomposition $v = P_U v + (v - P_U v)$ and identify where the residual lies. To show orthogonality, compute the inner product directly and simplify before normalizing anything. $\square$

Solution. It is the y -axis. $\square$

Exercise 13.24. Verify the parallelogram identity.

Hint. Find $U^\perp$ by imposing $\langle x, u_j \rangle = 0$ for a basis u_j of U . $\square$

Solution. Expand $\|u + v\|^2 + \|u - v\|^2$; the cross terms cancel, leaving $2\|u\|^2 + 2\|v\|^2$. $\square$

Exercise 13.25. Why is complex conjugation necessary in the standard complex inner product?

Hint. Write the relevant inner products explicitly and use bilinearity or sesquilinearity before simplifying norms or angles. For Cauchy–Schwarz, consider $\|u - tv\|^2 \geq 0$ and choose t to minimize the quadratic. $\square$

Solution. Without conjugation, $\langle v, v \rangle$ need not be real or positive; conjugation gives $\sum |v_i|^2$. $\square$

Exercise 13.26. Show $U \cap U^\perp = \{0\}$.

Hint. Use the orthogonal decomposition $v = P_U v + (v - P_U v)$ and identify where the residual lies. Equality in Cauchy–Schwarz occurs exactly when the two vectors are linearly dependent. $\square$

Solution. If v lies in both, then $\langle v, v \rangle = 0$, hence $v = 0$ by positive definiteness. $\square$

Exercise 13.27. Prove the triangle inequality from Cauchy–Schwarz.

Hint. Write the relevant inner products explicitly and use bilinearity or sesquilinearity before simplifying norms or angles. Use orthogonality to kill the cross term when expanding $\|u + w\|^2$. $\square$

Solution. Expand $\|u + v\|^2$ and bound the real part of $\langle u, v \rangle$ by $\|u\|\|v\|$. $\square$

Exercise 13.28. Give an example of two different inner products on $\mathbb{R}^2$.

Hint. Write the relevant inner products explicitly and use bilinearity or sesquilinearity before simplifying norms or angles. For a direct-sum decomposition, prove both orthogonality and that the dimensions add correctly. $\square$

Solution. The standard product and $\langle x, y \rangle = 2x_1y_1 + x_2y_2$ are distinct inner products. $\square$

13.12 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 13.29 (Weighted norm). For $\langle x, y \rangle = 2x_1y_1 + 3x_2y_2$, compute the norm of $(1, 2)$.

Hint. Evaluate $\langle x, x \rangle$ and take the nonnegative square root. $\square$

Solution. $\|(1, 2)\|^2 = 2 + 12 = 14$, so the norm is $\sqrt{14}$. $\square$

Exercise 13.30 (Angle). Find the angle between $(1, 1, 0)$ and $(1, 0, 1)$.

Hint. Compute the dot product and both norms, then use $\cos \theta = \langle u, v \rangle / (\|u\| \|v\|)$. $\square$

Solution. The dot product is 1, both norms are $\sqrt{2}$, so $\cos \theta = 1/2$ and $\theta = \pi/3$. $\square$

Exercise 13.31 (Orthogonal complement). Find $\text{span}(1, 2, 2)^\perp \subset \mathbb{R}^3$.

Hint. Solve $x + 2y + 2z = 0$ and parametrize the solution plane. $\square$

Solution. For example $x = -2s - 2t$, so a basis is $(-2, 1, 0), (-2, 0, 1)$. $\square$

Exercise 13.32 (Orthogonal decomposition). Decompose $v = (3, 1)$ into components parallel and perpendicular to $u = (1, 1)$.

Hint. Project onto the line through u , then subtract. $\square$

Solution. The parallel component is $(2, 2)$; the perpendicular residual is $(1, -1)$. $\square$

Exercise 13.33 (Complex product). With $\langle z, w \rangle = \sum z_i \overline{w_i}$, compute $\langle (1, i), (i, 1) \rangle$.

Hint. Conjugate the entries of the second vector. $\square$

Solution. $1(-i) + i(1) = 0$, so the vectors are orthogonal. $\square$

Proofs

Exercise 13.34 (Orthogonal independence). Prove every nonzero orthogonal list is linearly independent.

Hint. Take an inner product of a dependence relation with one list vector. $\square$

Solution. All cross terms vanish, leaving $a_j \|v_j\|^2 = 0$; positivity forces every $a_j = 0$. $\square$

Exercise 13.35 (Triangle inequality). Derive the triangle inequality from Cauchy–Schwarz.

Hint. Expand $\|u + v\|^2$ and bound the real part of the cross term. $\square$

Solution. $\|u + v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$. $\square$

Exercise 13.36 (Double orthogonal complement). For finite-dimensional V , prove $(U^\perp)^\perp = U$.

Hint. Use $U \subseteq (U^\perp)^\perp$ and compare dimensions. $\square$

Solution. Both inclusions and $\dim U + \dim U^\perp = \dim V$ give equal dimensions, hence equality. $\square$

Exercise 13.37 (Pythagoras converse). Over a real inner-product space, prove $\|u + v\|^2 = \|u\|^2 + \|v\|^2$ iff $u \perp v$.

Hint. Expand the left side and compare cross terms. $\square$

Solution. The difference is $2\langle u, v \rangle$; equality holds exactly when that inner product is zero. $\square$

Counterexamples and hypothesis testing

Exercise 13.38 (Positive weights matter). Show $\langle x, y \rangle = x_1y_1 - x_2y_2$ is not an inner product on $\mathbb{R}^2$.

Hint. Test positive definiteness on a coordinate vector. $\square$

Solution. For e_2 , $\langle e_2, e_2 \rangle = -1 < 0$, so positivity fails. $\square$

Exercise 13.39 (Norm without inner product). Show the ℓ^1 -norm on $\mathbb{R}^2$ does not come from an inner product.

Hint. Test the parallelogram identity on e_1, e_2 . $\square$

Solution. $\|e_1 + e_2\|_1^2 + \|e_1 - e_2\|_1^2 = 8$, while $2\|e_1\|_1^2 + 2\|e_2\|_1^2 = 4$. $\square$

Exercise 13.40 (Conjugation placement). Explain why $\sum z_i w_i$ is not the standard complex inner product.

Hint. Evaluate the would-be norm of $(i, 0, \dots)$. $\square$

Solution. It gives $i^2 = -1$, violating positive definiteness. $\square$

Applications and investigations

Exercise 13.41 (Distance to a plane). Find the distance from $v = (1, 2, 3)$ to the plane $x + y + z = 0$.

Hint. The normal vector is $n = (1, 1, 1)$; the distance is the magnitude of the projection onto its unit direction. $\square$

Solution. The distance is $|1 + 2 + 3|/\sqrt{3} = 2\sqrt{3}$. $\square$

Exercise 13.42 (Correlation geometry). For centered data vectors x, y , interpret $\langle x, y \rangle / (\|x\| \|y\|)$.

Hint. Recognize it as the cosine of the angle between the data vectors. $\square$

Solution. It is normalized correlation: 1 means same direction, 0 orthogonality, and -1 opposite direction. $\square$

Challenge problems

Exercise 13.43 (Equality in triangle inequality). Characterize equality in $\|u+v\| \leq \|u\| + \|v\|$.

Hint. Trace equality through Cauchy–Schwarz and the real-part bound. □

Solution. Equality holds when u, v point in the same nonnegative scalar direction (with the zero-vector cases included). □

Exercise 13.44 (Dimension lower bound). If $U, W \subset \mathbb{R}^7$ have dimensions 5 and 4, prove there is a nonzero vector in $U \cap W$, then interpret orthogonally.

Hint. Use the dimension formula and $\dim(U + W) \leq 7$. □

Solution. $\dim(U \cap W) \geq 5 + 4 - 7 = 2$. Equivalently, their orthogonal complements have dimensions 2, 3 and cannot account for all seven directions independently. □

Chapter summary

Inner products add geometry to linear algebra. Length, angle, orthogonality and best approximation all come from a positive definite sesquilinear form. The complex case forces careful placement of conjugation, but the resulting theory unifies Euclidean geometry with finite-dimensional Hilbert-space methods. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 14

Gram–Schmidt and Orthogonal Projection

Gram–Schmidt converts an arbitrary independent list into an orthonormal list spanning the same nested subspaces. Orthogonal projection then gives canonical decompositions and solves least-squares problems. This chapter connects exact geometry with the numerical algorithms QR factorization and normal equations.

Learning goals

After this chapter, the reader should be able to:

- perform the Gram–Schmidt process on an independent list and verify that successive spans are preserved;
- compute orthogonal projections onto lines and higher-dimensional subspaces from an orthonormal basis;
- prove the best-approximation property by decomposing the error into orthogonal components;
- construct a projection matrix and verify that it is idempotent and self-adjoint;
- derive a reduced QR factorization from Gram–Schmidt;
- solve a least-squares problem using projection, normal equations, or QR and compare the numerical consequences;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

14.1 Orthonormal lists

A list is orthonormal when its vectors have norm one and are pairwise orthogonal. Coordinates in an orthonormal basis are inner products.

14.2 Gram–Schmidt process

Successively subtract from each vector its projections onto the previously constructed orthonormal vectors, then normalize.

Example 14.1 (Gram–Schmidt with a verification). Start with

$$v_1 = (1, 1, 0), \quad v_2 = (1, 0, 1).$$

Set $e_1 = v_1/\sqrt{2}$. Then

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

Since $\|u_2\| = \sqrt{3/2}$, set $e_2 = u_2/\sqrt{3/2}$. A direct dot-product check gives $\langle e_1, e_2 \rangle = 0$, and both have norm 1. The construction also preserves the span because $u_2 = v_2 - \frac{1}{2}v_1$.

14.3 Projection onto a subspace

For an orthonormal basis $e_1, \dots, e_r$ of U , $P_U v = \sum_i \langle v, e_i \rangle e_i$.

Example 14.2 (Projection matrix onto a plane). Let $U \subset \mathbb{R}^3$ have orthonormal basis

$$q_1 = (1, 0, 0), \quad q_2 = (0, 1/\sqrt{2}, 1/\sqrt{2}).$$

With $Q = [q_1 \ q_2]$, the orthogonal projector is

$$P = QQ^T.$$

For $v = (2, 3, 1)$,

$$Pv = 2q_1 + \frac{4}{\sqrt{2}}q_2 = (2, 2, 2).$$

The residual $(0, 1, -1)$ has zero inner product with both q_1, q_2 , checking that it lies in $U^\perp$. Also $P^2 = P$, as every orthogonal projector must satisfy.

14.4 Best approximation

$P_U v$ is the unique vector in U minimizing $\|v - u\|$.

14.5 QR factorization

Applying Gram–Schmidt to independent columns of A produces $A = QR$, with Q having orthonormal columns and R upper triangular.

14.6 Least squares

For full-column-rank A , the least-squares solution satisfies $A^*Ax = A^*b$ and can be computed more stably using QR.

Example 14.3 (Least squares from QR and residual geometry). Fit $ax \approx b$ with

$$A = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}.$$

The unit column is $q = A/\sqrt{3}$, so the least-squares fitted vector is

$$qq^T b = \frac{7}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Thus $x = 7/3$. The residual is

$$r = b - Ax = \left(-\frac{4}{3}, -\frac{1}{3}, \frac{5}{3} \right)^T,$$

and $A^T r = 0$, which verifies the normal-equation condition. The method therefore agrees with the geometric statement that the residual is orthogonal to the column space.

14.7 Core structural results

Theorem 14.4 (Gram–Schmidt theorem). *Every finite independent list in an inner-product space can be replaced by an orthonormal list with the same successive spans.*

Proof. At step k , subtract the projection of v_k onto the span already constructed. Independence guarantees the remainder is nonzero; normalization finishes the step. $\square$

Theorem 14.5 (Projection theorem). *For finite-dimensional U , every v decomposes uniquely as $v = P_U v + (v - P_U v)$ with the first term in U and the second in $U^\perp$.*

Proof. Use an orthonormal basis of U to define $P_U v$. Direct calculation shows the residual is orthogonal to every basis vector and hence to U . $\square$

14.8 Worked examples

Example 14.6 (Two planar vectors). Starting from $(1, 0)$ and $(1, 1)$, Gram–Schmidt returns $(1, 0)$ and $(0, 1)$.

Example 14.7 (Projection onto a line). Projection onto the line spanned by unit u is $P(v) = \langle v, u \rangle u$.

Example 14.8 (Least squares line fitting). The columns 1 and t form a design matrix; projection of data onto their column space gives the best affine fit.

14.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

14.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 14.9 (Gram–Schmidt by hand). Apply Gram–Schmidt to $v_1 = (1, 1, 0)$ and $v_2 = (1, 0, 1)$.

Solution. Set $e_1 = v_1/\sqrt{2}$. The projection coefficient is $\langle v_2, e_1 \rangle = 1/\sqrt{2}$, so $u_2 = v_2 - \frac{1}{2}(1, 1, 0) = (1/2, -1/2, 1)$. Its norm is $\sqrt{3/2}$, hence $e_2 = u_2/\sqrt{3/2}$. $\square$

Problem 14.10 (Projection onto a line). Project $v = (2, 3)$ onto the line spanned by $u = (1, 1)$.

Solution. Using the unit vector $e = u/\sqrt{2}$, $P(v) = \langle v, e \rangle e = (5/\sqrt{2})(1/\sqrt{2}, 1/\sqrt{2}) = (5/2, 5/2)$. $\square$

Problem 14.11 (Projection matrix). Find the projection matrix onto the line spanned by a unit vector u .

Solution. $P = uu^T$ in the real case. Indeed $Pv = u(u^T v)$ is exactly the scalar projection coefficient times u . $\square$

Problem 14.12 (Best approximation). Why is $P_U v$ the unique closest vector in U to v ?

Solution. For $u \in U$, write $v - u = (v - P_U v) + (P_U v - u)$ with orthogonal summands. Pythagoras gives $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2$, minimized uniquely at $u = P_U v$. $\square$

Problem 14.13 (QR factorization). What does Gram–Schmidt do to the columns of a full-rank matrix A ?

Solution. It produces orthonormal columns Q and expresses each original column as a combination of the current and previous orthonormal columns. The coefficient matrix is upper triangular R , giving $A = QR$. $\square$

Problem 14.14 (Normal equations). Derive the least-squares normal equations.

Solution. At the minimizer, the residual $b - Ax$ is orthogonal to the column space of A . Therefore $A^*(b - Ax) = 0$, which rearranges to $A^*Ax = A^*b$. $\square$

Problem 14.15 (Why QR is stabler). Why is solving least squares through QR often preferable to normal equations?

Solution. Forming A^*A squares the spectral condition number, magnifying sensitivity. Orthogonal transformations used in QR have condition number 1 and preserve norms, so they avoid that amplification. $\square$

Problem 14.16 (Projection idempotence). Prove an orthogonal projection satisfies $P^2 = P$.

Solution. After one projection, Pv lies in the target subspace. Projecting a vector already in that subspace leaves it fixed, hence $P(Pv) = Pv$. $\square$

Problem 14.17 (Complementary projection). Show $I - P_U$ is the orthogonal projection onto $U^\perp$.

Solution. The orthogonal decomposition is $v = P_U v + (v - P_U v)$. The second term lies in $U^\perp$ and equals $(I - P_U)v$. Vectors already in $U^\perp$ are fixed. $\square$

Problem 14.18 (Orthonormal coordinates). If $e_1, \dots, e_n$ is orthonormal, why is the coefficient of e_j in v equal to $\langle v, e_j \rangle$?

Solution. Write $v = \sum a_i e_i$ and inner-product with e_j . Orthonormality kills all terms except $a_j \langle e_j, e_j \rangle = a_j$. $\square$

Problem 14.19 (Rank-deficient least squares). What changes if the columns of A are dependent?

Solution. The least-squares projection onto the column space still exists uniquely, but the coefficient vector x need not be unique because nonzero kernel vectors can be added without changing Ax . The pseudoinverse selects the minimum-norm solution. $\square$

Problem 14.20 (Projection synthesis). Connect Gram–Schmidt, QR, and least squares in one sentence.

Solution. Gram–Schmidt constructs an orthonormal basis of the column space, QR records that construction algebraically, and least squares projects data onto that same column space using the orthonormal coordinates. $\square$

14.11 Exercises with complete solutions

Exercise 14.21. Apply Gram–Schmidt to $(1, 1)$ and $(1, -1)$.

Hint. At the next Gram–Schmidt step, subtract projections onto the orthonormal vectors already obtained before normalizing the residual. At each Gram–Schmidt step, subtract only the components along the orthonormal vectors already constructed. $\square$

Solution. They are already orthogonal; normalize each by $\sqrt{2}$. $\square$

Exercise 14.22. Project $(2, 3)$ onto the x -axis.

Hint. For projection onto a line spanned by unit u , start with $P(v) = \langle v, u \rangle u$. $\square$

Solution. The projection is $(2, 0)$. $\square$

Exercise 14.23. Why is the projection residual orthogonal to the subspace?

Hint. Write the relevant inner products explicitly and use bilinearity or sesquilinearity before simplifying norms or angles. Compute the residual $v - P_U v$ and test its inner product with every basis vector of U . $\square$

Solution. Subtracting all orthonormal-basis components makes its inner product with each basis vector zero. $\square$

Exercise 14.24. Prove the best-approximation property.

Hint. Decompose the error into the projection residual and a vector inside the subspace; orthogonality turns the norm square into a Pythagorean sum. Decompose $v - u = (v - P_U v) + (P_U v - u)$; then use orthogonality and Pythagoras. $\square$

Solution. For $u \in U$, write $v - u = (v - P_U v) + (P_U v - u)$; the two terms are orthogonal, so Pythagoras is minimized exactly when $u = P_U v$. $\square$

Exercise 14.25. What are the dimensions of Q and R for an $m \times n$ matrix of full column rank?

Hint. Choose a basis adapted to the invariant flag; the vanishing entries below the diagonal encode preservation of the successive subspaces. In reduced QR for an $m \times n$ full-column-rank matrix, count the orthonormal columns of Q before deciding dimensions. $\square$

Solution. In reduced QR, Q is $m \times n$ with orthonormal columns and R is invertible $n \times n$ upper triangular. $\square$

Exercise 14.26. Why can normal equations be less stable than QR?

Hint. Use that the least-squares residual must be orthogonal to the column space, so apply A^* to $b - Ax$. Compare $\kappa(A^*A)$ with $\kappa(A)$; forming the normal equations squares the condition number. $\square$

Solution. Forming A^*A squares the condition number, amplifying numerical sensitivity. $\square$

Exercise 14.27. Show an orthogonal projection satisfies $P^2 = P$.

Hint. After the first projection, the vector already lies in the target subspace; project that vector one more time. After one projection the vector is already in U ; apply P once more. $\square$

Solution. Once a vector lies in U , projecting again leaves it unchanged; hence $P(Pv) = Pv$. $\square$

Exercise 14.28. Show $I - P_U$ projects onto $U^\perp$.

Hint. Use the orthogonal decomposition $v = P_U v + (v - P_U v)$ and identify where the residual lies. Use $v = P_U v + (v - P_U v)$ and identify the second term as the $U^\perp$ component. $\square$

Solution. The decomposition theorem gives $v - P_U v \in U^\perp$, and vectors already in $U^\perp$ are fixed by $I - P_U$. $\square$

14.12 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 14.29 (Gram–Schmidt pair). Apply Gram–Schmidt to $(1, 1, 0)$ and $(1, 0, 1)$.

Hint. Normalize the first vector, subtract its projection from the second, then normalize. $\square$

Solution. One obtains $e_1 = (1, 1, 0)/\sqrt{2}$ and $e_2 = (1, -1, 2)/\sqrt{6}$. $\square$

Exercise 14.30 (Projection onto a line). Project $v = (3, 1)$ onto $\text{span}(1, 2)$.

Hint. Use $P_U v = \langle v, u \rangle u / \langle u, u \rangle$. $\square$

Solution. The coefficient is $5/5 = 1$, so the projection is $(1, 2)$. $\square$

Exercise 14.31 (Projection matrix). Find the matrix projecting $\mathbb{R}^2$ onto $\text{span}(1, 1)$.

Hint. Use the unit vector $u = (1, 1)/\sqrt{2}$ and $P = uu^T$. $\square$

Solution. $P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. $\square$

Exercise 14.32 (Reduced QR). Find a reduced QR factorization of $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$.

Hint. Apply Gram–Schmidt to the two columns. □

Solution. Here $q_1 = e_1$, $u_2 = (0, 1, 0)$, $q_2 = e_2$; $Q = [e_1, e_2]$, $R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. □

Exercise 14.33 (Normal equation). Solve the least-squares problem for $A = (1, 1, 1)^T$, $b = (1, 2, 4)^T$.

Hint. Solve $A^T A x = A^T b$. □

Solution. $3x = 7$, so $x = 7/3$. □

Proofs

Exercise 14.34 (Gram–Schmidt span preservation). Prove each Gram–Schmidt step preserves the span of the vectors processed so far.

Hint. The new residual differs from the new original vector by a combination of earlier vectors. □

Solution. Each residual lies in the old span, while the original vector is recovered from the residual plus earlier projection terms; hence the successive spans agree. □

Exercise 14.35 (Projection idempotence). Prove an orthogonal projection satisfies $P^2 = P$.

Hint. After one projection the vector already lies in the target subspace. □

Solution. For every v , $Pv \in U$, and projection fixes every vector of U ; hence $P(Pv) = Pv$. □

Exercise 14.36 (Projection self-adjointness). Prove $P_U^* = P_U$.

Hint. Use an orthonormal basis q_i and the formula $P_U v = \sum \langle v, q_i \rangle q_i$. □

Solution. Expanding both $\langle P_U v, w \rangle$ and $\langle v, P_U w \rangle$ gives the same sum. □

Exercise 14.37 (Best approximation). Prove $P_U v$ uniquely minimizes $\|v - u\|$ over $u \in U$.

Hint. Decompose $v - u = (v - P_U v) + (P_U v - u)$. □

Solution. The two terms are orthogonal, so Pythagoras gives a fixed residual term plus $\|P_U v - u\|^2$, minimized only at $u = P_U v$. □

Counterexamples and hypothesis testing

Exercise 14.38 (Dependent Gram–Schmidt input). What fails if Gram–Schmidt is applied to a dependent list?

Hint. At some step the residual can become zero. □

Solution. For $v_2 = 2v_1$, subtracting the projection of v_2 onto v_1 leaves zero, which cannot be normalized. □

Exercise 14.39 (Normal equations conditioning). Why can forming $A^T A$ be a poor numerical choice?

Hint. Compare singular values of A and $A^T A$. □

Solution. The condition number is squared: $\kappa_2(A^T A) = \kappa_2(A)^2$ for full column rank. □

Exercise 14.40 (Oblique idempotent). Give an idempotent matrix that is not an orthogonal projector.

Hint. Try an upper-triangular rank-one projection that is not symmetric. □

Solution. $P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ satisfies $P^2 = P$ but $P^T \neq P$, so it is not an orthogonal projection. □

Applications and investigations

Exercise 14.41 (Least-squares line). Explain how fitting $y = a + bt$ to data becomes a projection problem.

Hint. Use design columns 1 and t . □

Solution. The fitted data vector is the orthogonal projection of the observation vector onto the column space of the design matrix. □

Exercise 14.42 (QR stability). Why are orthogonal transformations used in numerical least squares?

Hint. Recall their Euclidean condition number and norm preservation. □

Solution. Orthogonal transformations have condition number 1, so they reorganize the problem without amplifying Euclidean perturbations. □

Challenge problems

Exercise 14.43 (Projection characterization). Prove a real matrix P is an orthogonal projector iff $P^2 = P$ and $P^T = P$.

Hint. Use the image/kernel decomposition and show $\ker P = (\operatorname{im} P)^\perp$. □

Solution. Idempotence gives $V = \operatorname{im} P \oplus \ker P$; symmetry gives orthogonality of these spaces, so P is the orthogonal projection onto its image. The converse is standard. □

Exercise 14.44 (Least-squares solution set). If A is rank deficient and x_0 is one least-squares solution, describe all coefficient vectors giving the same fitted vector.

Hint. Add vectors in $\ker A$. □

Solution. All are $x_0 + z$ with $z \in \ker A$; they produce the same Ax . The pseudoinverse selects the minimum-norm member. □

Chapter summary

Gram–Schmidt converts an arbitrary independent list into an orthonormal list spanning the same nested subspaces. Orthogonal projection then gives canonical decompositions and solves least-squares problems. This chapter connects exact geometry with the numerical algorithms QR factorization and normal equations. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 15

Orthogonal and Unitary Operators

Orthogonal and unitary operators are the linear symmetries of inner-product geometry. They preserve lengths, angles and orthogonality, and their matrices satisfy $Q^*Q = I$. Reflections, rotations and Householder transformations show how geometric symmetries become structured matrices.

Learning goals

After this chapter, the reader should be able to:

- test whether a real or complex matrix is orthogonal or unitary using $A^*A = I$;
- prove that orthogonal and unitary operators preserve inner products, norms, and angles;
- compute the inverse of an orthogonal or unitary matrix from its adjoint;
- classify simple planar orthogonal maps as rotations or reflections;
- show that every eigenvalue of a unitary operator has modulus one;
- construct examples distinguishing orthogonal/unitary operators from merely diagonalizable ones;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition	→	structural theorem	→	coordinate calculation	→	application.
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Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

15.1 Isometries

A linear map is an isometry when it preserves norms, equivalently inner products.

15.2 Orthogonal and unitary matrices

A real matrix is orthogonal when $Q^T Q = I$; a complex matrix is unitary when $Q^* Q = I$.

Example 15.1 (Testing a matrix by orthonormal columns). Consider

$$Q = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}.$$

Its columns have norm 1 and inner product 0, hence $Q^T Q = I$. Therefore Q is orthogonal and

$$Q^{-1} = Q^T.$$

Its determinant is 1, so it is orientation preserving. The entries match $\cos \theta = 3/5$, $\sin \theta = 4/5$, identifying the map as a rotation. The norm and determinant checks agree with the geometric interpretation.

15.3 Equivalent characterizations

For a square matrix, orthonormal columns, orthonormal rows and inverse equal to adjoint are equivalent.

15.4 Determinant and orientation

A real orthogonal matrix has determinant ± 1 . Determinant $+1$ gives orientation-preserving transformations.

15.5 Reflections and Householder maps

For unit u , $H = I - 2uu^*$ reflects across the hyperplane $u^\perp$.

Example 15.2 (A Householder reflection constructed from a normal). Let $u = (1, 1)/\sqrt{2}$. Then

$$H = I - 2uu^T = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

We have

$$Hu = -u,$$

while every vector orthogonal to u , such as $(1, -1)$, is fixed. Also $H^T H = H^2 = I$. Thus the algebraic formula and the geometry agree: H reflects across the line $u^\perp = \text{span}(1, -1)$.

15.6 Rotations and planar blocks

Real orthogonal operators decompose into invariant one- and two-dimensional pieces, with planar rotation blocks on the latter.

Example 15.3 (Classifying a planar orthogonal map). Let

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its columns are orthonormal, so $Q \in O(2)$, but

$$\det Q = -1.$$

Hence it cannot be a rotation. The vector $(1, 1)$ is fixed and $(1, -1)$ changes sign, so Q is reflection across the line $y = x$.

This illustrates a reliable planar classification: first verify orthogonality, then use determinant and fixed directions to distinguish rotations from reflections.

15.7 Core structural results

Theorem 15.4 (Inner-product preservation criterion). *A linear map preserves inner products if and only if it preserves norms.*

Proof. One direction is immediate. For the converse, recover the inner product from the norm using the polarization identity. $\square$

Theorem 15.5 (Adjoint inverse criterion). *A matrix Q is unitary exactly when $Q^{-1} = Q^*$.*

Proof. This is simply a rearrangement of $Q^*Q = I$, with invertibility automatic. $\square$

15.8 Worked examples

Example 15.6 (Planar rotation). The standard rotation matrix is orthogonal with determinant 1.

Example 15.7 (Reflection). $\text{diag}(1, -1)$ reflects across the x -axis and has determinant -1 .

Example 15.8 (Permutation matrix). Every permutation matrix is orthogonal because its columns are a rearrangement of the standard orthonormal basis.

15.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

15.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 15.9 (Orthogonal inverse). If $Q^T Q = I$, show $Q^{-1} = Q^T$.

Solution. The equation already says Q^T is a left inverse. For square matrices a left inverse is also a right inverse, so Q is invertible and $Q^{-1} = Q^T$. $\square$

Problem 15.10 (Norm preservation). Show an orthogonal matrix preserves Euclidean norm.

Solution. $\|Qx\|^2 = (Qx)^T(Qx) = x^T Q^T Q x = x^T x = \|x\|^2$. $\square$

Problem 15.11 (Determinant sign). Why does a real orthogonal matrix have determinant ± 1 ?

Solution. Take determinants of $Q^T Q = I$: $(\det Q)^2 = 1$. Hence $\det Q = \pm 1$. $\square$

Problem 15.12 (Householder reflection). For a unit vector u , show $H = I - 2uu^T$ is orthogonal.

Solution. Let $P = uu^T$, so $P^2 = P$ and $P^T = P$. Then $H^T H = H^2 = (I - 2P)^2 = I - 4P + 4P^2 = I$. $\square$

Problem 15.13 (Reflection geometry). What does $H = I - 2uu^T$ do to u and to vectors orthogonal to u ?

Solution. $Hu = u - 2u = -u$. If $v \perp u$, then $u^T v = 0$ and $Hv = v$. Thus H reflects across the hyperplane $u^\perp$. $\square$

Problem 15.14 (Unitary eigenvalues). Why must every eigenvalue of a unitary matrix have modulus 1?

Solution. If $Uv = \lambda v$ and $v \neq 0$, norm preservation gives $\|v\| = \|Uv\| = |\lambda|\|v\|$, so $|\lambda| = 1$. $\square$

Problem 15.15 (Permutation matrices). Why is every permutation matrix orthogonal?

Solution. Its columns are a reordering of the standard orthonormal basis, so they remain orthonormal. Hence $P^T P = I$. $\square$

Problem 15.16 (Orientation). What distinguishes rotations from reflections inside $O(n)$?

Solution. Both preserve inner products, but orientation-preserving orthogonal maps have determinant +1 and form $SO(n)$. Reflections have determinant -1. $\square$

Problem 15.17 (Product of orthogonal matrices). Show the product of two orthogonal matrices is orthogonal.

Solution. If $Q^T Q = I$ and $R^T R = I$, then $(QR)^T (QR) = R^T Q^T QR = R^T R = I$. $\square$

Problem 15.18 (Columns and rows). Why do orthonormal columns of a square matrix imply orthonormal rows?

Solution. Orthonormal columns give $Q^* Q = I$, so Q is invertible with inverse Q^* . Hence also $QQ^* = I$, which says the rows are orthonormal. $\square$

Problem 15.19 (Condition number). What is the Euclidean condition number of an orthogonal matrix?

Solution. All singular values equal 1, so $\kappa_2(Q) = 1$. Orthogonal maps neither stretch nor compress, making them optimally conditioned. $\square$

Problem 15.20 (Symmetry synthesis). Why are orthogonal/unitary transformations the natural coordinate changes for inner-product problems?

Solution. They preserve the inner product, so lengths, angles, orthogonality, numerical norms, and spectral geometry are unchanged. This lets one simplify coordinates without distorting the geometric structure under study. $\square$

15.11 Exercises with complete solutions

Exercise 15.21. Show an orthogonal matrix preserves Euclidean norm.

Hint. Compute $A^* A$ and compare it with I ; this single identity packages preservation of the inner product. Compute $A^* A$; orthogonal or unitary means this product is exactly the identity. $\square$

Solution. $\|Qx\|^2 = x^T Q^T Q x = x^T x$. $\square$

Exercise 15.22. Why do orthogonal matrices have determinant ± 1 ?

Hint. Reduce to triangular form while recording how each row operation changes the determinant. Use $\langle Ux, Uy \rangle = \langle x, y \rangle$ to deduce norm and angle preservation. $\square$

Solution. Taking determinants in $Q^T Q = I$ gives $(\det Q)^2 = 1$. $\square$

Exercise 15.23. Verify a Householder matrix is orthogonal.

Hint. After the first projection, the vector already lies in the target subspace; project that vector one more time. For a unitary matrix, the inverse is A^* ; verify both products equal I . $\square$

Solution. With $P = uu^T$ and $P^2 = P$, $(I - 2P)^2 = I$. Since it is symmetric, its transpose is its inverse. $\square$

Exercise 15.24. What does a unitary operator do to an orthonormal basis?

Hint. Compute A^*A and compare it with I ; this single identity packages preservation of the inner product. In the plane, use determinant ± 1 together with the trace or fixed directions to distinguish rotations from reflections. $\square$

Solution. It sends it to another orthonormal basis because all pairwise inner products are preserved. $\square$

Exercise 15.25. Show the inverse of a unitary matrix is unitary.

Hint. Compute A^*A and compare it with I ; this single identity packages preservation of the inner product. If $Uv = \lambda v$, compare $\|Uv\|$ with $\|v\|$ to constrain $|\lambda|$. $\square$

Solution. If $Q^{-1} = Q^*$, then $(Q^{-1})^* = Q$ and $(Q^{-1})^* Q^{-1} = QQ^* = I$. $\square$

Exercise 15.26. Give an orthogonal matrix with determinant -1 in $\mathbb{R}^3$.

Hint. Compute A^*A and compare it with I ; this single identity packages preservation of the inner product. Columns of a unitary matrix form an orthonormal basis; test them with inner products. $\square$

Solution. For example $\text{diag}(1, 1, -1)$. $\square$

Exercise 15.27. Why is every eigenvalue of a complex unitary matrix of modulus one?

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Products of unitary operators remain unitary because the adjoint reverses multiplication order. $\square$

Solution. If $Qv = \lambda v$, norm preservation gives $\|v\| = \|Qv\| = |\lambda|\|v\|$, so $|\lambda| = 1$. $\square$

Exercise 15.28. Explain why orthogonal change of coordinates is numerically attractive.

Hint. Write the relevant inner products explicitly and use bilinearity or sesquilinearity before simplifying norms or angles. To disprove unitarity, it is enough to find one vector whose norm is not preserved. $\square$

Solution. Its inverse is the transpose and its condition number in the Euclidean norm is 1, so it does not amplify relative errors. $\square$

15.12 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 15.29 (Orthogonality test). Determine whether $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ is orthogonal.

Hint. Compute $Q^T Q$. □

Solution. $Q^T Q = I$, so Q is orthogonal. □

Exercise 15.30 (Inverse). Find the inverse of an orthogonal matrix Q .

Hint. Use the defining equation $Q^T Q = I$. □

Solution. $Q^{-1} = Q^T$. □

Exercise 15.31 (Householder matrix). Construct the Householder reflection for $u = e_1 \in \mathbb{R}^3$.

Hint. Use $H = I - 2uu^T$. □

Solution. $H = \text{diag}(-1, 1, 1)$. □

Exercise 15.32 (Rotation angle). For $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, identify the rotation angle.

Hint. Compare with $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. □

Solution. $\theta = \pi/2$ modulo 2π . □

Exercise 15.33 (Complex unitary). Check that $U = \text{diag}(i, -i)$ is unitary.

Hint. Compute $U^* U$. □

Solution. $U^* = \text{diag}(-i, i)$, and $U^* U = I$. □

Proofs

Exercise 15.34 (Norm preservation). Prove $Q^* Q = I$ implies $\|Qx\| = \|x\|$.

Hint. Compute the squared norm using the inner product. □

Solution. $\|Qx\|^2 = x^* Q^* Q x = x^* x$. □

Exercise 15.35 (Determinant modulus). Prove a real orthogonal matrix has determinant ± 1 .

Hint. Take determinants of $Q^T Q = I$. □

Solution. $(\det Q)^2 = 1$, hence $\det Q = \pm 1$. □

Exercise 15.36 (Unitary eigenvalue modulus). Prove every eigenvalue of a unitary operator has modulus one.

Hint. Apply norm preservation to an eigenvector. □

Solution. If $Uv = \lambda v$, then $\|v\| = \|Uv\| = |\lambda|\|v\|$; since $v \neq 0$, $|\lambda| = 1$. □

Exercise 15.37 (Product closure). Prove the product of unitary matrices is unitary.

Hint. Remember $(UV)^* = V^*U^*$. □

Solution. $(UV)^*(UV) = V^*U^*UV = V^*V = I$. □

Counterexamples and hypothesis testing

Exercise 15.38 (Diagonalizable not orthogonal). Give a diagonalizable real matrix that is not orthogonal.

Hint. Use a non-unit scaling diagonal matrix. □

Solution. $\text{diag}(2, 1)$ is diagonal but does not preserve norm. □

Exercise 15.39 (Determinant one is insufficient). Give a real matrix with determinant 1 that is not orthogonal.

Hint. Use a shear. □

Solution. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has determinant 1 but its columns are not orthonormal. □

Exercise 15.40 (Norm of one vector). Explain why preserving the norm of one nonzero vector does not prove a map is orthogonal.

Hint. Construct a map fixing one direction but stretching another. □

Solution. $\text{diag}(1, 2)$ preserves $\|e_1\|$ but doubles e_2 . □

Applications and investigations

Exercise 15.41 (Rigid coordinate change). Why are orthogonal changes of coordinates natural in Euclidean geometry?

Hint. List the geometric quantities preserved by the inner product. □

Solution. They preserve lengths, distances, angles and orthogonality, so coordinate simplification does not distort geometry. □

Exercise 15.42 (Householder QR). Why are Householder reflections effective for QR factorization?

Hint. Use their ability to map a vector to a coordinate axis while preserving norm. □

Solution. A Householder reflector can zero all entries below a chosen pivot in one stable orthogonal step. □

Challenge problems

Exercise 15.43 (Planar classification). Prove every real 2×2 orthogonal matrix with determinant 1 has rotation form.

Hint. Let the first unit column be (c, s) and determine the second using orientation. $\square$

Solution. Orthogonality forces the second column to be $(-s, c)$, with $c^2 + s^2 = 1$, so $c = \cos \theta, s = \sin \theta$. $\square$

Exercise 15.44 (Reflection eigenstructure). For a Householder reflection $H = I - 2uu^T$, determine all eigenvalues and eigenspaces.

Hint. Split space into $\text{span}(u) \oplus u^\perp$. $\square$

Solution. -1 occurs on $\text{span}(u)$; $+1$ occurs on $u^\perp$. $\square$

Chapter summary

Orthogonal and unitary operators are the linear symmetries of inner-product geometry. They preserve lengths, angles and orthogonality, and their matrices satisfy $Q^*Q = I$. Reflections, rotations and Householder transformations show how geometric symmetries become structured matrices. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 16

The Spectral Theorem

The spectral theorem is the central structure theorem for self-adjoint, symmetric and normal operators. It says that the right compatibility with the inner product upgrades triangularization to orthonormal diagonalization. Functional calculus and positive operators then become simple operations on eigenvalues.

Learning goals

After this chapter, the reader should be able to:

- recognize self-adjoint, normal, and unitary operators from matrix identities;
- find an orthonormal eigenbasis for a concrete self-adjoint or normal matrix;
- unitarily or orthogonally diagonalize an operator covered by the spectral theorem;
- write a spectral decomposition in terms of eigenspace projections;
- use orthogonal eigenspaces to simplify powers and functions of a normal operator;
- identify precisely which hypothesis fails when an operator does not admit an orthonormal eigenbasis;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition	→	structural theorem	→	coordinate calculation	→	application.
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Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

16.1 Adjoints

The adjoint T^* is characterized by $\langle Tv, w \rangle = \langle v, T^*w \rangle$.

16.2 Self-adjoint and normal operators

T is self-adjoint when $T = T^*$ and normal when $TT^* = T^*T$.

Example 16.1 (Self-adjoint, normal, or neither?). For

$$A = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix},$$

the conjugate transpose is

$$A^* = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix} = A.$$

Hence A is self-adjoint and therefore normal. Its characteristic polynomial has real coefficients and real roots, as the spectral theorem predicts. The first check should always be the structural identity $A^* = A$, because it immediately unlocks real spectrum and orthonormal diagonalization.

16.3 Reality of the spectrum

Eigenvalues of self-adjoint operators are real, and eigenspaces for distinct eigenvalues are orthogonal.

16.4 Spectral theorem

A finite-dimensional complex operator is normal if and only if it is unitarily diagonalizable. Over $\mathbb{R}$, symmetric operators are orthogonally diagonalizable.

Example 16.2 (Full orthogonal spectral decomposition). For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

the eigenvalues are 3 and 1, with orthonormal eigenvectors

$$q_1 = \frac{1}{\sqrt{2}}(1, 1), \quad q_2 = \frac{1}{\sqrt{2}}(1, -1).$$

Thus

$$A = 3q_1q_1^T + 1q_2q_2^T.$$

Multiplying the two rank-one projectors out recovers A , which checks the decomposition. The formula separates the operator into mutually orthogonal spectral modes.

16.5 Functional calculus

If $T = UDU^*$, define $f(T) = Uf(D)U^*$ for functions defined on the spectrum.

Example 16.3 (Functional calculus without repeated multiplication). With the same matrix A , compute A^{10} . The spectral decomposition gives

$$A^{10} = 3^{10}q_1q_1^T + q_2q_2^T.$$

No ten-fold matrix multiplication is needed. More generally, for any function defined on the spectrum,

$$f(A) = f(3)q_1q_1^T + f(1)q_2q_2^T.$$

This is the computational force of the spectral theorem: once eigenspaces are known, powers, exponentials, square roots and other functions act only on eigenvalues.

16.6 Positive operators

A self-adjoint operator is positive semidefinite when $\langle Tv, v \rangle \geq 0$; spectrally this means all eigenvalues are nonnegative.

16.7 Core structural results

Theorem 16.4 (Orthogonality of self-adjoint eigenspaces). *Eigenvectors of a self-adjoint operator associated with distinct eigenvalues are orthogonal.*

Proof. If $Tv = \lambda v$ and $Tw = \mu w$, then $\lambda \langle v, w \rangle = \langle Tv, w \rangle = \langle v, Tw \rangle = \mu \langle v, w \rangle$. $\square$

Theorem 16.5 (Finite-dimensional spectral theorem). *Every self-adjoint operator on a finite-dimensional inner-product space has an orthonormal basis of eigenvectors.*

Proof. Choose an eigenvector, take its orthogonal complement, show that complement is invariant under self-adjointness, and induct on dimension. $\square$

16.8 Worked examples

Example 16.6 (Real symmetric matrix). $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has orthogonal eigenvectors $(1, 1)$ and $(1, -1)$ with eigenvalues 3 and 1.

Example 16.7 (Positive square root). For positive $A = UDU^*$, set $A^{1/2} = UD^{1/2}U^*$; then $A^{1/2}$ is positive and squares to A .

Example 16.8 (Normal but not self-adjoint). A nontrivial complex unitary rotation is normal but may not be self-adjoint.

16.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

16.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 16.9 (Adjoint matrix). In the standard complex inner product, what matrix represents the adjoint of A ?

Solution. The adjoint is represented by the conjugate transpose $A^* = \overline{A}^T$, because $\langle Ax, y \rangle = x^* A^* y = \langle x, A^* y \rangle$. $\square$

Problem 16.10 (Real eigenvalues). Prove an eigenvalue of a self-adjoint operator is real.

Solution. If $Tv = \lambda v$, then $\lambda\|v\|^2 = \langle Tv, v \rangle = \langle v, Tv \rangle = \bar{\lambda}\|v\|^2$. Since $v \neq 0$, $\lambda = \bar{\lambda}$. $\square$

Problem 16.11 (Orthogonal eigenspaces). Why are eigenspaces of a self-adjoint operator for distinct eigenvalues orthogonal?

Solution. If $Tv = \lambda v$ and $Tw = \mu w$, self-adjointness gives $\lambda\langle v, w \rangle = \langle Tv, w \rangle = \langle v, Tw \rangle = \mu\langle v, w \rangle$. Distinctness forces the inner product to vanish. $\square$

Problem 16.12 (Diagonalizing a symmetric matrix). Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ orthogonally.

Solution. Eigenvalues are 3 and 1 with eigenvectors $(1, 1)$ and $(1, -1)$. Normalizing them gives an orthogonal matrix Q with $Q^T A Q = \text{diag}(3, 1)$. $\square$

Problem 16.13 (Positive square root). How does the spectral theorem construct $A^{1/2}$ for positive semidefinite A ?

Solution. Write $A = U \text{diag}(\lambda_i) U^*$ with $\lambda_i \geq 0$. Define $A^{1/2} = U \text{diag}(\sqrt{\lambda_i}) U^*$. It is positive and squares to A . $\square$

Problem 16.14 (Normal operator). Why is every unitary operator normal?

Solution. For unitary U , $U^*U = UU^* = I$, so $UU^* = U^*U$. $\square$

Problem 16.15 (Polynomial functional calculus). If $T = UDU^*$ is normal, compute $p(T)$.

Solution. $p(T) = Up(D)U^*$, where $p(D)$ is obtained by applying p to each diagonal eigenvalue. This follows because powers of T preserve the same diagonalizing basis. $\square$

Problem 16.16 (Positivity and spectrum). Why is a self-adjoint operator positive semidefinite iff all eigenvalues are nonnegative?

Solution. In an orthonormal eigenbasis, $\langle Tv, v \rangle = \sum_i \lambda_i |v_i|^2$. This is nonnegative for all v exactly when each $\lambda_i \geq 0$. $\square$

Problem 16.17 (Invariant orthogonal complement). If U is invariant under self-adjoint T , show $U^\perp$ is invariant.

Solution. For $w \in U^\perp$ and $u \in U$, $\langle Tw, u \rangle = \langle w, Tu \rangle = 0$ because $Tu \in U$. Hence $Tw \in U^\perp$. $\square$

Problem 16.18 (Spectral norm). For self-adjoint T , what is $\|T\|_2$ in terms of eigenvalues?

Solution. Because the operator is unitarily diagonalizable, its singular values are $|\lambda_i|$. Therefore the operator norm is $\max_i |\lambda_i|$. $\square$

Problem 16.19 (Spectral projector). How can the projection onto an eigenspace be expressed when eigenvalues are distinct?

Solution. Use Lagrange interpolation: $P_i = \prod_{j \neq i} (T - \lambda_j I) / (\lambda_i - \lambda_j)$. It acts as 1 on the λ_i eigenspace and 0 on the others. $\square$

Problem 16.20 (Spectral synthesis). Why is the spectral theorem stronger than ordinary diagonalization?

Solution. It provides an orthonormal eigenbasis, not merely an arbitrary eigenbasis. Consequently the change-of-basis matrix is unitary, preserving the inner product and making geometric and numerical conclusions immediately available. $\square$

16.11 Exercises with complete solutions

Exercise 16.21. Show eigenvalues of a self-adjoint operator are real.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Check the adjoint identity first: self-adjoint means $A^* = A$, normal means $A^*A = AA^*$. $\square$

Solution. If $Tv = \lambda v$, then $\lambda\|v\|^2 = \langle Tv, v \rangle = \langle v, Tv \rangle = \bar{\lambda}\|v\|^2$. $\square$

Exercise 16.22. Diagonalize $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ orthogonally.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. For a self-adjoint matrix, compute eigenspaces and then orthonormalize within repeated-eigenvalue eigenspaces. $\square$

Solution. Normalize eigenvectors $(1, 1)$ and $(1, -1)$ to form Q ; then $Q^T A Q = \text{diag}(3, 1)$. $\square$

Exercise 16.23. Why is every unitary operator normal?

Hint. Compute A^*A and compare it with I ; this single identity packages preservation of the inner product. Eigenvectors for distinct eigenvalues of a self-adjoint operator are orthogonal; use that before Gram-Schmidt. $\square$

Solution. $UU^* = U^*U = I$. $\square$

Exercise 16.24. Show a positive semidefinite operator has nonnegative eigenvalues.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Write the operator as $\sum_{\lambda} \lambda P_{\lambda}$, where P_{λ} projects onto the eigenspace. $\square$

Solution. For an eigenvector v , $\langle Tv, v \rangle = \lambda\|v\|^2 \geq 0$. $\square$

Exercise 16.25. Define T^2 using spectral functional calculus.

Hint. To compute $f(T)$, apply f to each eigenvalue in the spectral decomposition. $\square$

Solution. If $T = UDU^*$ then $T^2 = UD^2U^*$, corresponding to $f(t) = t^2$. $\square$

Exercise 16.26. Why is the orthogonal complement of an invariant eigenspace also invariant for a self-adjoint operator?

Hint. Use the orthogonal decomposition $v = P_U v + (v - P_U v)$ and identify where the residual lies. Normality is exactly the hypothesis for unitary diagonalization over $\mathbb{C}$. $\square$

Solution. If w is orthogonal to the eigenspace and v is in it, then $\langle Tw, v \rangle = \langle w, Tv \rangle = \lambda\langle w, v \rangle = 0$. $\square$

Exercise 16.27. What extra conclusion does normality give beyond triangularization?

Hint. Compute A^*A and compare it with I ; this single identity packages preservation of the inner product. Check that the chosen eigenvectors are not merely independent but orthonormal. $\square$

Solution. A normal operator can be diagonalized by a unitary basis change, so the basis can be chosen orthonormal. $\square$

Exercise 16.28. Relate spectral theorem to quadratic forms.

Hint. Find the eigenspaces first and use orthogonality between distinct eigenspaces before assembling an orthonormal eigenbasis. If a matrix is triangular and normal, compare off-diagonal norm contributions to show it must be diagonal. $\square$

Solution. A real symmetric matrix can be orthogonally diagonalized, reducing its quadratic form to a sum of scalar multiples of squares. $\square$

16.12 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 16.29 (Adjoint). Compute A^* for $A = \begin{pmatrix} 1 & i \\ 2 & 3i \end{pmatrix}$.

Hint. Transpose and complex-conjugate every entry. $\square$

Solution. $A^* = \begin{pmatrix} 1 & 2 \\ -i & -3i \end{pmatrix}$. $\square$

Exercise 16.30 (Self-adjoint test). Is $\begin{pmatrix} 2 & i \\ -i & 1 \end{pmatrix}$ self-adjoint?

Hint. Compare the matrix with its conjugate transpose. $\square$

Solution. Yes; the off-diagonal entries are complex conjugates. $\square$

Exercise 16.31 (Orthogonal diagonalization). Orthogonally diagonalize $\begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$.

Hint. Use eigenvectors $(1, 1)$ and $(1, -1)$ and normalize them. $\square$

Solution. The eigenvalues are 6, 2; with normalized eigenvectors as columns of Q , $Q^T A Q = \text{diag}(6, 2)$. $\square$

Exercise 16.32 (Spectral projector). For distinct eigenvalues 2, 5, write the projector onto the 5-eigenspace as a polynomial in T .

Hint. Use Lagrange interpolation. $\square$

Solution. $P_5 = (T - 2I)/(5 - 2) = (T - 2I)/3$. $\square$

Exercise 16.33 (Positive square root). If $A = Q \text{diag}(9, 4)Q^T$ is positive, write $A^{1/2}$.

Hint. Apply the square root to eigenvalues. $\square$

Solution. $A^{1/2} = Q \text{diag}(3, 2)Q^T$. $\square$

Proofs

Exercise 16.34 (Real spectrum). Prove every eigenvalue of a self-adjoint operator is real.

Hint. Compare $\langle Tv, v \rangle$ with its conjugate using self-adjointness. □

Solution. For $Tv = \lambda v$, $\lambda\|v\|^2 = \overline{\lambda}\|v\|^2$, so $\lambda = \overline{\lambda}$. □

Exercise 16.35 (Orthogonal eigenspaces). Prove eigenspaces of a self-adjoint operator for distinct eigenvalues are orthogonal.

Hint. Compute $\langle Tv, w \rangle$ in two ways. □

Solution. $(\lambda - \mu)\langle v, w \rangle = 0$; distinct eigenvalues force $\langle v, w \rangle = 0$. □

Exercise 16.36 (Invariant orthogonal complement). If U is invariant under self-adjoint T , prove $U^\perp$ is invariant.

Hint. For $w \in U^\perp, u \in U$, move T across the inner product. □

Solution. $\langle Tw, u \rangle = \langle w, Tu \rangle = 0$ because $Tu \in U$. □

Exercise 16.37 (Positivity from spectrum). Prove a self-adjoint operator is positive semidefinite iff all eigenvalues are nonnegative.

Hint. Expand a vector in an orthonormal eigenbasis. □

Solution. $\langle Tv, v \rangle = \sum \lambda_i |c_i|^2$, which is nonnegative for all coefficients exactly when every $\lambda_i \geq 0$. □

Counterexamples and hypothesis testing

Exercise 16.38 (Normal not self-adjoint). Give a normal complex matrix that is not self-adjoint.

Hint. Use a non-real unitary diagonal matrix. □

Solution. $\text{diag}(i, 1)$ is unitary and normal but not equal to its adjoint. □

Exercise 16.39 (Diagonalizable not normal). Give a diagonalizable matrix that is not normal.

Hint. Conjugate a real diagonal matrix by a non-orthogonal basis change. □

Solution. For example $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ has distinct eigenvalues and is diagonalizable, but $AA^T \neq A^T A$. □

Exercise 16.40 (Real spectral theorem limit). Why does the complex normal spectral theorem not imply every real normal matrix is orthogonally diagonalizable over $\mathbb{R}$?

Hint. Consider a nontrivial planar rotation. □

Solution. A 90° rotation is real normal but has no real eigenvalues; over $\mathbb{R}$ it has a rotation block rather than a real diagonal form. □

Applications and investigations

Exercise 16.41 (Matrix exponential). If $A = QDQ^T$ is symmetric, how do you compute e^A ?

Hint. Apply e^t to each eigenvalue. □

Solution. $e^A = Qe^DQ^T$, where e^D is diagonal with entries e^{λ_i} . □

Exercise 16.42 (Covariance matrix). Why does the spectral theorem underlie principal-component analysis for a covariance matrix?

Hint. A covariance matrix is symmetric positive semidefinite. □

Solution. It has orthonormal eigenvectors; leading eigenvectors give orthogonal directions of greatest variance. □

Challenge problems

Exercise 16.43 (Normal triangular matrix). Prove an upper-triangular normal complex matrix is diagonal.

Hint. Compare squared norms of the first row and first column, then induct. □

Solution. Normality forces the off-diagonal entries in the first row to vanish; remove that coordinate and iterate, leaving only diagonal entries. □

Exercise 16.44 (Spectral norm). For self-adjoint T , prove $\|T\|_2 = \max_i |\lambda_i|$.

Hint. Expand a unit vector in an orthonormal eigenbasis and bound the resulting weighted sum. □

Solution. $\|Tv\|^2 = \sum |\lambda_i|^2 |c_i|^2 \leq (\max |\lambda_i|)^2$; equality occurs on an eigenvector attaining the maximum. □

Chapter summary

The spectral theorem is the central structure theorem for self-adjoint, symmetric and normal operators. It says that the right compatibility with the inner product upgrades triangularization to orthonormal diagonalization. Functional calculus and positive operators then become simple operations on eigenvalues. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 17

Quadratic Forms

Quadratic forms encode second-order geometry. Symmetric bilinear forms and symmetric matrices represent the same finite-dimensional data once a basis is chosen. Congruence, not similarity, is the natural basis-change relation, and Sylvester's law of inertia identifies the basis-independent signature.

Learning goals

After this chapter, the reader should be able to:

- associate a symmetric or Hermitian matrix with a quadratic form and recover the form from the matrix;
- diagonalize a quadratic form by an orthogonal or unitary change of coordinates when appropriate;
- classify a real quadratic form as positive definite, negative definite, semidefinite, or indefinite;
- apply eigenvalues or Sylvester-type criteria to decide definiteness;
- compute the signature and identify its invariance under congruence;
- transform a quadratic expression to principal-axis form and interpret the resulting geometry;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition $\longrightarrow$ structural theorem $\longrightarrow$ coordinate calculation $\longrightarrow$ application.

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

17.1 Quadratic and bilinear forms

A quadratic form on a real vector space has the form $q(v) = B(v, v)$ for a symmetric bilinear form B .

17.2 Matrix representation

In coordinates, $q(x) = x^T A x$ with A symmetric. Cross terms appear twice the corresponding off-diagonal entry.

Example 17.1 (Extracting the symmetric matrix correctly). For

$$q(x, y) = 3x^2 + 4xy + 2y^2,$$

the symmetric matrix is

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 2 \end{pmatrix},$$

because

$$[x \ y] A \begin{pmatrix} x \\ y \end{pmatrix} = 3x^2 + (2 + 2)xy + 2y^2.$$

The factor of two in the cross term is the main bookkeeping point. Multiplying out $x^T A x$ is the quickest check that the matrix has been extracted correctly.

17.3 Polarization

Over characteristic not 2, the symmetric bilinear form is recovered from q by $B(u, v) = \frac{1}{2}(q(u + v) - q(u) - q(v))$.

17.4 Congruence

Under coordinate change $x = Py$, the matrix transforms as $A \mapsto P^T A P$.

Example 17.2 (Congruence and completing the square). Consider

$$q(x, y) = x^2 + 4xy + y^2.$$

Set $u = x + 2y$, so $x = u - 2y$. Then

$$q = (u - 2y)^2 + 4(u - 2y)y + y^2 = u^2 - 3y^2.$$

Thus an invertible change of variables converts the form to one positive and one negative square. The inertia is therefore $(1, 1, 0)$.

This is congruence geometry, not similarity: both copies of the coordinate vector change in $x^T A x$.

17.5 Sylvester law of inertia

Every real symmetric quadratic form is congruent to a diagonal form with entries $1, -1, 0$, and the counts of positive, negative and zero entries are invariant.

17.6 Definiteness

A symmetric form is positive definite exactly when all eigenvalues are positive; equivalent criteria include positivity of leading principal minors.

Example 17.3 (Definiteness by two independent tests). For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

the leading principal minors are $2 > 0$ and $3 > 0$, so Sylvester's criterion says the form is positive definite. Independently, the eigenvalues are 3 and 1, both positive.

The two tests agree. The eigenvalue method explains the geometry after orthogonal diagonalization; the principal-minor method can be faster when the matrix is already explicit.

17.7 Core structural results

Theorem 17.4 (Polarization identity). *For a real quadratic form $q(v) = B(v, v)$ with symmetric bilinear B , $B(u, v) = \frac{1}{2}(q(u+v) - q(u) - q(v))$.*

Proof. Expand $q(u+v) = B(u, u) + 2B(u, v) + B(v, v)$ and rearrange. □

Theorem 17.5 (Sylvester law of inertia). *The numbers of positive, negative and zero squares in a real diagonalization by congruence are basis-independent.*

Proof. A congruence change is an invertible change of variables; maximal dimensions of subspaces on which the form is positive or negative definite are intrinsic and determine the counts. □

17.8 Worked examples

Example 17.6 (Conic form). $x^2 + 2xy + y^2 = (x+y)^2$ has rank one and is positive semidefinite.

Example 17.7 (Indefinite form). $x^2 - y^2$ has inertia $(1, 1, 0)$ and vanishes on the two lines $x = \pm y$.

Example 17.8 (Hessian viewpoint). At a critical point of a smooth function, the Hessian defines a quadratic form whose signature describes local second-order behavior.

17.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

17.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 17.9 (Matrix of a quadratic form). Find the symmetric matrix for $q(x, y) = 3x^2 + 4xy + 2y^2$.

Solution. $q = [x \ y] \begin{pmatrix} 3 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ because the cross term is $2a_{12}xy = 4xy$. $\square$

Problem 17.10 (Polarization). Recover the symmetric bilinear form B from a real quadratic form q .

Solution. $B(u, v) = \frac{1}{2}(q(u+v) - q(u) - q(v))$. Expanding $q(u+v) = B(u, u) + 2B(u, v) + B(v, v)$ proves the formula. $\square$

Problem 17.11 (Congruence versus similarity). Why does a basis change send a quadratic-form matrix to P^TAP ?

Solution. If old coordinates are $x = Py$, then $q = x^T Ax = y^T P^T AP y$. Both copies of the coordinate vector change, producing congruence rather than similarity. $\square$

Problem 17.12 (Definiteness by eigenvalues). How can the spectral theorem test positive definiteness of a real symmetric matrix?

Solution. Orthogonally diagonalize $A = QDQ^T$. Then $x^T Ax = y^T Dy = \sum_i \lambda_i y_i^2$. This is positive for all nonzero x exactly when every eigenvalue is positive. $\square$

Problem 17.13 (Indefinite form). Classify $q(x, y) = x^2 - y^2$.

Solution. It takes positive values, such as at $(1, 0)$, and negative values, such as at $(0, 1)$, so it is indefinite. Its inertia is $(1, 1, 0)$. $\square$

Problem 17.14 (Rank-one semidefinite form). Classify $q(x, y) = (x + y)^2$.

Solution. The form is nonnegative and vanishes on the nonzero line $x + y = 0$, so it is positive semidefinite but not definite. Its matrix has rank 1. $\square$

Problem 17.15 (Diagonalizing a cross term). Diagonalize $2xy$ by an orthogonal change of variables.

Solution. Set $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$. Then $x = (u + v)/\sqrt{2}$, $y = (u - v)/\sqrt{2}$, so $2xy = u^2 - v^2$. $\square$

Problem 17.16 (Inertia invariance). What intrinsic meaning do the positive and negative inertia indices have?

Solution. They are the maximal dimensions of subspaces on which the quadratic form is positive definite or negative definite. Because those maximal dimensions are basis-independent, the inertia is invariant under congruence. $\square$

Problem 17.17 (Hessian interpretation). At a critical point, how does the Hessian quadratic form classify a nondegenerate point?

Solution. Positive definite Hessian gives a strict local minimum, negative definite gives a strict local maximum, and an indefinite Hessian gives a saddle point. The signature captures the second-order geometry. $\square$

Problem 17.18 (Principal-minor criterion). For $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, state the positive-definite criterion using leading principal minors.

Solution. Sylvester's criterion gives $a > 0$ and $ac - b^2 > 0$. These conditions are equivalent to both eigenvalues being positive. $\square$

Problem 17.19 (Null directions). What does a zero eigenvalue of a symmetric quadratic-form matrix mean geometrically?

Solution. It gives a nonzero direction v with $Av = 0$, hence $q(v) = 0$ and no quadratic curvature in that direction. The form is degenerate. $\square$

Problem 17.20 (Quadratic-form synthesis). Why is signature more appropriate than individual matrix entries as geometric data?

Solution. Entries depend strongly on coordinates, while signature is invariant under invertible changes of variables. It records the intrinsic counts of positive, negative, and null directions. $\square$

17.11 Exercises with complete solutions

Exercise 17.21. Find the symmetric matrix for $q(x, y) = 2x^2 + 6xy + 5y^2$.

Hint. Write the quadratic form as $x^T Ax$ with A symmetric by averaging the mixed-term coefficients. $\square$

Solution. $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$ because the cross term is $2a_{12}xy$. $\square$

Exercise 17.22. Recover the bilinear form from a quadratic form by polarization.

Hint. Pass to the symmetric or Hermitian matrix of the form and inspect its eigenvalues or a valid definiteness criterion. Diagonalize the symmetric matrix orthogonally; the diagonal entries are the principal quadratic coefficients. $\square$

Solution. Use $B(u, v) = \frac{1}{2}(q(u + v) - q(u) - q(v))$. $\square$

Exercise 17.23. Why does a basis change act by congruence rather than similarity?

Hint. Write each new basis vector in the old coordinates and use those coordinate columns to build the conversion matrix. For definiteness, inspect the signs of all eigenvalues rather than individual matrix entries. $\square$

Solution. Both copies of the coordinate vector change in $x^T Ax$, producing $y^T P^T A P y$. $\square$

Exercise 17.24. Determine the definiteness of $x^2 - y^2$.

Hint. Use leading principal minors only when the hypotheses of Sylvester's criterion apply. $\square$

Solution. It is indefinite because it takes both positive and negative values. $\square$

Exercise 17.25. What is the inertia of $x^2 + y^2$?

Hint. Under a congruence transformation $A \mapsto P^T A P$, count positive and negative squares rather than eigenvalues themselves. $\square$

Solution. $(2, 0, 0)$: two positive directions, no negative or null directions. $\square$

Exercise 17.26. Show positive definite implies nonsingular.

Hint. Pass to the symmetric or Hermitian matrix of the form and inspect its eigenvalues or a valid definiteness criterion. Complete the square one variable at a time if an orthogonal diagonalization is unnecessary. $\square$

Solution. A nonzero kernel vector would satisfy $q(v) = v^T A v = 0$, contradicting positive definiteness. $\square$

Exercise 17.27. Diagonalize $2xy$ by a linear change of variables.

Hint. The signature is the number of positive squares minus the number of negative squares after diagonalization. $\square$

Solution. Set $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$; then $2xy = u^2 - v^2$. $\square$

Exercise 17.28. Explain the role of the spectral theorem for symmetric quadratic forms.

Hint. Find the eigenspaces first and use orthogonality between distinct eigenspaces before assembling an orthonormal eigenbasis. Test the form on strategically chosen vectors to refute positive or negative definiteness quickly. $\square$

Solution. Orthogonal diagonalization of the symmetric matrix expresses the form as a weighted sum of coordinate squares. $\square$

17.12 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 17.29 (Matrix extraction). Find the symmetric matrix of $q(x, y) = 5x^2 - 6xy + 2y^2$.

Hint. Remember the off-diagonal entry contributes twice. $\square$

Solution. $A = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}$. $\square$

Exercise 17.30 (Evaluate a form). For $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, compute $q(1, -1)$.

Hint. Evaluate $x^T A x$. $\square$

Solution. $q(1, -1) = 3$. $\square$

Exercise 17.31 (Complete the square). Diagonalize $x^2 + 2xy + 2y^2$ by completing the square.

Hint. Group $x^2 + 2xy = (x + y)^2 - y^2$. $\square$

Solution. $q = (x + y)^2 + y^2$, so the form is positive definite. $\square$

Exercise 17.32 (Sylvester criterion). Classify $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$.

Hint. Check leading principal minors. $\square$

Solution. $3 > 0$ and $\det A = 2 > 0$, so A is positive definite. $\square$

Exercise 17.33 (Inertia). Find the inertia of $q = x_1^2 + x_2^2 - 4x_3^2$.

Hint. Count positive, negative and zero diagonal coefficients. $\square$

Solution. The inertia is $(2, 1, 0)$. $\square$

Proofs

Exercise 17.34 (Polarization). Prove $B(u, v) = \frac{1}{2}(q(u+v) - q(u) - q(v))$ for a real symmetric bilinear form.

Hint. Expand $q(u+v) = B(u+v, u+v)$. □

Solution. Symmetry makes the two cross terms equal, giving $q(u+v) = q(u) + 2B(u, v) + q(v)$. □

Exercise 17.35 (Congruence law). Derive $A \mapsto P^T A P$ under $x = Py$.

Hint. Substitute the coordinate change into $x^T A x$. □

Solution. $q = y^T P^T A P y$. □

Exercise 17.36 (Positive definite nonsingular). Prove a positive-definite symmetric matrix is invertible.

Hint. A nonzero kernel vector would make the quadratic form vanish. □

Solution. If $Av = 0$ with $v \neq 0$, then $v^T Av = 0$, contradicting positive definiteness. □

Exercise 17.37 (Eigenvalue criterion). Prove a real symmetric matrix is positive definite iff all eigenvalues are positive.

Hint. Use orthogonal diagonalization. □

Solution. In eigen-coordinates $q = \sum \lambda_i y_i^2$, which is positive for all nonzero y exactly when all $\lambda_i > 0$. □

Counterexamples and hypothesis testing

Exercise 17.38 (Positive diagonal entries). Show positive diagonal entries do not guarantee positive definiteness.

Hint. Choose a symmetric matrix with a large off-diagonal term. □

Solution. $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has positive diagonal entries but eigenvalues $3, -1$, so it is indefinite. □

Exercise 17.39 (Similarity is wrong relation). Explain why similarity is not the natural coordinate change for quadratic forms.

Hint. Track both appearances of the coordinate vector in $x^T A x$. □

Solution. Substitution $x = Py$ gives $P^T A P$, not $P^{-1} A P$. □

Exercise 17.40 (Semidefinite not definite). Give a nonzero positive-semidefinite quadratic form that is not positive definite.

Hint. Use a square of one linear form. □

Solution. $(x+y)^2$ is nonnegative but vanishes on the nonzero line $x+y=0$. □

Applications and investigations

Exercise 17.41 (Hessian classification). At a nondegenerate critical point, what do positive, negative and mixed Hessian eigenvalues indicate?

Hint. Interpret the quadratic Taylor term. □

Solution. All positive gives a local minimum, all negative a local maximum, and mixed signs a saddle. □

Exercise 17.42 (Principal axes). Why does orthogonal diagonalization reveal principal axes of a conic or quadric?

Hint. Eigenvectors remove cross terms without distorting Euclidean angles. □

Solution. In orthonormal eigen-coordinates the form is a weighted sum of squares, so coordinate axes align with intrinsic principal directions. □

Challenge problems

Exercise 17.43 (Signature invariance). Explain why the numbers of positive and negative squares cannot change under invertible coordinate changes.

Hint. Interpret them as maximal dimensions of positive- and negative-definite subspaces. □

Solution. Those maximal dimensions are intrinsic properties of the form, so congruence cannot alter them; this is Sylvester's law of inertia. □

Exercise 17.44 (Degenerate form). For $q(x, y, z) = (x + y)^2 + (y + z)^2$, find the null space and rank.

Hint. Set both squares to zero. □

Solution. $x = -y$ and $z = -y$, so the null space is $\text{span}(-1, 1, -1)$; in dimension 3, rank is 2. □

Chapter summary

Quadratic forms encode second-order geometry. Symmetric bilinear forms and symmetric matrices represent the same finite-dimensional data once a basis is chosen. Congruence, not similarity, is the natural basis-change relation, and Sylvester's law of inertia identifies the basis-independent signature. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 18

Singular-Value Decomposition

The singular-value decomposition is the universal orthogonal factorization of a linear map between finite-dimensional inner-product spaces. Unlike eigenvalue decomposition, it applies to every rectangular matrix. Singular values quantify stretching, rank, conditioning and optimal low-rank approximation.

Learning goals

After this chapter, the reader should be able to:

- compute singular values from the eigenvalues of A^*A ;
- construct left and right singular vectors and assemble a singular-value decomposition;
- read rank, operator norm, and null-space information from the singular values;
- compute the Moore–Penrose pseudoinverse from an SVD;
- solve a least-squares problem with the pseudoinverse, including rank-deficient cases;
- form and interpret the best low-rank approximation obtained by truncating the SVD;

Conceptual roadmap

The chapter is organized around a repeated pattern:

definition	→	structural theorem	→	coordinate calculation	→	application.
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Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

18.1 Singular values

The singular values of A are the nonnegative square roots of the eigenvalues of A^*A .

Example 18.1 (Constructing an SVD from $A^T A$). Let

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$A^T A = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix},$$

so the singular values are 3 and 1, with right singular vectors $v_1 = e_1, v_2 = e_2$. The left singular vectors are

$$u_1 = \frac{Av_1}{3} = e_1, \quad u_2 = \frac{Av_2}{1} = e_2.$$

Extending by $u_3 = e_3$ gives $A = U\Sigma V^T$. Multiplying the factors back recovers A , providing the final construction check.

18.2 SVD theorem

Every $m \times n$ matrix has a factorization $A = U\Sigma V^*$ with U, V unitary (orthogonal over $\mathbb{R}$) and Σ diagonal rectangular with nonnegative entries.

18.3 Construction from the spectral theorem

Diagonalize A^*A to obtain right singular vectors; normalize their images under A to obtain left singular vectors.

18.4 Rank and norms

The rank is the number of positive singular values. The operator norm is the largest singular value, and the Frobenius norm squared is the sum of squared singular values.

18.5 Pseudoinverse

Replacing each nonzero singular value by its reciprocal gives the Moore–Penrose pseudoinverse.

Example 18.2 (Pseudoinverse and a rank-deficient least-squares problem). For

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

the rank is one. Write

$$A = 2uu^T, \quad u = \frac{1}{\sqrt{2}}(1, 1).$$

Hence

$$A^+ = \frac{1}{2}uu^T = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

For $b = (2, 0)^T$,

$$x_* = A^+b = (1/2, 1/2)^T.$$

The fitted vector $Ax_* = (1, 1)^T$ is the projection of b onto $\text{im } A$, and among all coefficient vectors producing that fit, x_* has minimum norm.

18.6 Low-rank approximation

Truncating the SVD after the largest r singular values gives a best rank- r approximation in operator and Frobenius norms.

Example 18.3 (Low-rank approximation and error read directly from singular values). Suppose a matrix has singular values

$$10, \quad 3, \quad 1, \quad 0.2.$$

The best rank-two approximation keeps only the first two SVD terms. The optimal spectral-norm error is

$$\sigma_3 = 1,$$

while the Frobenius error is

$$\sqrt{1^2 + 0.2^2} = \sqrt{1.04}.$$

Thus the singular-value tail quantifies exactly what is lost by compression. There is no need to search over all rank-two matrices: the truncated SVD is already optimal.

18.7 Core structural results

Theorem 18.4 (Singular-value decomposition). *For every matrix $A \in \mathbb{F}^{m \times n}$, there exist unitary U, V and a rectangular diagonal Σ with nonnegative entries such that $A = U\Sigma V^*$.*

Proof. Apply the spectral theorem to the positive semidefinite matrix A^*A . For each positive eigenvalue σ_i^2 , set $u_i = Av_i/\sigma_i$; extend the resulting orthonormal lists to bases. $\square$

Theorem 18.5 (Rank from singular values). *The number of nonzero singular values equals rank A .*

Proof. The kernels of A and A^*A coincide, so they have the same nullity. Rank-nullity then shows A^*A has as many positive eigenvalues as the rank of A . $\square$

18.8 Worked examples

Example 18.6 (Diagonal matrix). For a real diagonal matrix, singular values are the absolute values of its diagonal entries.

Example 18.7 (Rank-one matrix). If $A = uv^*$, there is a single nonzero singular value $\|u\|\|v\|$.

Example 18.8 (Condition number). For invertible A , the Euclidean condition number is $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$.

18.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

18.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 18.9 (Singular values of a diagonal matrix). Find the singular values of $\text{diag}(3, -2, 0)$.

Solution. $A^T A = \text{diag}(9, 4, 0)$, so the singular values are the nonnegative square roots 3, 2, 0. $\square$

Problem 18.10 (Kernel identity). Prove $\ker(A^* A) = \ker A$.

Solution. If $Av = 0$, then $A^* Av = 0$. Conversely, if $A^* Av = 0$, then $0 = \langle A^* Av, v \rangle = \langle Av, Av \rangle = \|Av\|^2$, so $Av = 0$. $\square$

Problem 18.11 (Rank from singular values). Why does the number of nonzero singular values equal $\text{rank } A$?

Solution. $A^* A$ has the same kernel as A , hence the same nullity. Rank-nullity then implies the number of positive eigenvalues of $A^* A$, i.e. nonzero singular values, equals the rank of A . $\square$

Problem 18.12 (Rank-one SVD). If $A = uv^*$ with nonzero u, v , what is its nonzero singular value?

Solution. Write $A = (u/\|u\|)(\|u\|\|v\|)(v/\|v\|)^*$. This is already a rank-one SVD, so the sole nonzero singular value is $\|u\|\|v\|$. $\square$

Problem 18.13 (Operator norm). Why is $\|A\|_2$ the largest singular value?

Solution. In an SVD $A = U\Sigma V^*$, unitary factors preserve norm. Thus maximizing $\|Ax\|$ over unit x reduces to maximizing $\|\Sigma y\|$, whose largest stretch is the largest diagonal singular value. $\square$

Problem 18.14 (Frobenius norm). Express $\|A\|_F^2$ in terms of singular values.

Solution. Unitary invariance of the Frobenius norm gives $\|A\|_F^2 = \|\Sigma\|_F^2 = \sum_i \sigma_i^2$. $\square$

Problem 18.15 (Pseudoinverse). How is the Moore–Penrose pseudoinverse constructed from an SVD?

Solution. If $A = U\Sigma V^*$, define $A^+ = V\Sigma^+ U^*$, where each nonzero singular value σ_i is replaced by $1/\sigma_i$ and zeros remain zero. $\square$

Problem 18.16 (Condition number). For invertible A , why is $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$?

Solution. $\|A\|_2 = \sigma_{\max}$ and $\|A^{-1}\|_2 = 1/\sigma_{\min}$. Their product is the stated ratio. $\square$

Problem 18.17 (Rectangular geometry). Why does the SVD use two different orthonormal bases?

Solution. A rectangular map has different domain and codomain spaces. Right singular vectors form an orthonormal basis of input directions; left singular vectors form orthonormal output directions. The diagonal Σ records the stretch between matched directions. $\square$

Problem 18.18 (Low-rank truncation). What is the rank of the matrix obtained by keeping only the first r nonzero SVD terms?

Solution. Each term $\sigma_i u_i v_i^*$ has rank one and the corresponding left/right vectors are orthogonal. Keeping r positive terms produces rank exactly r . $\square$

Problem 18.19 (Best approximation principle). State the Eckart–Young conclusion for truncated SVD.

Solution. The matrix obtained by keeping the largest r singular values is a best rank- r approximation in both spectral and Frobenius norms. The spectral error is σ_{r+1} and the squared Frobenius error is $\sum_{i>r} \sigma_i^2$. $\square$

Problem 18.20 (SVD synthesis). How does SVD unify rank, conditioning, least squares, and data compression?

Solution. Nonzero singular values determine rank; their ratio controls conditioning; the pseudoinverse solves least-squares/minimum-norm problems; and decay of singular values quantifies how accurately the map can be compressed to low rank. One factorization exposes all four phenomena. $\square$

18.11 Exercises with complete solutions

Exercise 18.21. Find the singular values of $\text{diag}(3, -2)$.

Hint. First determine λ from $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$ for the eigenspace. Compute A^*A ; the singular values are the square roots of its nonnegative eigenvalues. $\square$

Solution. They are 3 and 2, the square roots of eigenvalues 9 and 4 of $A^T A$. $\square$

Exercise 18.22. Why is A^*A positive semidefinite?

Hint. Choose orthonormal eigenvectors of A^*A as right singular vectors, then set $u_i = Av_i/\sigma_i$ for $\sigma_i > 0$. $\square$

Solution. $\langle A^*Av, v \rangle = \langle Av, Av \rangle = \|Av\|^2 \geq 0$. $\square$

Exercise 18.23. Show $\ker(A^*A) = \ker A$.

Hint. Count the nonzero singular values; that number is the rank. $\square$

Solution. If $Av = 0$ then clearly $A^*Av = 0$. Conversely, $A^*Av = 0$ implies $0 = \langle A^*Av, v \rangle = \|Av\|^2$. $\square$

Exercise 18.24. What is the rank of a matrix with singular values 5, 2, 0, 0?

Hint. Start from A^*A : its orthonormal eigenvectors are right singular vectors and its eigenvalues are the squared singular values. The operator norm induced by the Euclidean norm is the largest singular value. $\square$

Solution. The rank is 2, the number of nonzero singular values. $\square$

Exercise 18.25. Describe the pseudoinverse of a diagonal singular-value matrix.

Hint. Start from A^*A : its orthonormal eigenvectors are right singular vectors and its eigenvalues are the squared singular values. For the pseudoinverse, reciprocate only the nonzero singular values and transpose the SVD factors. $\square$

Solution. Transpose the rectangular shape and replace each positive σ_i by $1/\sigma_i$, leaving zeros unchanged. $\square$

Exercise 18.26. Why is SVD available for rectangular matrices while diagonalization is not?

Hint. Find the eigenspaces first and use orthogonality between distinct eigenspaces before assembling an orthonormal eigenbasis. In a rank-deficient least-squares problem, the pseudoinverse selects the minimum-norm coefficient vector. $\square$

Solution. SVD uses the square self-adjoint matrices A^*A and AA^* and two possibly different orthonormal bases; ordinary diagonalization requires an endomorphism. $\square$

Exercise 18.27. What does the largest singular value measure?

Hint. Start from A^*A : its orthonormal eigenvectors are right singular vectors and its eigenvalues are the squared singular values. For best rank- k approximation, keep the k largest singular values and discard the rest. $\square$

Solution. It is $\max_{\|x\|=1} \|Ax\|$, the maximum stretching factor. $\square$

Exercise 18.28. Explain truncated SVD conceptually.

Hint. Start from A^*A : its orthonormal eigenvectors are right singular vectors and its eigenvalues are the squared singular values. Verify the SVD by multiplying $U\Sigma V^*$ and by checking that the columns of U and V are orthonormal. $\square$

Solution. Keep only the dominant orthogonal input/output modes and their singular values; the discarded modes contribute the smallest possible approximation error for the prescribed rank. $\square$

18.12 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 18.29 (Diagonal SVD). Find the singular values of $\text{diag}(-4, 2, 0)$.

Hint. Take square roots of eigenvalues of $A^T A$. $\square$

Solution. The singular values are 4, 2, 0. $\square$

Exercise 18.30 (Rank from singular values). A matrix has singular values 7, 2, 0, 0. Find its rank.

Hint. Count positive singular values. $\square$

Solution. The rank is 2. $\square$

Exercise 18.31 (Operator norm). For singular values 5, 3, 1, compute $\|A\|_2$.

Hint. The Euclidean operator norm is the largest singular value. $\square$

Solution. $\|A\|_2 = 5$. $\square$

Exercise 18.32 (Pseudoinverse diagonal). Find the pseudoinverse of $\Sigma = \text{diag}(4, 0, 2)$.

Hint. Reciprocate nonzero singular values only. $\square$

Solution. $\Sigma^+ = \text{diag}(1/4, 0, 1/2)$. $\square$

Exercise 18.33 (Condition number). An invertible matrix has singular values 12, 4, 3. Find $\kappa_2(A)$.

Hint. Use $\sigma_{\max}/\sigma_{\min}$. □

Solution. $\kappa_2(A) = 12/3 = 4$. □

Proofs

Exercise 18.34 (Kernel identity). Prove $\ker(A^*A) = \ker A$.

Hint. Use $\langle A^*Av, v \rangle = \|Av\|^2$. □

Solution. One inclusion is immediate; the reverse follows because $A^*Av = 0$ implies $\|Av\|^2 = 0$. □

Exercise 18.35 (Rank from SVD). Prove the number of positive singular values equals rank A .

Hint. Use the kernel identity or the diagonal form $U\Sigma V^*$. □

Solution. Unitary factors preserve rank, and the rank of Σ is exactly the number of its nonzero diagonal entries. □

Exercise 18.36 (Operator norm). Prove $\|A\|_2 = \sigma_{\max}$.

Hint. Move to SVD coordinates, where unitary factors preserve norms. □

Solution. For $A = U\Sigma V^*$, maximizing $\|Ax\|$ over unit x reduces to maximizing $\|\Sigma y\|$, achieved in the largest-singular-value coordinate. □

Exercise 18.37 (Minimum-norm pseudoinverse solution). Why does A^+b give the minimum-norm solution when $Ax = b$ is consistent?

Hint. Decompose coefficient space into $(\ker A)^\perp \oplus \ker A$. □

Solution. The pseudoinverse solution lies in $(\ker A)^\perp$; every other solution adds a kernel vector orthogonal to it, increasing norm by Pythagoras. □

Counterexamples and hypothesis testing

Exercise 18.38 (Eigenvalues versus singular values). Show singular values are not generally absolute values of eigenvalues.

Hint. Use a nonnormal nilpotent matrix. □

Solution. For $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, both eigenvalues are 0, but singular values are 1, 0. □

Exercise 18.39 (Normal equations sensitivity). Why can A^TA hide numerical accuracy even though its eigenvalues give singular values squared?

Hint. Compare condition numbers. □

Solution. $\kappa_2(A^TA) = \kappa_2(A)^2$, so relative sensitivity is squared. □

Exercise 18.40 (Truncating arbitrary modes). Why is deleting arbitrary columns or coordinates not equivalent to optimal low-rank approximation?

Hint. SVD modes are selected by singular value and orthogonal input/output directions. $\square$

Solution. Arbitrary coordinate truncation depends on basis and need not minimize either spectral or Frobenius error. $\square$

Applications and investigations

Exercise 18.41 (Data compression). How does singular-value decay justify low-rank compression?

Hint. Compare the energy/error in the discarded singular-value tail. $\square$

Solution. If the tail singular values are small, truncated SVD stores only dominant modes while guaranteeing small optimal approximation error. $\square$

Exercise 18.42 (Ill-conditioning). What does a very small nonzero $\sigma_{\min}$ mean for solving $Ax = b$?

Hint. Relate $\|A^{-1}\|_2$ to the smallest singular value. $\square$

Solution. $\|A^{-1}\|_2 = 1/\sigma_{\min}$ is large, so perturbations in data can be strongly amplified. $\square$

Challenge problems

Exercise 18.43 (Eckart–Young error). A matrix has singular values 9, 4, 2, 1. State the best rank-two errors in spectral and Frobenius norms.

Hint. Discard the singular values after the second. $\square$

Solution. The spectral error is 2; the Frobenius error is $\sqrt{2^2 + 1^2} = \sqrt{5}$. $\square$

Exercise 18.44 (SVD and polar decomposition). For invertible $A = U\Sigma V^*$, derive a polar decomposition $A = QH$ with Q unitary and H positive.

Hint. Group UV^* and $V\Sigma V^*$. $\square$

Solution. Take $Q = UV^*$ and $H = V\Sigma V^* = (A^*A)^{1/2}$; then $QH = A$. $\square$

Chapter summary

The singular-value decomposition is the universal orthogonal factorization of a linear map between finite-dimensional inner-product spaces. Unlike eigenvalue decomposition, it applies to every rectangular matrix. Singular values quantify stretching, rank, conditioning and optimal low-rank approximation. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.